

AP EAMCET-2019-Engineering-21 Apr-Shift-1 Mathematics

1. Let $D = \mathbb{R} - \{0, 1\}$ and $f : D \rightarrow D$, $g : D \rightarrow D$ and $h : D \rightarrow D$ be three functions defined by $f(x) = \frac{1}{x}$; $g(x) = 1 - x$ and $h(x) = \frac{1}{1-x}$. If $j : D \rightarrow D$ is such that $(gojof)(x) = f(x)$ for all $x \in D$ then which of the following is $j(x)$?

(1) $(fog)(x)$

(2) $f(x)$

(3) $g(x)$

(4) $(goh)(x)$

2. The maximum value of function

$$f(x) = \tan\left(x + \frac{2\pi}{3}\right) - \tan\left(x + \frac{\pi}{6}\right) + \cos\left(x + \frac{\pi}{6}\right) \text{ in } \left[-\frac{5\pi}{12}, \frac{-\pi}{3}\right]$$

is

(1) $\frac{11\sqrt{2}}{6}$

$$(2) \frac{11\sqrt{3}}{6}$$

$$(3) 3$$

$$(4) 1$$

$$3. \quad 1^2 + (1^2 + 2^2) + (1^2 + 2^2 + 3^2) + \dots + (1^2 + 2^2 + \dots + n^2) =$$

$$(1) \frac{n(n+1)(n+2)}{12}$$

$$(2) \frac{n(n+1)(2n+1)}{6}$$

$$(3) \frac{n(n+1)^2(n+2)}{12}$$

$$(4) \frac{n(n+1)(n+2)(n+3)}{12}$$

$$4. \quad \text{If } \begin{bmatrix} a+b+2c & a & b \\ c & 2a+b+c & b \\ c & a & a+2b+c \end{bmatrix} = 2 \text{ then}$$

$$a^3 + b^3 + c^3 - 3abc =$$

$$(1) 1 - 3ab - 3bc - 3ca$$

$$(2) 0$$

$$(3) 1 - 2ab - 2bc - 2ca$$

(4) 1

5. If k is one of the roots of the equation $x^2 - 25x + 24 = 0$ such

that $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & k \end{bmatrix}$ is a non singular matrix then $A^{-1} =$

(1) $-\frac{1}{46} \begin{bmatrix} -90 & -94 & 8 \\ -138 & 46 & 0 \\ 2 & 2 & -8 \end{bmatrix}$

(2) $-\frac{1}{92} \begin{bmatrix} 45 & -41 & 4 \\ -69 & 23 & 0 \\ 1 & 1 & 4 \end{bmatrix}$

(3) $-\frac{1}{46} \begin{bmatrix} 45 & -47 & 4 \\ -69 & 23 & 0 \\ 1 & 1 & -4 \end{bmatrix}$

(4) $-\frac{1}{92} \begin{bmatrix} 90 & -94 & 8 \\ -138 & 46 & 0 \\ 2 & 2 & -8 \end{bmatrix}$

6. A value of b for which the rank of matrix $A = \begin{bmatrix} 1 & 1 & -1 & 0 \\ 4 & 4 & -3 & 1 \\ b & 2 & 2 & 2 \\ 9 & 9 & b & 3 \end{bmatrix}$ is

3, is

- (1) -2
- (2) -4
- (3) -6
- (4) 3

7. If $z = x - iy$ and $z^{\frac{1}{3}} = a + ib$, then $\frac{\left(\frac{x}{a} + \frac{y}{b}\right)}{a^2 + b^2} =$

- (1) -2
- (2) -1
- (3) 1
- (4) 2

8. The equation whose solution are the non zero solutions of the equation $\bar{z} = iz^2$, is

- (1) $z^3 + i = 0$
- (2) $z^3 + z + 1 = 0$

(3) $z^3 - i = 0$

(4) $z^3 + iz + 1 = 0$

9. If $x, y \in R$ and $x^2 + y + 4i$ and $-3 + x^2 yi$ are conjugate to each other, then $(|x| + |y|)^2 =$

(1) 17

(2) 16

(3) 25

(4) 9

10. $\sum_{k=1}^6 \left(\sin \frac{2\pi k}{7} - i \cos \frac{2\pi k}{7} \right) =$

(1) -1

(2) 0

(3) i

(4) $-i$

11. If α satisfies the equation $\sqrt{\frac{x}{2x+1}} + \sqrt{\frac{2x+1}{x}} = 2$, then the roots

of the equation $\alpha^2 x^2 + 4\alpha x + 3 = 0$ are

(1) 1, 3

- (2) $-1, 1$
- (3) $2, -3$
- (4) $3, 4$

12. Let $f(x) = x^2 + 2x + 2$, $g(x) = -x^2 + 2x - 1$ and a, b be the extreme values of $f(x), g(x)$ respectively. If c is the extreme value of $\frac{f}{g}(x)$ (for $x \neq 1$), then $a + 2b + 5c + 4 =$

- (1) 2
- (2) 1
- (3) 4
- (4) 3

13. The set of real number satisfying the equation $x^2 - |x + 2| + x > 0$ is

- (1) $[-2, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
- (2) $(-\infty, -2) \cup (2, \infty)$
- (3) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
- (4) $(-\infty, -2) \cup (\sqrt{2}, \infty)$

14. If α, β, γ are the roots of $x^3 - 6x^2 + 11x - 6 = 0$ then the equation having the roots $\alpha^2 + \beta^2, \beta^2 + \gamma^2, \gamma^2 + \alpha^2$ is

- (1) $x^3 - 28x^2 + 245x - 650 = 0$
- (2) $x^3 - 28x^2 + 245x + 650 = 0$
- (3) $x^3 + 28x^2 - 245x - 650 = 0$
- (4) $x^3 + 28x^2 + 245x - 650 = 0$

15. If the number of elements in set G and A are 3 and 4 respectively, then match the items in List I with those in List II

List-I	List-II
(A) The number of non bijective functions from $G \times G$ to G (I)	24
(B) The number of bijective functions from A to A (II)	0
(C) The number of functions from G to $G \times A$ (III)	1728
(D) The number of surjective functions from A to $A \times A$ (IV)	12
	(V) 19683

The correct match is

(1) A B C D
V I III II

(2) A B C D
V III IV II

(3) A B C D
III IV V II

(4) A B C D
I II III IV

16. There are 20 straight lines in a plane such that no two of them are parallel and no three of them are concurrent. If their point of intersection are joined, then the number of new line segments formed are,

(1) 3420

(2) 14535

(3) 2907

(4) 17955

17. Let $a_0, a_1, a_2, \dots, a_n \in R$ be an arithmetic progression and let

$C_0, C_1, C_2, \dots, C_n$ be the binomial coefficient then $\sum_{k=0}^n a_k C_k =$

$$(1) \frac{1}{2}(a_0 + a_n)$$

$$(2) (a_0 + a_n)2^{n-1}$$

$$(3) a_0 + a_n$$

$$(4) 0$$

18. If $x = \frac{3}{10} + \frac{3 \cdot 7}{10 \cdot 15} + \frac{3 \cdot 7 \cdot 9}{10 \cdot 15 \cdot 20} + \dots$, then $5x + 8 =$

$$(1) \frac{5\sqrt{5}}{3\sqrt{3}}$$

$$(2) \frac{5\sqrt{5}}{\sqrt{3}}$$

$$(3) \frac{3\sqrt{3}}{\sqrt{5}}$$

$$(4) \frac{25\sqrt{5}}{3\sqrt{3}}$$

19. If $\frac{x^4}{(x-1)(x-2)(x-3)} = x + k + \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$ then

$$k + A - B + C =$$

$$(1) 104$$

$$(2) 52$$

(3) 63

(4) $\frac{127}{2}$

20. $\operatorname{cosec}48^\circ + \operatorname{cosec}96^\circ + \operatorname{cosec}192^\circ + \operatorname{cosec}384^\circ =$

(1) -2

(2) -1

(3) 0

(4) $\frac{\sqrt{3}}{2}$

21. $\left[\sin^4 \frac{\pi}{8} + \sin^4 \frac{2\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{4\pi}{8} + \sin^4 \frac{5\pi}{8} \right. \\ \left. + \sin^4 \frac{6\pi}{8} + \sin^4 \frac{7\pi}{8} \right] =$

(1) $\frac{3}{2}$

(2) $\frac{5}{2}$

(3) 3

(4) $\frac{7}{2}$

22. If $x : y : z = \tan\left(\frac{\pi}{15} + \alpha\right) : \tan\left(\frac{\pi}{15} + \beta\right) : \tan\left(\frac{\pi}{15} + \gamma\right)$, then

$$\frac{z+x}{z-x} \sin^2(\gamma - \alpha) + \frac{x+y}{x-y} \sin^2(\alpha - \beta) + \frac{y+z}{y-z} \sin^2(\beta - \gamma) =$$

(1) $\sin^2 \theta$

(2) $\cos^2 \theta$

(3) 0

(4) 1

23. Let $[x]$ denotes the largest integer $\leq x$. if the number of solutions

of $\sin x \sqrt{4 \cos^2 x} = \frac{2+x-[x]}{1-x+[x]}$ is k then for $x \in \left[\frac{\pi}{4}, \frac{\pi}{3}\right]$, the value

of $k^{\tan^2 x}$

(1) is equal to 1

(2) lies in between 2^1 to 2^3

(3) is equal to zero

(4) lies in between $\frac{1}{2^3}$ and $\frac{1}{2}$

24. If α and β are the least and greatest values of

$f(x) = (\sin^{-1} x)^2 + (\cos^{-1} x)^2$ for all $x \in R$ respectively, then

$8(\alpha + \beta) =$

(1) π^2

(2) $11\pi^2$

(3) $9\pi^2$

(4) $25\pi^2$

25. If $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, then $\log \sec x =$

(1) $2 \operatorname{cosech}^{-1} \left(\cot^2 \frac{x}{2} - 1 \right)$

(2) $2 \operatorname{cosech}^{-1} \left(\cot^2 \frac{x}{2} + 1 \right)$

(3) $2 \cot h^{-1} \left(\operatorname{cosec}^2 \frac{x}{2} - 1 \right)$

(4) $2 \cot h^{-1} \left(\operatorname{cosec}^2 \frac{x}{2} + 1 \right)$

26. The area (in square units) of ΔABC if $\angle A = 75^\circ, \angle B = 45^\circ$ and

$$a = 2(\sqrt{3} + 1) \text{ is}$$

(1) 6

(2) $2\sqrt{3}$

(3) $6 - 2\sqrt{3}$

(4) $6 + 2\sqrt{3}$

27. In a ΔABC , If $3a = b + c$, then $\cot \frac{B}{2} \cot \frac{C}{2} =$

(1) 1

(2) 2

(3) $\frac{1}{3}$

(4) $\frac{1}{2}$

28. In a ΔABC , If $\frac{2r_2 r_3}{r_2 - r_1} = r_3 - r_1$, then $\frac{r_1(r_2 + r_3)}{\sqrt{r_1 r_2 + r_2 r_3 + r_3 r_1}} =$

(1) $\frac{a^2 + b^2 + c^2}{\Delta^2}$

(2) $b - c$

(3) $\frac{1}{2R}$

(4) $2R$

29. If $3\vec{i} - 2\vec{j} - \vec{k}, 2\vec{i} + 3\vec{j} - 4\vec{k}, -\vec{i} + \vec{j} + 2\vec{k}, 4\vec{i} + 5\vec{j} + \lambda\vec{k}$ are respectively the position vectors of four coplanar points P, Q, R and S then $\lambda =$

(1) $\frac{46}{17}$

(2) $-\frac{46}{17}$

(3) $\frac{146}{17}$

(4) $-\frac{146}{17}$

30. If $\vec{OA} = 2\vec{i} + 2\vec{j} + \vec{k}, \vec{OB} = 2\vec{i} + 4\vec{j} + 4\vec{k}$ and the length of the internal bisector of $\angle BOA$ of triangle AOB is k , then $9k^2 =$

(1) $\sqrt{225}$

(2) 136

(3) 712

(4) 20

31. If $\bar{a} + x\bar{b} + y\bar{c} = 0$, $\bar{a} \times \bar{b} + \bar{b} \times \bar{c} + \bar{c} \times \bar{a} = 6(\bar{b} \times \bar{c})$, then the locus of the point (x, y) is

- (1) $x^2 + y^2 = 1$
- (2) $x + y - 5 = 0$
- (3) $2x + 6y = 5$
- (4) $x + y + 6 = 0$

32. Let $A = (\alpha, 1, 2\alpha)$, $B = (3, 1, 2)$ and $\bar{C} = 4\bar{i} - \bar{j} + 3\bar{k}$. If

$$\overline{AB} \times \bar{C} = 6\bar{i} + 9\bar{j} - 5\bar{k} \text{ then } \alpha^2 + \alpha + 5 =$$

- (1) 11
- (2) 7
- (3) 9
- (4) 5

33. The shortest distance between the skew lines

$$\bar{r} = (6\bar{i} + 2\bar{j} + 2\bar{k}) + t(\bar{i} - 2\bar{j} + 2\bar{k}) \text{ and}$$

$$\bar{r} = (-4\bar{i} - \bar{k}) + s(3\bar{i} - 2\bar{j} - 2\bar{k}) \text{ is}$$

- (1) 9

(2) $\frac{40}{7}$

(3) 108

(4) 120

34. If $\vec{a}$ makes an acute angle with $\vec{b}$, $\vec{r} \cdot \vec{a} = 0$ and $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$, then $\vec{r} =$

(1) $\vec{a} \times \vec{c} - \vec{b}$

(2) $\vec{c} \times \vec{a}$

(3) $\vec{c} - \left(\frac{\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a}} \right) \vec{b}$

(4) $\vec{c} + \left(\frac{\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a}} \right) \vec{b}$

35. For a data consisting of 15 observations $x_i, i = 1, 2, 3, \dots, 15$ the following results are obtained:

$$\sum_{i=1}^{15} x_i = 170, \sum_{i=1}^{15} x_i^2 = 2830. \text{ If one of the observations namely 20}$$

was found wrong and was replaced by its correct value 30, then the corrected variance is

(1) 80

(2) 78

(3) 76

(4) 75

36. The mean deviation about the mean for the following data is

Class-Interval	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70
Frequency	4	6	16	28	16	6	4

(1) 35

(2) 10

(3) 15

(4) 12

37. A and B each select one number at random from the distinct numbers $1, 2, 3, \dots, n$ and the probability that the number selected by A is less than the number selected by B is $\frac{1009}{2019}$.

Now, the probability that the number selected by B is the number immediately next to the number selected by A is

(1) $\frac{2018}{2019}$

(2) $\frac{2018}{(2019)^2}$

$$(3) \frac{2000}{(2019)}$$

$$(4) \frac{2000}{(2019)^2}$$

38. There are 3 bags A, B and C. Bag A contains 2 white and 3 black balls, bag B contains 4 white and 2 black balls and Bag C contains 3 white and 2 black balls. If a ball is drawn at random from a randomly chosen bag, then the probability that the ball drawn is black, is

$$(1) \frac{2}{3}$$

$$(2) \frac{4}{9}$$

$$(3) \frac{5}{9}$$

$$(4) \frac{1}{9}$$

39. The following table shows the probability of selecting the boxes A, B and C and the number of balls of different colours contained in them

Box	Number of balls			Probability
	White	Green	Red	
A	1	2	3	$\frac{1}{2}$
B	2	3	1	$\frac{1}{3}$
C	3	1	2	$\frac{1}{6}$

A box is selected at random and a ball is drawn from it. If it is given that the ball drawn is green, then the probability it has come from box C is

(1) $\frac{1}{13}$

(2) $\frac{6}{13}$

(3) $\frac{5}{13}$

(4) $\frac{7}{13}$

40. If a random variable X has the probability distribution given by $P(X=0)=3C^3$, $P(X=2)=5C-10C^2$ and $P(X=4)=4C-1$, then the variance of that distribution is

(1) $\frac{68}{9}$

(2) $\frac{22}{9}$

(3) $\frac{612}{81}$

(4) $\frac{128}{81}$

41. A box contains 30 toys of same size in which 10 toys are white and all the remaining toys are blue. A toy is drawn from the box and it is replaced in the box after noting down its colour. If 5 toys are drawn in this way, then the probability of getting atmost 2 white toys is

(1) $\left(\frac{6}{9}\right)^2$

(2) $\left(\frac{8}{9}\right)^2$

(3) $\left(\frac{7}{9}\right)^2$

(4) $\left(\frac{2}{3}\right)^5$

42. For any value of θ if the straight lines $x \sin \theta + (1 - \cos \theta)y = a \sin \theta$ and $x \sin \theta - (1 + \cos \theta)y + a \sin \theta$ intersect at $P(\theta)$, then the locus of $P(\theta)$ is a

- (1) Straight line
- (2) circle
- (3) parabola
- (4) hyperbola

43. A line L has intercepts a and b on the coordinate axes. When the axes are rotated through a given angle θ keeping the origin fixed, this line L has the intercepts p and q . Then

- (1) $a^2 + b^2 = p^2 + q^2$
- (2) $a^2 + p^2 = b^2 + q^2$
- (3) $\frac{1}{a^2} + \frac{1}{p^2} = \frac{1}{b^2} + \frac{1}{q^2}$
- (4) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2} + \frac{1}{q^2}$

44. If O is the origin and A and B are points on the line $3x - 2y + 25 = 0$ such that $\overline{OA} = \overline{OB} = 13$, then the area of ΔOAB (in square units) is

- (1) 30
- (2) 120
- (3) 60
- (4) 65

45. If $P(\alpha, \beta)$ be a point on the line $3x + y = 0$ such that the point P and the point $Q(1, 1)$ lie on the either side of the line $3x = 4y + 8$, then

- (1) $\alpha < \frac{8}{15}, \beta < \frac{-8}{5}$
- (2) $\alpha > \frac{8}{15}, \beta < \frac{-8}{5}$
- (3) $\alpha > \frac{8}{15}, \beta > \frac{-8}{5}$
- (4) $\alpha < \frac{8}{15}, \beta > \frac{-8}{5}$

46. Two vertices of a triangle are $(5, -1)$ and $(-2, 3)$. If the origin is the orthocenter of this triangle, then the coordinates of this triangle, then the coordinates of the third vertex of that triangle are

(1) $(4, 7)$

(2) $\left(-2, \frac{-7}{2}\right)$

(3) $(-4, -7)$

(4) $(-2, 3)$

47. The distance from the origin to the orthocenter of the triangle formed by the lines $x + y - 1 = 0$ and $6x^2 - 13xy + 5y^2 = 0$ is

(1) $\frac{11\sqrt{2}}{2}$

(2) 13

(3) 11

(4) $\frac{11\sqrt{2}}{24}$

48. The combined equation of line L and L_1 is $2x^2 + axy + 3y^2 = 0$ and the combined equation of two lines L and L_2 is

$2x^2 + bxy - 3y^2 = 0$. If L_1 and L_2 are perpendicular, then

$$a^2 + b^2 =$$

(1) 26

(2) 29

(3) 13

(4) 85

49. The power of the point $B(-1, 1)$ with respect to the circle $S \equiv x^2 + y^2 - 2x - 4y + 3 = 0$ is p . If the length of the tangent drawn from B to the circle $S \equiv 0$ is t , then the point $(2, 3)$ with respect to the circle $S' \equiv 0$ having centre at (p, t^2) and passing through the origin

(1) lies inside the circle $S' \equiv 0$

(2) lies outside the circle $S' \equiv 0$

(3) lies on the circle $S' \equiv 0$

(4) is the centre of the circle $S' \equiv 0$

50. If tangents are drawn to the circle $x^2 + y^2 = 12$ at the points where it intersects the circle $x^2 + y^2 - 5x + 3y - 2 = 0$, then the coordinate of the point of intersection of those tangents are

$$(1) \left(-6, \frac{18}{5} \right)$$

$$(2) \left(6, \frac{18}{5} \right)$$

$$(3) \left(-6, -\frac{18}{5} \right)$$

$$(4) \left(6, -\frac{18}{5} \right)$$

51. If the point of intersection of the pair of the transverse common tangents and that of the pair of direct common tangents drawn to the circles $x^2 + y^2 - 14x + 6y + 33 = 0$ and $x^2 + y^2 + 30x - 2y + 1 = 0$ are T and D respectively, then the centre of the circle having $\overline{TD}$ as diameter is

$$(1) \left(\frac{39}{2}, \frac{-7}{4} \right)$$

$$(2) \left(\frac{39}{4}, \frac{7}{2} \right)$$

$$(3) \left(\frac{39}{4}, \frac{-7}{2} \right)$$

$$(4) \left(\frac{39}{2}, \frac{-7}{2} \right)$$

52. If the circles $x^2 + y^2 + 2\lambda x + 2 = 0$ and $x^2 + y^2 + 4y + 2 = 0$ touch each other, then $\lambda =$

- (1) ± 1
- (2) ± 2
- (3) ± 3
- (4) ± 4

53. The equation of the circle whose diameter is the common chord of the circles $x^2 + y^2 + 2x + 3y + 1 = 0$ and $x^2 + y^2 + 4x + 3y + 2 = 0$ is

- (1) $2x^2 + 2y^2 + x + 3y + 2 = 0$
- (2) $2x^2 + 2y^2 + 2x + 6y + 1 = 0$
- (3) $2x^2 + 2y^2 + 4x - 3y - 1 = 0$
- (4) $x^2 + y^2 + 2x + 6y - 2 = 0$

54. If the focus of a parabola divides a focal chord of the parabola into segments of lengths 5, 3 units, then the lengths of the latus rectum of that parabola is

- (1) $\frac{15}{4}$

(2) 20

(3) $\frac{25}{2}$

(4) $\frac{15}{2}$

55. The angle between the tangent drawn to the parabola $y^2 = 4x$ from the point $(1, 4)$ is

(1) $\frac{\pi}{4}$

(2) $\frac{\pi}{3}$

(3) $\frac{2\pi}{5}$

(4) $\frac{\pi}{6}$

56. If the tangent drawn to the parabola $y^2 = 4x$ at $(t^2, 2t)$ is the normal to the ellipse $4x^2 + 5y^2 = 20$ at $(\sqrt{5} \cos \theta, 2 \sin \theta)$, then

(1) $5t^4 + 4t^2 = 1$

(2) $\frac{5}{t^4} + \frac{100}{t^2} = 1$

(3) $t = \sin \theta$

(4) $\cos \theta = t + 1$

57. If the tangents drawn from a point P to the ellipse $4x^2 + 9y^2 - 24x + 36y = 0$ are perpendicular, then the locus of P is

(1) $x^2 + y^2 - 6x + 4y + 13 = 0$

(2) $x^2 + y^2 - 6x + 4y - 13 = 0$

(3) $x^2 + y^2 = 26$

(4) $x^2 + y^2 + 6x - 4y - 13 = 0$

58. The locus of the mid-point of the chord of the circle $x^2 + y^2 = 16$ which are the tangents to the hyperbola $9x^2 - 16y^2 = 144$ is

(1) $3x^2 - 4y^2 = 16(x^2 + y^2)$

(2) $4x^2 - 3y^2 = 9(x^2 + y^2)$

(3) $16x^2 - 9y^2 = (x^2 + y^2)^2$

(4) $16x^2 - 9y^2 = 4(x^2 + y^2)$

59. $A(3, 2, -1)$, $B(4, 1, 1)$, $C(6, 2, 5)$ are three points. If D , E , F are three points which divide BC , CA , AB respectively in the same ratio $2:1$, then the centroid of $\triangle DEF$ is

(1) $\left(\frac{13}{3}, \frac{5}{3}, \frac{5}{3}\right)$

(2) $(13, 5, 5)$

(3) $(4, 2, 1)$

(4) $\left(\frac{11}{3}, \frac{4}{3}, \frac{1}{3}\right)$

60. If $A(1, 8, 4)$, $B(2, 3, -1)$. then the direction cosines of a normal to the plane AOB is

(1) $\frac{2}{\sqrt{78}}, \frac{5}{\sqrt{78}}, \frac{-7}{\sqrt{78}}$

(2) $\frac{2\sqrt{10}}{9}, \frac{7\sqrt{10}}{90}, \frac{-19\sqrt{10}}{90}$

(3) $\frac{4}{\sqrt{218}}, \frac{9}{\sqrt{218}}, \frac{-11}{\sqrt{218}}$

(4) $\frac{2}{11}, \frac{6}{11}, \frac{-9}{11}$

61. If the two lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$ have a point in common, then $k =$

(1) $\frac{2}{9}$

(2) $-\frac{2}{9}$

(3) $\frac{9}{2}$

(4) 0

62. $\lim_{x \rightarrow 0} \frac{x^2 (\tan 2x - 2 \tan x)^2}{(1 - \cos 2x)^4} =$

(1) 4

(2) 2

(3) $\frac{1}{2}$

(4) $\frac{1}{4}$

63. $\lim_{x \rightarrow -\infty} \left(\frac{6x^2 - \cos 3x}{x^2 + 5} - \frac{5x^3 + 3}{\sqrt{x^6 + 2}} \right) =$

(1) 11

(2) 0

(3) -1

(4) 1

64. The number of discontinuities in $\mathbb{R}$ for the function

$$f(x) = \frac{x-1}{x^3 + 6x^2 + 11x + 6} \text{ is}$$

(1) 3

(2) 2

(3) 1

(4) 0

65. $\frac{d}{dx} \left(\log \left(\sqrt{x + \sqrt{x^2 + x^2}} \right) \right) =$

(1) $\sqrt{x^2 + a^2}$

(2) $\frac{1}{\sqrt{x^2 + a^2}}$

(3) $\frac{1}{2\sqrt{x^2 + a^2}}$

$$(4) \frac{1}{2(x + \sqrt{x^2 + a^2})}$$

66. If $f(x) = \cot^{-1}\left(\frac{x^x - x^{-x}}{2}\right)$, then $f'(1) =$

(1) $-\log 2$

(2) $\log 2$

(3) 1

(4) -1

67. If $a \neq 0$, $x = a(t + \sin t)$ and $y = a(1 - \cos t)$, then $\frac{d^2y}{dx^2}$ at $t = \frac{2\pi}{3}$

is

(1) $\frac{4}{a}$

(2) $\frac{1}{4a}$

(3) $4a$

(4) $\frac{a}{4}$

68. The number of tangents to the curve

$y^2(x-a) = x^2(x+a) (a > 0)$ that are parallel to the X-axis is

(1) infinitely many

(2) 0

(3) 1

(4) 2

69. If $f(x) = (2k+1)x - 3 - ke^{-x} + 2e^x$ is normally increasing for all

$x \in \mathbb{R}$ then the least value of k is

(1) 1

(2) 0

(3) $-\frac{1}{2}$

(4) -1

70. If the function $f(x) = ax^3 + bx^2 + 11x - 6$ satisfy the conditions

of Rolle's theorem in $[1, 3]$ and $f'\left(2 + \frac{1}{\sqrt{3}}\right) = 0$, then $a + b =$

(1) -5

(2) -3

(3) 4

(4) 7

71. For $a > 0$, If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$ attains its maximum value at p and minimum value at q such that $p^2 = q$, then $a =$

(1) $\frac{1}{2}$

(2) 1

(3) 2

(4) 4

72. If $\int \cos x \cdot \cos 2x \cdot \cos 5x dx = A \sin 2x + B \sin 4x + C \sin 6x + D \sin 8x + k$

(where k is the arbitrary constant of integration), then $\frac{1}{B} + \frac{1}{C} =$

(1) $\frac{1}{A} - \frac{1}{D}$

(2) $\frac{1}{A} + \frac{1}{D}$

(3) 1

(4) 0

73. If $\int e^x \left(\frac{x+2}{x+4} \right)^2 dx = f(x) + \text{arbitrary constant}$, then $f(x) =$

(1) $\frac{xe^x}{x+4}$

(2) $\frac{e^x}{x+4}$

(3) $\frac{xe^x}{(x+4)^2}$

(4) $\frac{e^x}{(x+4)^2}$

74. $\int \frac{dx}{\sin x + \sin 2x} =$

(1) $\frac{1}{2} \log_e |1 + \cos x| + \frac{1}{6} \log_e |1 - \cos x| - \frac{2}{3} \log_e |1 + 2 \cos x| + c$

(2) $\frac{1}{3} \log_e |1 + \cos x| - \frac{2}{3} \log_e |1 - \cos x| + \frac{1}{2} \log_e |1 + 2 \cos x| + c$

(3) $\frac{1}{2} \log_e |1 + \sin x| - \frac{1}{3} \log_e |1 - \sin x| - \frac{1}{3} \log_e |1 + \cos x| + c$

(4) $\frac{1}{3} \log_e |1 - \sin x| + \frac{1}{2} \log_e |1 + \cos x| - \frac{2}{3} \log_e |1 - 2 \cos x| + c$

75. If $I_n = \int \frac{\sin nx}{\sin x} dx$, for $n = 1, 2, 3, \dots$ then $I_6 =$

(1) $\frac{3}{5} \sin 3x + \frac{8}{3} \sin^5 x - \sin x + c$

(2) $\frac{2}{5} \sin 5x - \frac{5}{3} \sin^3 x - 2 \sin x + c$

(3) $\frac{2}{3} \sin 5x - \frac{8}{3} \sin^5 x + 4 \sin x + c$

(4) $\frac{2}{5} \sin 5x - \frac{8}{3} \sin^3 x + 4 \sin x + c$

76. If $\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right) \dots \left(1 + \frac{n^2}{n^2}\right) \right]^{\frac{1}{n}} = k$ then $\log k =$

(1) $\log 4 + \frac{\pi}{2} - 1$

(2) $\log 2 + \frac{\pi}{2} + 1$

(3) $\log 2 + \frac{\pi}{2} - 2$

(4) $\log 2 + \frac{\pi}{2} - 1$

77. $\int_0^{\frac{\pi}{2}} \frac{\sin^3 x \cos x dx}{\sin^4 x + \cos^4 x} =$

(1) π

(2) $\frac{\pi}{2}$

(3) $\frac{\pi}{4}$

(4) $\frac{\pi}{8}$

78. The curve $y = ax^2 + bx$ passes through the point $(1, 2)$ and lies above the X-axis for $0 \leq x \leq 8$. If the area enclosed by this curve, the X-axis and the line $x = 6$ is 108 square units, then $2b - a =$

(1) 2

(2) 0

(3) 1

(4) -1

79. The differential equation of all parabolas whose axes are parallel to Y-axis is

(1) $\frac{d^3 y}{dx^3} = 0$

$$(2) \frac{d^2 y}{dx^2} = 0$$

$$(3) \frac{d^2 y}{dx^2} + \frac{dy}{dx} = 0$$

$$(4) y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = 0$$

80. The solution of the equation $\frac{dy}{dx} + 2y \tan x = \sin x$ satisfying

$y = 0$ when $x = \frac{\pi}{3}$, is

$$(1) y = 2 \sin^2 x + \cos x - 2$$

$$(2) y = 2 \sin^2 x - \cos x - 2$$

$$(3) y = 2 \cos^2 x - \sin x + 2$$

$$(4) y = 2 \cos x - \sin^2 x - 1$$

AP EAMCET-2019-Engineering-21 Apr-Shift-1 Physics

81. Two intervals of time are measured as $\Delta t_1 = (2.00 \pm 0.02) \text{ s}$ and $\Delta t_2 = (4.00 \pm 0.02) \text{ s}$. The value of $\sqrt{(\Delta t_1)(\Delta t_2)}$ with correct significant figures and error is
- (1) $(2.828 \pm 0.01) \text{ s}$
 - (2) $(2.83 \pm 0.01) \text{ s}$
 - (3) $(2.828 \pm 0.0075) \text{ s}$
 - (4) $(2.83 \pm 0.0075) \text{ s}$
82. The speed of the particle changes from $\sqrt{5} \text{ ms}^{-1}$ to $2\sqrt{5} \text{ ms}^{-1}$ in a time t . If the magnitude of change in the velocity is 5 ms^{-1} , the angle between the initial and final velocities of the particle is
- (1) 30°
 - (2) 45°
 - (3) 60°
 - (4) 90°

83. The maximum height and range of the projectile are 3 m and 4 m respectively then the velocity of the projectile is

(Acceleration due to gravity = 10 ms^{-2})

(1) $20\sqrt{\frac{6}{5}} \text{ ms}^{-1}$

(2) $10\sqrt{\frac{3}{2}} \text{ ms}^{-1}$

(3) $10\sqrt{\frac{2}{3}} \text{ ms}^{-1}$

(4) $20\sqrt{\frac{5}{6}} \text{ ms}^{-1}$

84. A body is projected at an angle other than 90° with the horizontal with same velocity. If the time of ascent of the body is 1 s then the maximum height it can reach is

(Acceleration due to gravity = 10 ms^{-2})

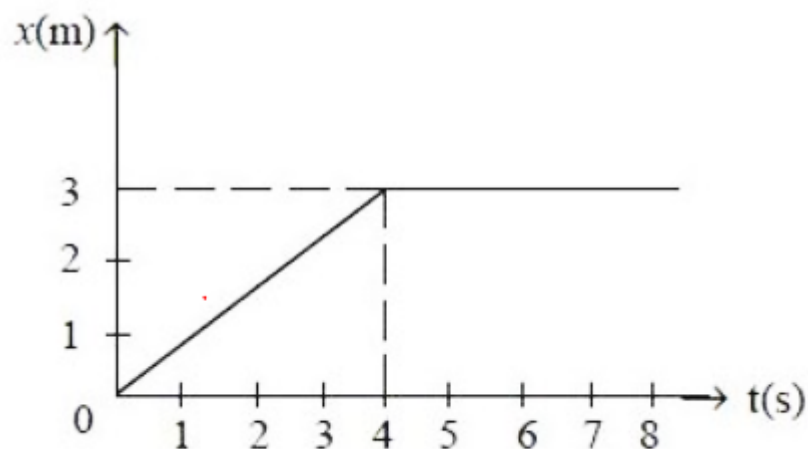
(1) 5 m

(2) 10 m

(3) 2.5 m

(4) 75 m

85. The position – time ($x-t$) graph of a moving body of mass 2 kg is shown in the figure. The impulse on the body at $t = 4$ s is

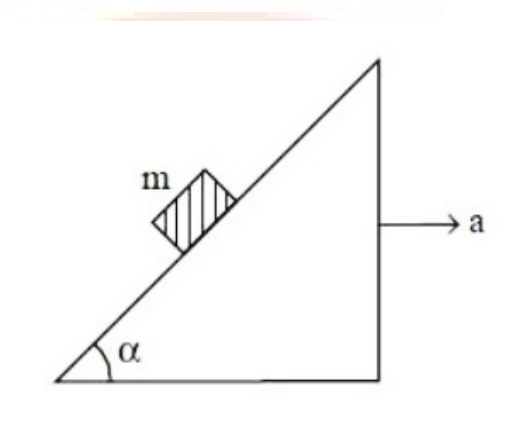


- (1) $1.5 \text{ kg} \cdot \text{ms}^{-1}$
- (2) $-1.5 \text{ kg} \cdot \text{ms}^{-1}$
- (3) $1 \text{ kg} \cdot \text{ms}^{-1}$
- (4) $2 \text{ kg} \cdot \text{ms}^{-1}$

86. A block of mass ' m ' is lying on a rough inclined plane having an inclination $\alpha = \tan^{-1}\left(\frac{1}{5}\right)$. The inclined plane is moving horizontally with constant acceleration of $a = 2 \text{ ms}^{-2}$ as shown in the figure. The minimum value of the coefficient of friction,

so that the block remains stationary with respect to the inclined plane is

(Acceleration due to gravity = 10 ms^{-2})



(1) $\frac{2}{9}$

(2) $\frac{5}{12}$

(3) $\frac{1}{5}$

(4) $\frac{2}{5}$

87. Potential energy of a body of mass 1 kg free to move along the

X-axis is given by $U(x) = \left(\frac{x^2}{2} - x \right) \text{ J}$. If the total mechanical

energy of the body is 2 J, then the maximum speed of the body is (Assume only conservative force acts in the body)

(1) $\sqrt{5} \text{ ms}^{-1}$

(2) 5 ms^{-1}

(3) 3.5 ms^{-1}

(4) $\sqrt{8} \text{ ms}^{-1}$

88. A cylindrical well of radius 2.5 m has water up to a height of 14 m from bottom. If the water level is at a depth of 6 m from the top of the well, then time taken in minutes to empty the well using a motor of 10 HP is approximately

(Acceleration due to gravity = 10 ms^{-2})

(1) 30

(2) 80

(3) 98

(4) 90

89. A fly wheel of mass 1 kg and radius vector $(2\hat{i} + \hat{j} + 2\hat{k})$ m is at rest. When a force $(3\hat{i} + 2\hat{j} - 4\hat{k})$ N acts on it tangentially, it can rotate freely. Then. Its angular velocity after 4.5 s is

(1) $\frac{2}{9}\sqrt{261} \text{ rad s}^{-1}$

(2) $\frac{3}{2}\sqrt{261} \text{ rad s}^{-1}$

(3) $\sqrt{261} \text{ rad s}^{-1}$

(4) $\frac{5}{9}\sqrt{261} \text{ rad s}^{-1}$

90. Three identical spheres each of diameter $2\sqrt{3}$ are kept on a horizontal surface such that each sphere touches the other two spheres. If one of the sphere is removed, then the shift in the position of the centre of mass of the system is

(1) 12 m

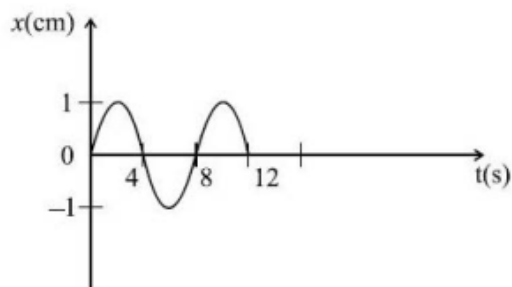
(2) 1 m

(3) 2 m

(4) $\frac{3}{2}$ m

91. For a particle executing simple harmonic motion, the displacement – time ($x-t$) graph is as shown in the figure. The

acceleration of the particle $t = \frac{4}{3}$ s is



(1) $-\frac{\sqrt{3}}{32}\pi^2 \text{ cm s}^{-2}$

(2) $-\frac{32}{\sqrt{3}}\pi^2 \text{ cm s}^{-2}$

(3) $+\frac{\sqrt{3}}{32}\pi \text{ cm s}^{-2}$

(4) $+\frac{32}{\sqrt{3}}\pi \text{ cm s}^{-2}$

92. Two masses of 90 kg and 160 kg are separated by a distance of 5 m. The magnitude of intensity of the gravitational field at a point which is at a distance 3 m from the 90 kg mass and 4m from the 160 kg mass is

(Universal gravitational constant = $6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$).

(1) $94.3 \times 10^{-10} \text{ N kg}^{-1}$

(2) $9.43 \times 10^{-10} \text{ N kg}^{-1}$

(3) $9.43 \times 10^{-12} \text{ N kg}^{-1}$

(4) $94.3 \times 10^{-12} \text{ N kg}^{-1}$

93. The following four wires are made of the same material. If same tension is applied to each, the wire having largest extension is

(1) length 0.5 m, diameter 0.5 mm.

(2) length 1 m, diameter 1 mm.

(3) length 2 m, diameter 2 mm.

(4) length 3 m, diameter 3 mm.

94. A liquid drop of density ρ is floating half immersed in a liquid of surface tension S and density $\frac{\rho}{2}$. If the surface tension S of the liquid is numerically equal to 10 times of acceleration due to gravity, then the diameter of the drop is :

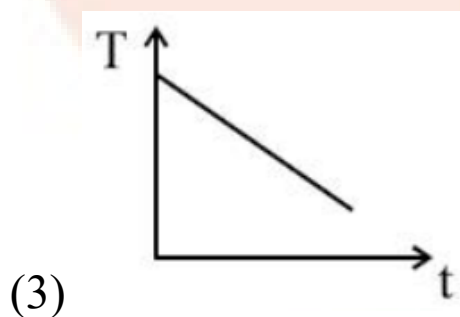
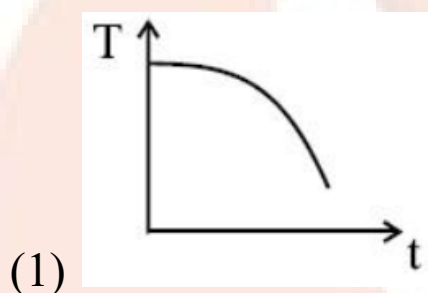
(1) $\sqrt{\frac{20}{\rho}}$

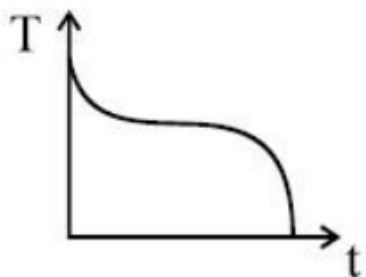
(2) $\sqrt{\frac{80}{\rho}}$

(3) $\sqrt{\frac{60}{\rho}}$

$$(4) \sqrt{\frac{40}{\rho}}$$

95. The block of metal is heated to a temperature much higher than the room temperature and placed in an evacuated cavity. The curve which correctly represents the rate of cooling (T is temperature of the block and t is the time)





(4)

96. A solid copper sphere of density ρ , specific heat capacity C and radius r is initially at 200 K. It is suspended inside a chamber whose walls are at 0 K. The time required (in μs) for the temperature of the sphere to drop to 100 K is

(σ is Stefan's constant and all the quantities are in SI units)

(1) $48 \frac{r\rho C}{\sigma}$

(2) $\frac{1}{48} \frac{r\rho C}{\sigma}$

(3) $\frac{27}{7} \frac{r\rho C}{\sigma}$

(4) $\frac{7}{27} \frac{r\rho C}{\sigma}$

97. Match the temperatures of the source and sink (T_1 and T_2 respectively) of a Carnot heat engine given in List-I with the corresponding efficiencies in List-II.

List-I

(A) $T_1 = 500 \text{ K}, T_2 = 300 \text{ K}$

(B) $T_1 = 500 \text{ K}, T_2 = 350 \text{ K}$

(C) $T_1 = 800 \text{ K}, T_2 = 400 \text{ K}$

(D) $T_1 = 450 \text{ K}, T_2 = 360 \text{ K}$

List-II

(I) 0.2

(I) 0.3

(III) 0.4

(IV) 0.5

The correct answer is

A	B	C	D
(1) (III)	(IV)	(II)	(I)
(2) (IV)	(III)	(II)	(I)
(3) (III)	(I)	(IV)	(II)
(4) (III)	(II)	(IV)	(I)

98. A hammer of mass 200 kg strikes a steel block of mass 200 g with a velocity 8 ms^{-1} . If 23% of the energy is utilized to heat the steel block, the rise in temperature of the block is (specified heat capacity of the steel = $460 \text{ J kg}^{-1} \text{ K}^{-1}$)

- (1) 8°C
- (2) 16°C
- (3) 12°C
- (4) 24°C

99. At the temperature of 314 K and a pressure of 100 kPa, the speed of the sound in a gas is 1380 ms^{-1} . The radius of each gas molecule is $0.5 \text{ \AA}$. The frequency of sound wave at which the wavelength of sound wave in the gas becomes equal to the mean free path of the gas molecules is

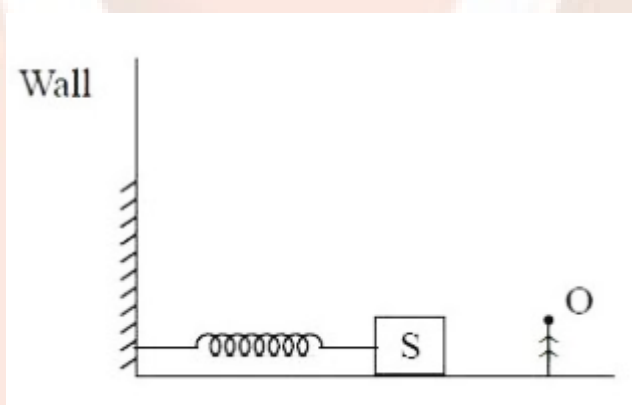
(Boltzmann constant = $1.38 \times 10^{-23} \text{ JK}^{-1}$)

- (1) 1000 MHz
- (2) $1000\sqrt{2}$ MHz
- (3) $\frac{1000}{\sqrt{2}}$ MHz
- (4) 500 MHz

100. At a temperature of 27°C , two identical organ pipe produces a note of frequency 140 Hz. If the temperature of one pipe is raised to 57.75°C , then the number of beats produced per second is

- (1) 7
- (2) 5
- (3) 3
- (4) 9

101. A source of sound S in the form of a block kept on smooth horizontal surface is connected to a spring as shown in the figure. If the spring oscillates with an amplitude of 50 cm along the horizontal between the wall and the observer O , the maximum frequency heard by the observer is 12.5 % more than the minimum frequency heard by him. If the mass of the source of sound is 100 g, the force constant of the spring is (Speed of sound in air is 340 ms^{-1})



- (1) 40 N m^{-1}
- (2) 80 N m^{-1}
- (3) 160 N m^{-1}
- (4) 320 N m^{-1}

102. A girl of height 150 cm with her eye level at 140 cm stands in front of plane mirror of height 75 cm fixed to a wall. The lower

edge of the mirror is at a height of 85 cm above her feet level.

The height of her image the girl can see the mirror is

- (1) 130 cm
- (2) 140 cm
- (3) 120 cm
- (4) 150 cm

103. Unpolarised light from the air incidents on the surface of a transparent medium of refractive index 1.414 such that the reflected light is completely polarised. Match the angle given in the List-I with the corresponding values given in the List-II.

List-I

List-II

(A) Angle of reflection

(I) $2\sin^{-1}\left(\sqrt{\frac{2}{3}}\right)$

(B) Angle of refraction

(II) $2\sin^{-1}\left(\sqrt{\frac{2}{3}}\right) - \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(C) Angle between incident and completely polarised rays

(III) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(D) Angle of deviation of the incident ray

(IV) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

The correct answer is

A	B	C	D
(1) (II)	(III)	(I)	(IV)
(2) (II)	(III)	(IV)	(I)
(3) (IV)	(I)	(III)	(II)
(4) (IV)	(III)	(I)	(II)

104. The electric field intensity at a point on the axis of an electric dipole in air is 4 NC^{-1} . Then the electric field intensity at a point on the equatorial line which is at the distance equal to twice the distance on the axial line and if the dipole is in a medium of dielectric constant 4 is

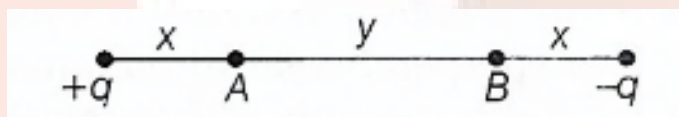
- (1) 1 NC^{-1}
- (2) $\frac{1}{8} \text{ NC}^{-1}$
- (3) 16 NC^{-1}
- (4) $\frac{1}{16} \text{ NC}^{-1}$

105. Two small spheres of each charge ' q ', mass ' m ' and material density ' d ' are suspended from a fixed point with the help of inextensible light thread. When the spheres are in air, the angle

between the threads is 90° . When the spheres are suspended in a liquid of density $\frac{2}{3}d$, the angle between the threads is 60° . The value of dielectric constant of the liquid is

- (1) $6\sqrt{3}$
- (2) $2\sqrt{5}$
- (3) $5\sqrt{3}$
- (4) $7\sqrt{2}$

106. The potential difference ($V_A - V_B$) in the arrangement shown in the figure is ($|q| = 1 \mu\text{C}$, $x = 2 \text{ cm}$, $y = 3 \text{ cm}$)



- (1) $5.4 \times 10^5 \text{ V}$
- (2) $2.7 \times 10^5 \text{ V}$
- (3) $5.4 \times 10^2 \text{ V}$
- (4) $2.7 \times 10^2 \text{ V}$

107. In a parallel plate capacitor the separation between the plates is $3x$. This separation is filled by two layers of dielectrics, in which

one layer has thickness x and dielectric constant $3 K$, the other layer is of thickness $2x$ and dielectric constant $5 K$. If the plates of the capacitor are connected to a battery, then the ratio of potential difference across the dielectric layers is

- (1) $\frac{1}{2}$
- (2) $\frac{4}{3}$
- (3) $\frac{3}{5}$
- (4) $\frac{5}{6}$

108. Assertion (A) : When a wire of aluminium and another wire of silicon are heated from room temperature to 80°C , the conductivity of aluminium increases and that of silicon decreases.

Reason (R) : Aluminium has positive temperature coefficient of resistivity and silicon has negative temperature coefficient of resistivity.

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct but (R) is correct.

109. The walls of a closed cubical box of edge 60 cm are made of material of thickness 1 mm and thermal conductivity $4 \times 10^{-4} \text{ cal s}^{-1} \text{ cm}^{-1} \text{ }^{\circ}\text{C}^{-1}$. The interior of the box is maintained 1000°C above the outside temperature by a heater placed inside the box and connected across 400 V DC supply. The resistance of the heater is

(1) 4.41Ω

(2) 44.1Ω

(3) 0.441Ω

(4) 441Ω

110. A galvanometer of resistance $G \Omega$, is shunted by resistance $S \Omega$. To keep the main current in the circuit unchanged, the resistance to be connected in series with the galvanometer is

(1) $\frac{G^2}{S + G}$

$$(2) \frac{S}{S+G}$$

$$(3) \frac{S^2}{S+G}$$

$$(4) \frac{SG}{S+G}$$

111. A proton and an α -particle are simultaneously projected in opposite direction into a region of uniform magnetic field of 2 mT perpendicular to the direction of the field. After some time it is found that the velocity of proton has changed in direction by 90° . Then at this time, The angle between the velocity vectors of proton and α particle is

$$(1) 60^\circ$$

$$(2) 90^\circ$$

$$(3) 45^\circ$$

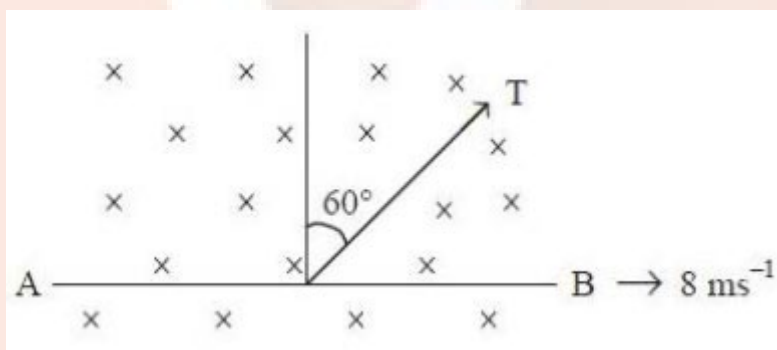
$$(4) 180^\circ$$

112. A bar magnet is placed in a uniform magnetic field making an angle θ with the field experiences a torque. If the angle made by the magnet with the field is doubled, the torque experienced by

the magnet increases by 41.4 %. The initial angle made by the magnet with the magnetic field is

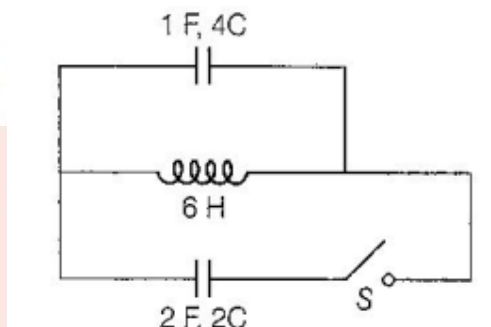
- (1) 60°
- (2) 30°
- (3) 90°
- (4) 45°

113. A metal rod AB of length 50 cm is moving at a velocity of 8 ms^{-1} in a magnetic field of 2 T. If the field is at 60° with the plane of motion as shown in the figure, then the potentials V_A and V_B are related by



- (1) $V_A - V_B = 8 \text{ V}$
- (2) $V_A - V_B = 4 \text{ V}$
- (3) $V_B - V_A = 8 \text{ V}$
- (4) $V_B - V_A = 4 \text{ V}$

114. In the given electrical circuit, if the switch S is closed then the maximum energy stored in the inductors is :



- (1) 3 J
- (2) 9 J
- (3) 12 J
- (4) 6 J

115. Which of the following is/are the property/properties of a monochromatic electromagnetic wave propagating in the free space?

- (a) Electric and magnetic field will have phase difference $\frac{\pi}{2}$.
- (b) The energy of the wave is distributed equally between electric and magnetic fields.
- (c) The pressure exerted by the wave is the product of its speed and energy density.

(d) The speed of the wave is equal to the ratio of the magnetic field to the electric field.

(1) (a) and (c)

(2) only (b)

(3) (b) and (c)

(4) only (d)

116. The maximum kinetic energy of a photoelectron liberated from the surface of lithium with the work function 2.35 eV by electromagnetic radiation whose electric component varies with the time as:

$$E = a[1 + \cos(2\pi f_1 t)] \cos 2\pi f_2 t \quad (\text{where 'a' is constant}) \text{ is (}$$

$$f_1 = 3.6 \times 10^{15} \text{ Hz} \quad \text{and} \quad f_2 = 1.2 \times 10^{15} \text{ Hz} \quad \text{and} \quad \text{Planck's constant } h = 6.6 \times 10^{-34} \text{ Js})$$

(1) 2.64 eV

(2) 7.55 eV

(3) 12.52 eV

(4) 17.45 eV

117. Magnetic moment due to the motion of the electron in the n^{th} energy state of hydrogen atom is proportional to _____

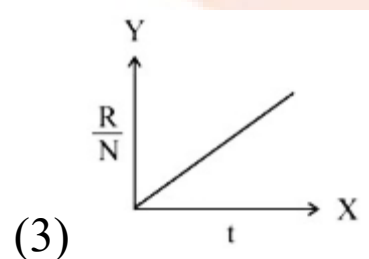
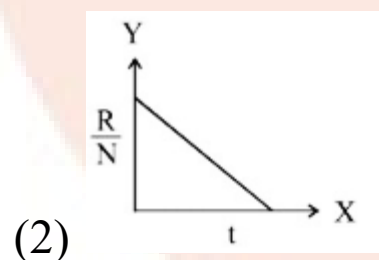
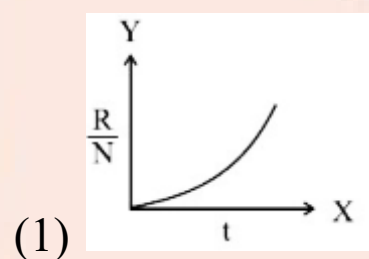
(1) n^{-2}

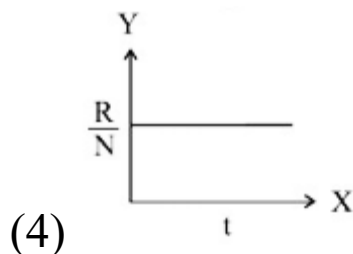
(2) n

(3) n^2

(4) n^3

118. The rate of disintegration of a radioactive sample is 'R' and the number of atoms present at any time 't' is 'N'. When $\frac{R}{N}$ is taken along Y-axis and 't' is taken along the X-axis, the correct graph is





119. For an LED to emit light in the visible region of the electromagnetic spectrum, it can have energy band gap in the range of

(Plank's constant $= 6.6 \times 10^{-34}$ Js and speed of light $= 3 \times 10^8$ ms⁻¹ in vacuum)

- (1) 0.1 eV to 0.4 eV
- (2) 0.9 eV to 1.6 eV
- (3) 1.7 eV to 3.1 eV
- (4) 0.5 eV to 0.8 eV

120. A transmitting antenna of height 20 m and the receiving antenna of height 'h' are separated by a distance of 40 km for satisfactory communication in the line of sight mode. Then the value of 'h' is

(Given Radius of earth is 6400 km)

- (1) 40 m

(2) 45 m

(3) 30 m

(4) 25 m



AP EAMCET-2019-Engineering-21 Apr-Shift-1 Chemistry

121. The energies of an electron in first orbit of He^+ and in third orbit of Li^{2+} in J are respectively

- (1) $-8.72 \times 10^{-18}, -2.18 \times 10^{-18}$
- (2) $-8.72 \times 10^{-18}, -1.96 \times 10^{-17}$
- (3) $-1.96 \times 10^{-17}, -2.18 \times 10^{-18}$
- (4) $-8.72 \times 10^{-17}, -1.96 \times 10^{-17}$

122. How many orbital/s is/are possible with $n = 3, l = 1, m_l = -1$?

- (1) 2
- (2) 3
- (3) 5
- (4) 1

123. If four elements with atomic numbers $Z - 2, Z - 1, Z$ and $Z + 1$ are forming isoelectronic ions, the atomic number of the ion having largest size is

- (1) $Z - 2$
- (2) $Z - 1$

(3) Z

(4) $Z+1$

124. Identify the molecule in which the arrangement of electron pairs around the central atom is octahedral and shape is not octahedral

(1) SF_6

(2) XeF_6

(3) BrF_5

(4) XeO_2F_4

125. The wave functions of $1s$ orbitals of two hydrogen atoms are ψ_A and ψ_B . ψ_A and ψ_B are linearly combined to form two molecular orbitals (σ and σ^*). Which of the following statements are correct?

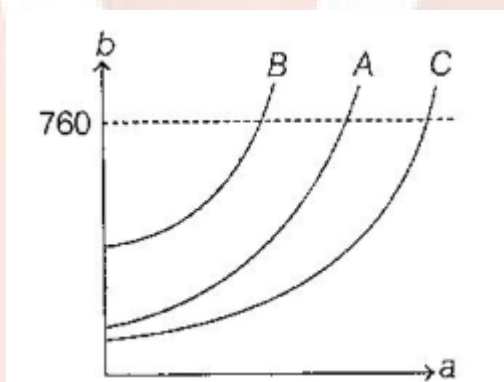
I) σ^* is equal to $(\psi_A - \psi_B)$

II) In σ orbital, one nodal plane is present in between two nuclei

III) The energy of σ orbital is lower than the energy of σ^* orbital.

- (1) I, II, III
- (2) I, II only
- (3) II, III only
- (4) I, III only

126. The variation of vapour pressure (b) as a function of temperature (a) is studied for $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$, CCl_4 and H_2O at 760 mm Hg and is shown in the figure below. The boiling temperatures of $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$, CCl_4 and H_2O are 308, 350 and 373 K respectively. Curves A, B, C respectively correspond to



- (1) H_2O , $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$, CCl_4
- (2) $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$, CCl_4 , H_2O
- (3) CCl_4 , $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$, H_2O
- (4) CCl_4 , H_2O , $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$

127. 30.0 mL of the given HCl solution requires 20.0 mL of 0.1 M sodium carbonate solution for complete neutralisation. What is the volume of this HCl solution required to neutralise 30.0 mL of 0.2 M NaOH solution?

- (1) 25 mL
- (2) 50 mL
- (3) 90 mL
- (4) 45 mL

128. The heat required to rise the temperature of 54 g of aluminium from 40°C to 60°C in J is (molar heat capacity of aluminium in this temperature range is $24 \text{ J mol}^{-1} \text{ K}^{-1}$; atomic weight of Al is 27)

- (1) 480
- (2) 800
- (3) 960
- (4) 1280

129. The equilibrium constant at 850 K for the reaction

$\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$ is 0.5625. The equilibrium concentration of $\text{NO}(\text{g})$ is $3.0 \times 10^{-3} \text{ M}$. If the equilibrium

concentrations of $\text{N}_2(\text{g})$ and $\text{O}_2(\text{g})$ are equal, the concentration of $\text{N}_2(\text{g})$ in M is-

- (1) 4.0×10^{-3}
- (2) 4.0×10^{-2}
- (3) 1.6×10^{-3}
- (4) 3.0×10^{-3}

130. The solubility product of a sparingly soluble salt A_2B is

3.2×10^{-11} . Its solubility in mol L^{-1} is

- (1) 4×10^{-4}
- (2) 2×10^{-4}
- (3) 6×10^{-4}
- (4) 3×10^{-4}

131. What is the volume (in mL) of 20 vol H_2O_2 required to completely react with 500 mL of 0.02 M acidified KMnO_4 solution?

- (1) 14.0
- (2) 7.0
- (3) 28.0

(4) 42.0

132. KO_2 , reacts with water to form A, B and C. B forms C when it reacts with iodine in basic medium. What are B and C respectively?

(1) KOH , H_2O_2

(2) K_2O_2 , H_2O_2

(3) KOH , O_2

(4) H_2O_2 , O_2

133. Identify the correct statements from the following:

I) Ga_2O_3 is an amphoteric oxide.

II) The dimer of aluminium chloride has three Al-Cl-Al bridge bonds.

III) Boron is very hard refractory solid of high melting temperature.

(1) I, II only

(2) I, III only

(3) II, III only

(4) I, II, III

134. Which one of the following methods is used to prepare carbon monoxide on commercial scale?

- (1) dehydration of formic acid with conc. H_2SO_4
- (2) direct oxidation of C in limited supply of oxygen
- (3) passing steam over hot coke
- (4) heating lime stone

135. Match the following.

List-I

- A) Insecticide
- B) $\text{K}_2\text{Cr}_2\text{O}_7 / 50\%\text{H}_2\text{SO}_4$
- C) Bleaching of cloths and paper
- D) Eye Irritant

List-II

- I) COD
- II) PAN
- III) Na_3AsO_3
- IV) BOD
- V) H_2O_2

A B C D
(1) III IV V II

A B C D
(2) III I V II

A B C D
(3) III I II V

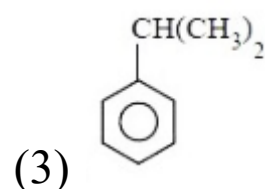
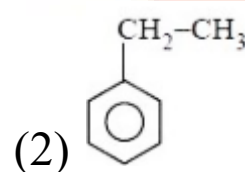
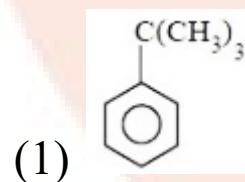
- A B C D
(4) V I III II

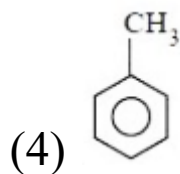
136. For which of the following, Kjeldahl's method is not used for the estimation of nitrogen?

Aniline Azobenzene Nitrobenzene Pyridine
(I) (II) (III) (IV)

- (1) II, III, IV
(2) II, III only
(3) III, IV only
(4) I, III, IV

137. Identify the compound, which has maximum number of no bond resonance structures





138. Which of the following statements are correct?

- I) In nitrating mixture nitric acid participates as an acid
- II) σ complex is the intermediate substance in electrophilic substitution of benzene
- III) Benzene on Friedel-Crafts alkylation with n-propyl chloride gives isopropyl benzene

- (1) II, III only
- (2) I, II only
- (3) I, III only
- (4) I, II, III

139. Which one of the following functional groups is not meta directing?

- (1) $-\text{COOH}$
- (2) $-\text{NO}_2$
- (3) $-\text{CHO}$
- (4) $-\text{OCH}_3$

140. If the radius of an atom of an element which forms a body centered cubic unit cell is 173.2 pm. the volume of unit cell in cm^3 is

(1) 3.12×10^{-23}

(2) 6.4×10^{-24}

(3) 3.2×10^{-24}

(4) 2.13×10^{-23}

141. A solution is prepared by dissolving 10 g of a non-volatile solute (molar mass. ' M ' g mol^{-1}) in 360 g of water. What is the molar mass in g mol^{-1} of solute if the relative lowering of vapour pressure of solution is 5×10^{-3} ?

(1) 199

(2) 99.5

(3) 299

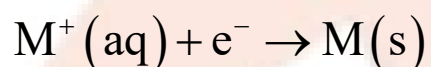
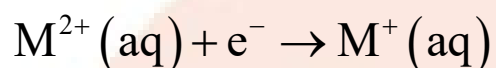
(4) 149.5

142. $\underline{x}$ g of MgSO_4 ($i = 1.8$) is in 2.5 L of solution has an osmotic pressure of 2.463 atm at 27°C . What is the value of $\underline{x}$ in g?

(1) 33.2

- (2) 6.6
- (3) 3.3
- (4) 16.6

143. The electrode potentials for



are +0.15 V and +0.50 V respectively. The value of $E_{M^{2+}/M}^0$ will be

- (1) 0.150 V
- (2) 0.300 V
- (3) 0.325 V
- (4) 0.650 V

144. The half life periods of a first order reaction at 300 K and 400 K are 50 s and 10 s respectively. The activation energy of the reaction in kJ mol^{-1} is ($\log 5 = 0.70$)

- (1) 4.0
- (2) 8.0
- (3) 16.10
- (4) 20.10

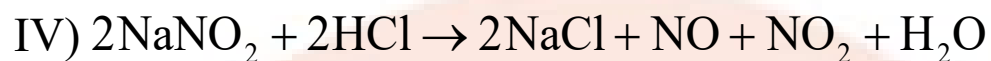
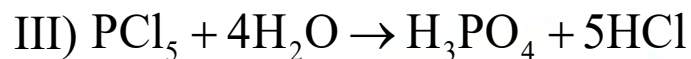
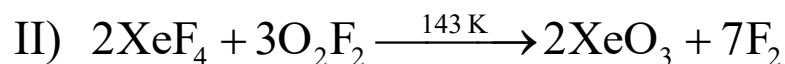
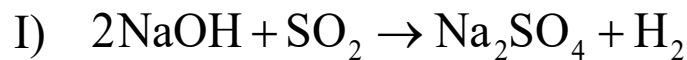
145. Which one of the following statement is correct for adsorption of solutes on solids in solutions?

- (1) The extent of adsorption increases with an increase in temperature.
- (2) The extent of adsorption decreases with an increase of surface area of the adsorbate.
- (3) The extent of adsorption decreases with an increase in temperature.
- (4) The extent of adsorption does not depend on the amount of the solute in solution.

146. Identify the metal which is **not** common to German Silver and Brass.

- (1) Cu
- (2) Zn
- (3) Fe
- (4) Ni

147. Which of the following reactions are correct with respect to the formation of products?



(1) II, IV

(2) III, IV

(3) I, III

(4) II, III

148. Chlorine oxidises sulphurdioxide in the presence of water to give an oxyacid A. Chlorine also oxidises iodine in the presence of water to give an oxyacid B. The oxidation states of S and I in A and B are respectively

(1) +4, +5

(2) +6, +3

(3) +6, +5

(4) +4, +7

149. White phosphorous is heated with concentrated NaOH in CO_2 , atmosphere to form a gas A and compound B. When A is

bubbled into aqueous CuSO_4 solution copper phosphide and C are formed. B and C are respectively.

- (1) $\text{PH}_3, \text{H}_2\text{SO}_4$
- (2) $\text{NaH}_2\text{PO}_2, \text{H}_2\text{SO}_4$
- (3) $\text{NaHPO}_2, \text{CuS}$
- (4) $\text{NaH}_2\text{PO}_2, \text{Cu}_2\text{S}$

150. Which of the following set of elements do not possess f-electrons?

- (1) La, U, Lr
- (2) La, Th, Lr
- (3) La, Ac, Th
- (4) Ce, Ac, Th

151. The Δ_0 of a coordination complex of a metal ion ($3d^1$) is 1000 kJ mol^{-1} . If the energy of t_{2g} orbitals is -400 kJ mol^{-1} , the energy (in kJ mol^{-1}) of e_g orbitals is

- (1) -600
- (2) 600
- (3) 1000

(4) 400

152. How many of the following polymers given, come under the category of condensation polymers?

Bakelite, Teflon, Nylon 6, Dacron, Polyisoprene, Melamine, Neoprene

(1) 4

(2) 3

(3) 5

(4) 6

153. Identify the correct set from the following.

Vitamin	Source	deficiency disease
(1) B ₆	Milk	convulsions

Vitamin	Source	deficiency disease
(2) K	Leaf vegetables	anaemia

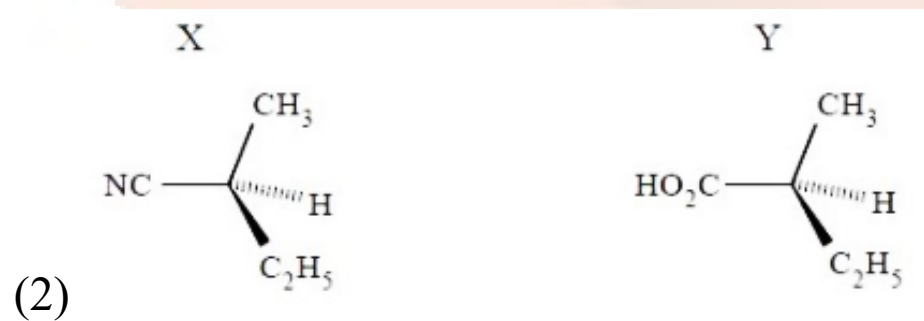
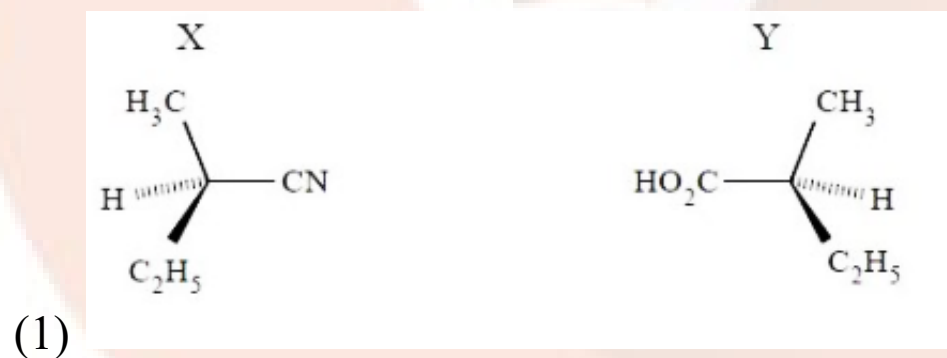
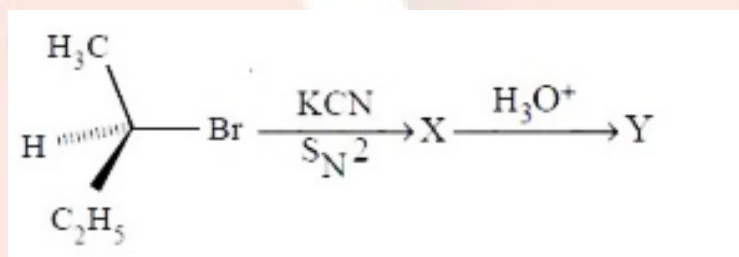
Vitamin	Source	deficiency disease
(3) C	Fish	scurvy

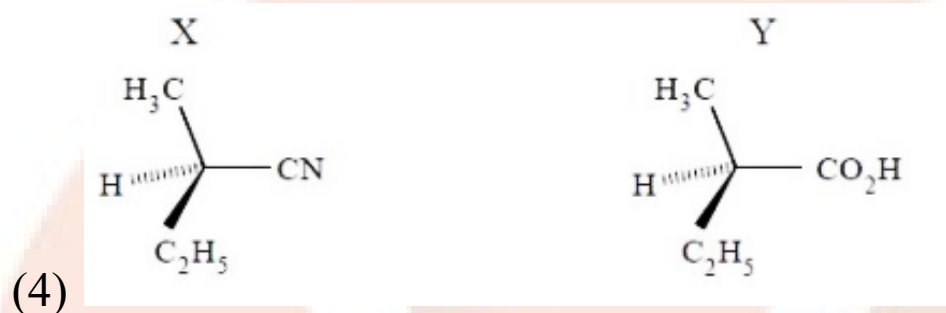
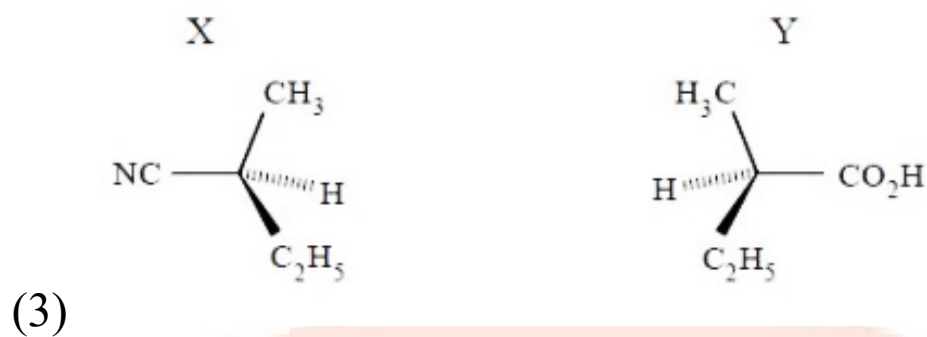
Vitamin	Source	deficiency disease
(4) D	Citrus fruits	ricket

154. Which one of the following contains $-\text{As} = \text{As} -$ in its structure?

- (1) Ranitidine
- (2) Saccharin
- (3) Salvarsan
- (4) Seldane

155. What are X and Y in the following reaction sequence?



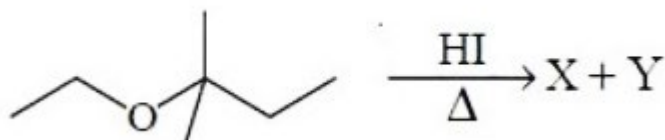


156. Which of the following sets is in the correct order regarding the property mentioned against them?

Sets		Property
I)	$\text{NCCH}_2\text{COOH} > \text{FCH}_2\text{COOH} > \text{H}_3\text{CCH}_2\text{COOH}$	acidity
II)	$\text{CH}_3\text{CH}_2\text{CHO} > \text{PhCOCH}_3 > \text{PhCHO}$	reactivity
III)	$\text{H}_3\text{COCH}_2\text{CH}_3 < \text{H}_3\text{CCH}_2\text{CHO} < \text{H}_3\text{CCH}_2\text{CH}_2\text{OH}$	boiling points

- (1) I, II only
 (2) I, III only
 (3) II, III only
 (4) I, II, III

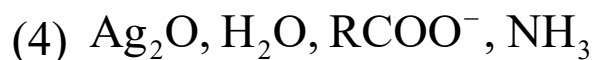
157. Identify the products (X, Y) and reaction mechanism (Z) of the following reaction?



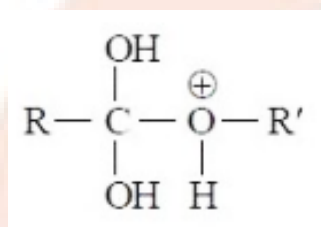
- | | X | Y | Z |
|-----|---|---|------------------------|
| (1) | | | $\text{S}_{\text{N}}1$ |
| (2) | | | $\text{S}_{\text{N}}2$ |
| (3) | | | $\text{S}_{\text{N}}1$ |
| (4) | | | $\text{S}_{\text{N}}2$ |

158. What are the products formed when an aldehyde (RCHO) is reacted with Tollens reagent?

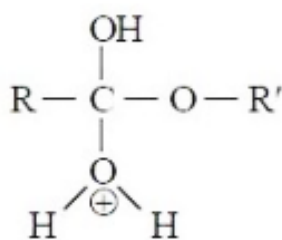
- (1) $\text{Ag}, \text{H}_2\text{O}, \text{RCH}_2\text{OH}, \text{NH}_3$
- (2) $\text{Ag}, \text{H}_2\text{O}, \text{RCOO}^-, \text{H}_2$



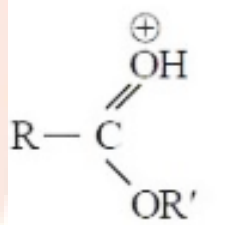
159. The following species are involved in the formation of an ester from a carboxylic acid in the presence of acid. The correct sequence of formation of these species is



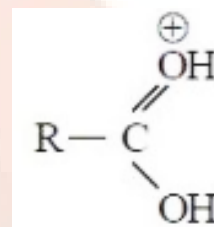
(1)



(2)



(3)



(4)

(1) 2, 1, 4, 3

(2) 4, 1, 2, 3

(3) 1, 4, 3, 2

(4) 2, 3, 1, 4

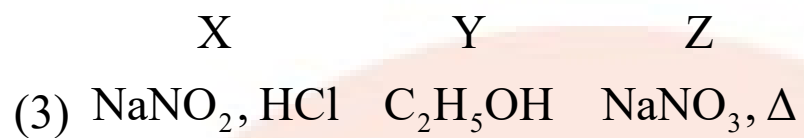
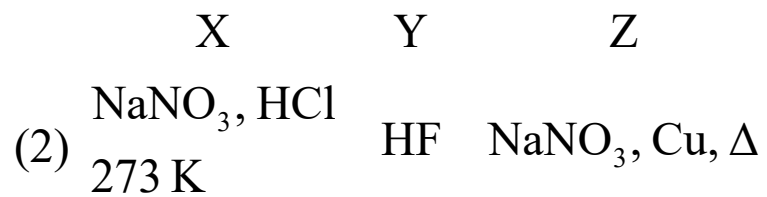
160. Identify the reagents (X, Y, Z) used in the conversion of 3-methylaniline to 3-nitrotoluene

X

Y

Z

(1) $\text{NaNO}_2, \text{HCl}$ HBF_4 $\text{NaNO}_2, \text{Cu}, \Delta$
 273 K



AP EAMCET-2019-Engineering-21 Apr-Shift-1
Answer Key

1	3	33	1	65	3	97	4	129	1
2	2	34	3	66	4	98	2	130	2
3	3	35	2	67	1	99	2	131	1
4	1	36	2	68	2	100	1	132	4
5	2	37	2	69	2	101	3	133	2
6	3	38	2	70	1	102	3	134	3
7	1	39	1	71	3	103	4	135	2
8	1	40	4	72	2	104	4	136	1
9	3	41	2	73	1	105	1	137	4
10	3	42	2	74	1	106	1	138	1
11	1	43	4	75	4	107	4	139	4
12	3	44	3	76	3	108	4	140	*
13	3	45	1	77	4	109	3	141	2
14	1	46	3	78	2	110	1	142	4
15	1	47	4	79	1	111	3	143	3
16	2	48	1	80	1	112	4	144	3
17	2	49	1	81	81	113	2	145	3
18	4	50	4	82	82	114	1	146	3

19	3	51	3	83	83	115	2	147	2
20	3	52	2	84	84	116	4	148	3
21	3	53	2	85	85	117	2	149	2
22	3	54	4	86	86	118	4	150	3
23	3	55	2	87	87	119	3	151	2
24	2	56	1	88	88	120	2	152	2
25	3	57	2	89	89	121	1	153	1
26	4	58	3	90	90	122	4	154	3
27	2	59	1	91	1	123	1	155	2
28	4	60	2	92	2	124	3	156	2
29	4	61	3	93	1	125	4	157	1
30	2	62	4	94	2	126	3	158	3
31	2	63	1	95	2	127	3	159	2
32	2	64	1	96	2	128	3	160	1

AP EAMCET-2019-Engineering-21 Apr-Shift-1
Solutions – Mathematics

1. Given,

$$(g \circ j \circ f)(x) = f(x) \text{ for all } x \in D$$

So,

$$g((j \circ f)(x)) = \frac{1}{x}$$

$$1 - (j \circ f)(x) = \frac{1}{x}$$

$$1 - j(f(x)) = \frac{1}{x}$$

Simply further,

$$1 - j\left(\frac{1}{x}\right) = \frac{1}{x}$$

$$f\left(\frac{1}{x}\right) = 1 - \frac{1}{x}$$

$$f(x) = 1 - x$$

$$f(x) = g(x)$$

2. Consider the given function.

$$f(x) = \tan\left(x + \frac{2\pi}{3}\right) - \tan\left(x + \frac{\pi}{6}\right) + \cos\left(x + \frac{\pi}{6}\right)$$

So,

$$\begin{aligned}
 f(x) &= \frac{\sin \frac{\pi}{2}}{\cos\left(x + \frac{2\pi}{3}\right)\cos\left(x + \frac{\pi}{6}\right)} + \cos\left(x + \frac{\pi}{6}\right) \\
 &= \frac{1 \times 2}{2\cos\left(x + \frac{2\pi}{3}\right)\cos\left(x + \frac{\pi}{6}\right)} + \cos\left(x + \frac{\pi}{6}\right) \\
 &= \cos(x + 30^\circ) - \frac{2}{\cos(2x - 30^\circ)}
 \end{aligned}$$

Differentiate the above function with respect to x .

$$f'(x) = -\sin(x + 30^\circ) - \frac{4\sin(2x - 30^\circ)}{\cos^2(2x - 30^\circ)}$$

$$f'(x) > 0 \text{ for all } -75^\circ < x < -60^\circ$$

$$f_{\max} \text{ at } x = -60^\circ$$

Find the value of the function.

$$\begin{aligned}
 f(-60^\circ) &= -\tan(60^\circ) - \tan(-30^\circ) + \cos(-30^\circ) \\
 &= \sqrt{3} + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \\
 &= \frac{6\sqrt{3} + 2\sqrt{3} + 3\sqrt{3}}{6} \\
 &= \frac{11\sqrt{3}}{6}
 \end{aligned}$$

3. Let,

$$S = \left[1^2 + (1^2 + 2^2) + (1^2 + 2^2 + 3^2) + \dots + (1^2 + 2^2 + \dots + n^2) \right]$$

So,

$$\begin{aligned} S &= \sum_{i=1}^n \sum_{j=1}^i j^2 \\ &= \sum_{i=1}^n \frac{2i^3 + 3i^2 + i}{6} \\ &= \frac{1}{3} \left[\frac{n(n+1)}{2} \right]^2 + \frac{1}{2} \left[\frac{n(n+1)(2n+1)}{6} \right] + \frac{1}{6} \left[\frac{n(n+1)}{2} \right] \end{aligned}$$

Simplify further.

$$\begin{aligned} S &= \frac{1}{3} \frac{n^2 (n+1)^2}{4} + \frac{1}{2} \frac{n(n+1)(2n+1)}{6} + \frac{1}{6} \frac{n(n+1)}{2} \\ &= \frac{n(n+1)}{12} (n^2 + n + 2n + 2) \\ &= \frac{n(n+1)}{12} [(n+1)(n+2)] \\ &= \frac{n(n+1)^2 (n+2)}{12} \end{aligned}$$

4. Consider the given matrix.

$$\begin{vmatrix} a+b+2c & a & b \\ c & 2a+b+c & b \\ c & a & a+2b+c \end{vmatrix} = 2$$

Apply the column operation, $C_1 \rightarrow C_1 + C_2 + C_3$ and take $2(a+b+c)$ common from C_1 .

$$2(a+b+c) \begin{vmatrix} 1 & a & b \\ 1 & b+c+2a & b \\ 1 & a & c+a+2b \end{vmatrix} = 2$$

Apply the row operation $R_3 \rightarrow R_3 - R_1$.

$$2(a+b+c) \begin{vmatrix} 1 & a & b \\ 0 & b+c+a & 0 \\ 0 & 0 & c+a+b \end{vmatrix} = 2$$

$$2(a+b+c)^3 = 2$$

$$(a+b+c)^3 = 1$$

$$a+b+c = 1$$

Thus,

$$\begin{aligned} a^3 + b^3 + c^3 - 3abc &= (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca) \\ &= 1 \cdot (a^2 + b^2 + c^2) - 2ab - 2bc - 2ca - ab - bc - ca \\ &= 1 - 2a - 2bc - 2ca - ab - bc - ca \\ &= 1 - 3ab - 3bc - 3ca \end{aligned}$$

5. Consider the equation.

$$x^2 - 25x + 24 = 0$$

$$x^2 - x - 24x + 24 = 0$$

$$x(x-1) - 24(x-1) = 0$$

$$x = 1, 24$$

Consider the matrix and substitute 1 for k .

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

Calculate the determinant.

$$\begin{aligned} |A| &= 1(2-3) - 2(3-3) + 1(3-2) \\ &= -1 + 0 + 1 \\ &= 0 \end{aligned}$$

$k = 1$ is not possible because the given matrix A is singular.

Substitute 24 for k .

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 24 \end{bmatrix}$$

Calculate the determinant.

$$\begin{aligned} [A] &= 1(48-3) - 2(27-3) + 1(3-2) \\ &= 45 - 138 + 1 \\ &= -92 \end{aligned}$$

The adjoint of the matrix is calculated as,

$$\begin{aligned}\text{adj}A &= \begin{bmatrix} 45 & -69 & 1 \\ -47 & 23 & 1 \\ 4 & 0 & -4 \end{bmatrix} \\ &= \begin{bmatrix} 45 & -47 & 4 \\ -69 & 23 & 0 \\ 1 & 1 & -4 \end{bmatrix}\end{aligned}$$

The inverse of the matrix is calculated as,

$$\begin{aligned}A^{-1} &= \frac{1}{|A|} \text{adj} A \\ &= -\frac{1}{92} \begin{bmatrix} 45 & -47 & 4 \\ -69 & 23 & 0 \\ 1 & 1 & -4 \end{bmatrix}\end{aligned}$$

6. Consider the given matrix.

$$A = \begin{bmatrix} 1 & 1 & -1 & 0 \\ 4 & 4 & -3 & 1 \\ b & 2 & 2 & 2 \\ 9 & 9 & b & 3 \end{bmatrix}$$

Apply the row operation $R_2 \rightarrow R_2 - 4R_1$ and $R_3 \rightarrow R_3 - 2R_1$.

$$A = \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ b-2 & 0 & 4 & 2 \\ 9 & 9 & b & 3 \end{bmatrix}$$

Apply the operation $R_4 \rightarrow R_4 - 9R_1$

$$A = \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ b-2 & 0 & 4 & 2 \\ 0 & 0 & b+9 & 3 \end{bmatrix}$$

Apply the operation $R_4 \rightarrow R_4 - 3R_2$.

$$a = \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ b-2 & 0 & 4 & 2 \\ 0 & 0 & b+9 & 3 \end{bmatrix}$$

If the rank is 3 then last row must have all elements 0.

$$b + 6 = 0$$

$$b = -6$$

7. Consider the given equation.

$$z = x + iy \text{ and } z^{\frac{1}{3}} = a + ib$$

So,

$$z^{\frac{1}{3}} = (a + ib)$$

$$z = (a + ib)^3$$

$$z = a^3 + (ib)^3 + 3a^2ib + 3a(ib)^2$$

$$x - iy = (a^3 - 3ab^2) - i(b^3 - 3a^2b)$$

The values are,

$$x = a^3 - 3ab^2$$

$$y = b^3 - 3a^2b$$

Find the value of function.

$$\begin{aligned} \frac{\left(\frac{x}{a} + \frac{y}{b}\right)}{a^2 + b^2} &= \frac{\left(\frac{a^3 - 3ab^2}{a} + \frac{b^3 - 3a^2b}{b}\right)}{a^2 + b^2} \\ &= -\frac{a^2 - 3b^2 + b^2 - 3a^2}{a^2 + b^2} \\ &= \frac{-2a - 2b^2}{a^2 + b^2} \\ &= -2 \end{aligned}$$

8. Consider the given equation.

$$\bar{z} = iz^2$$

So,

$$z = -i\bar{z}^2$$

$$z = -i(iz^2)^2$$

$$z = iz^4$$

$$z^4 = \frac{1}{i}z = -iz$$

Solve further,

$$z^4 + iz = 0$$

$$z(z^3 + i) = 0$$

$$\text{So, } z^3 + i = 0 \quad (\because z \neq 0)$$

9. Since $x^2 + y + 4i$ and $-3 + x^2yi$ are conjugate. So,

$$x^2 + y + 4i = -3 - x^2yi$$

$$(x^2 + y) + 4i = (-3) - x^2yi$$

Compare both the sides.

$$x^2 + y = -3 \quad \dots(1)$$

$$4 = -x^2y$$

$$y = \frac{4}{-x^2}$$

$$y = -\frac{4}{x^2} \dots(2)$$

Substitute the value of y in equation (1) and (2).

$$x^2 - \frac{4}{x^2} = -3$$

$$x^4 + 3x^2 - 4 = 0$$

$$(x^2 + 4)(x^2 - 1) = 0$$

$$x = \pm 1$$

Substitute the value of x in equation (2).

$$y = \frac{-4}{(1)^2}$$

$$y = -4$$

The value is calculated as,

$$\begin{aligned} (|x^2| + |y^2|) &= |x^2| + |y^2| + 2|x||y| \\ &= (1)^2 + (-4)^2 + 2(1)(-4) \\ &= 1 + 16 + 8 \\ &= 25 \end{aligned}$$

10. The value of the expression is calculated as,

$$\begin{aligned} \sum_{k=1}^6 \left(\sin \frac{2\pi k}{7} - i \cos \frac{2\pi k}{7} \right) &= -i \sum_{k=1}^6 \left[\cos \left(\frac{2\pi k}{7} \right) - \frac{1}{i} \sin \left(\frac{2\pi k}{7} \right) \right] \\ &= -i \sum_{k=1}^6 \left[\cos \left(\frac{2\pi k}{7} \right) + i \sin \left(\frac{2\pi k}{7} \right) \right] \\ &= -i \left[\sum_{k=1}^6 e^{\frac{i2\pi k}{7}} \right] \dots (1) \end{aligned}$$

Now from the given condition.

$$1 + e^{\frac{i2\pi}{7}} + e^{\frac{i4\pi}{7}} + e^{\frac{i6\pi}{7}} + e^{\frac{i8\pi}{7}} + e^{\frac{i10\pi}{7}} + e^{\frac{i12\pi}{7}} = 0$$

$$1 + \sum_{k=1}^6 e^{i(2\pi k/7)} = 0$$

$$\sum_{k=1}^6 e^{i(2\pi k/7)} = -1 \quad \dots(2)$$

From equation (1) and (2).

$$\left(\sin \frac{2\pi k}{7} - i \cos \frac{2\pi k}{7} \right) = -i(-1)$$

$$= i$$

11. Consider the expression,

$$\sqrt{\frac{x}{2x+1}} + \sqrt{\frac{2x+1}{x}} = 2 \quad (I)$$

Let, $\sqrt{\frac{x}{2x+1}} = y$ then,

$$y + \frac{1}{y} = 2$$

$$y^2 - 2y + 1 = 0$$

$$(y-1)(y-1) = 0$$

$$(y-1)^2 = 0$$

Simplify further,

$$y - 1 = 0$$

$$y = 1$$

$$\sqrt{\frac{x}{2x+1}} = 1$$

$$\frac{x}{2x+1} = 1$$

Solve further,

$$2x + 1 = x$$

$$x = -1$$

Here, α satisfies equation (I), so, $\alpha = -1$, now,

$$\alpha^2 x^2 + 4\alpha x + 3 = 0$$

$$(-1)^2 x^2 + 4(-1)x + 3 = 0$$

$$x^2 - 4x + 3 = 0$$

$$x = 1, 3$$

12. Consider the given function,

$$f(x) = x^2 + 2x + 2$$

So,

$$f(x) = x^2 + 2x + 1 + 1$$

$$= (x+1)^2 + 1$$

Here, $f(x) \in [1, \infty)$ and,

$$\begin{aligned}g(x) &= -x^2 + 2x - 1 \\&= -(x^2 - 2x + 1) \\&= -(x-1)^2\end{aligned}$$

Here, $g(x) \in (-\infty, 0]$

Then,

$$\begin{aligned}\frac{f}{g}(x) &= \frac{x^2 + 2x + 2}{-x^2 + 2x - 1} \\&= y\end{aligned}$$

So,

$$x^2 + 2x + 2 = -yx^2 + 2xy - y$$

$$x^2 + yx^2 + 2x - 2xy + 2 + y = 0$$

$$x^2(1+y) + (2-2y)x + 2+y = 0$$

Since, $D \geq 0$, so

$$(2-2y)^2 - 4(2+y)(1+y) \geq 0$$

$$4 + 4y^2 - 8y - 4(2+y)(1+y) \geq 0$$

$$1 + y^2 - 2y - (2+y)(1+y) \geq 0$$

$$1 + y^2 - 2y - 2 - 3y - y^2 \geq 0$$

Further simplify the above,

$$-5y - 1 \geq 0$$

$$y \leq -\frac{1}{5}$$

$$\text{Thus, } \frac{f}{g}(x) \in \left(-\infty, -\frac{1}{5}\right]$$

So,

$$a = 1, b = 0 \text{ and } c = -\frac{1}{5}$$

Hence,

$$\begin{aligned} a + 2b + 5c &= 1 + 0 + 5\left(-\frac{1}{5}\right) + 4 \\ &= 1 - 1 + 4 \\ &= 4 \end{aligned}$$

13. Consider the given expression,

$$x^2 - |x + 2| + x > 0$$

Case I :

$$\text{when } x + 2 \geq 0$$

Then,

$$x^2 - (x + 2) > 0$$

$$x^2 - 2 > 0$$

$$x \in \left[-2, -\sqrt{2}\right] \cup \left[\sqrt{2}, \infty\right]$$

Case II :

if $x + 2 < 0$ then,

$$x^2 + (x + 2) + x > 0$$

$$x^2 + 2x + 1 > 0$$

$$(x + 1)^2 + 1 > 0$$

$$x < -2$$

Therefore, the solution set is $x \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

14. Consider the given expression,

$$x^3 - 6x^2 + 11x - 6 = 0$$

So,

$$(x - 1)(x - 2)(x - 3) = 0$$

$$x = 1, 2, 3$$

Therefore, $\alpha = 1, \beta = 2, \gamma = 3$

Now,

$$\alpha^2 + \beta^2 = (1^2) + (2^2)$$

$$= 5$$

$$= \alpha' \quad (\text{say})$$

$$\beta^2 + \gamma^2 = 2^2 + 3^2$$

$$= 13$$

$$= \beta' \quad (\text{say})$$

$$\begin{aligned}
 \gamma^2 + \alpha^2 &= 3^2 + 1 \\
 &= 10 \\
 &= \gamma' \quad (\text{say})
 \end{aligned}$$

Thus, the expression of the equation that has roots α', β', γ' is,

$$x^3 - (\alpha' + \beta' + \gamma')x^2 + (\alpha'\beta' + \beta'\gamma' + \gamma'\alpha')x - \alpha'\beta'\gamma' = 0$$

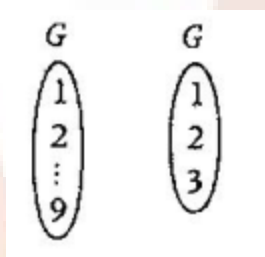
$$x^3 - (5 + 13 + 10)x^2 + (5 \times 13 + 13 \times 10 + 10 \times 5)x - 5 \times 13 \times 10 = 10$$

$$x^3 - 28x^2 + 245x - 650 = 0$$

15. The number of elements in sets G and A are 3 and 4 respectively.

For (A) :

$G \times G$ to G

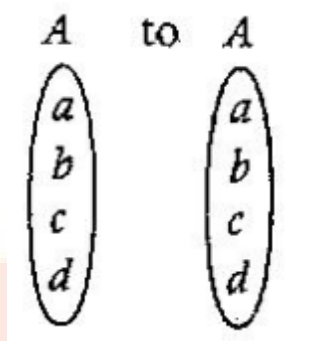


The number of non bijective functions is,

$$\begin{aligned}
 G \times G \text{ to } G &= 3^9 \\
 &= 19683
 \end{aligned}$$

For (B) :

A to A

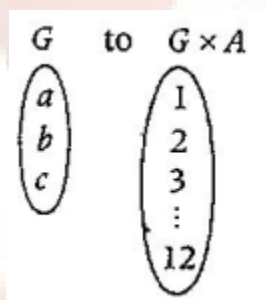


Here a has four choice, b has 3 choice c has 2 choice and d has one choice. So, number of bijective functions from A to A is,

$$4 \times 3 \times 2 \times 1 = 24$$

For (C) :

G to $G \times A$



The number of bijective functions from G to $G \times A$ is,

$$12^3 = 1728$$

For (D) :

A to $A \times A$

The number of surjective from A to $A \times A$ is, 4 to 16 is 0.

So, option (1) is correct.

16. The number of fresh lines introduced is,

$$\begin{aligned}
 \frac{1}{8}n(n-1)(n-2)(n-3) &= \frac{1}{8}(20)(20-1)(20-2)(20-3) \\
 &= \frac{1}{8}(20)(19)(18)(17) \\
 &= \frac{116280}{8} \\
 &= 14535
 \end{aligned}$$

17. Consider the expression,

$$\begin{aligned}
 \sum_{k=0}^n a_k \cdot C_k &= a_0 C_0 + a_1 C_1 + a_2 C_2 + \dots + a_n C_n \\
 &= a_0 C_0 + (a_0 + d) C_1 + (a_0 + 2d) C_2 + \dots + (a_0 + nd) C_n
 \end{aligned}$$

Here d is an common difference then,

$$\begin{aligned}
 \sum_{k=0}^n a_k \cdot C_k &= a_0 (C_0 + C_1 + \dots + C_n) + d (C_1 + 2C_2 + 3C_3 + \dots + n C_n) \\
 &= a_0 2^n + d (n 2^{n-1}) \\
 &= 2^{n-1} [2a_0 + nd] \\
 &= (a_0 + a_n) 2^{n-1}
 \end{aligned}$$

18. It is given that,

$$x = \frac{3}{10} + \frac{3 \cdot 7}{10 \cdot 15} + \frac{3 \cdot 7 \cdot 9}{10 \cdot 15 \cdot 20} + \dots$$

So,

$$x = \frac{3 \cdot 5}{5 \cdot 10} + \frac{3 \cdot 5 \cdot 7}{5 \cdot 10 \cdot 15} + \frac{3 \cdot 5 \cdot 7 \cdot 9}{5 \cdot 10 \cdot 15 \cdot 20} + \dots$$

The general term is,

$$\begin{aligned} T_{r+1} &= \frac{3 \cdot 5 \cdot 7 \dots (2r+1)}{5^r \cdot 1 \cdot 2 \cdot 3 \dots r} \\ &= \left(\frac{2}{5}\right)^r \frac{\frac{3}{2} \cdot \frac{5}{2} \cdot \frac{7}{2} \dots \left(r + \frac{1}{2}\right)}{r!} \\ &= \frac{\left(-\frac{3}{2}\right)\left(-\frac{3}{2}-1\right)\left(-\frac{3}{2}-2\right) \dots \left(-\frac{3}{2}-r+1\right)\left(-\frac{2}{5}\right)^r}{r!} \end{aligned}$$

Compare above with general terms of expansion $(1+x)^n$,

$$\frac{n(n-1)(n-2) \dots (n-r+1)x^r}{r!}$$

This gives,

$$n = -\frac{3}{2}, x = -\frac{2}{5}$$

Then,

$$\begin{aligned}x + \frac{8}{5} &= \left(1 - \frac{2}{5}\right)^{-\frac{3}{2}} \\&= \left(\frac{3}{5}\right)^{-\frac{3}{2}} \\&= \left(\frac{5}{3}\right)^{\frac{3}{2}} \\&= \frac{5\sqrt{5}}{3\sqrt{3}}\end{aligned}$$

So,

$$\begin{aligned}\frac{5x+8}{5} &= \frac{5\sqrt{5}}{3\sqrt{3}} \\5x+8 &= \frac{25\sqrt{5}}{3\sqrt{3}}\end{aligned}$$

19. It is given that,

$$\frac{x^4}{(x-1)(x-2)(x-3)} = x + k + \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$$

Compare the coefficient and put $x = 1$ then,

$$A = \frac{1}{2}$$

$$\text{At } x = 2, B = \frac{16}{-1} = -16$$

At $x = 3$, $C = \frac{81}{2}$

When $x = 0$,

$$0 = k(-1)(-2)(-3) + \frac{1}{2}(-2)(-3) - 16(-3)(-1) + \frac{81}{2}(-1)(-2)$$

$$6k = 3 - 48 + 81$$

$$k = 6$$

So,

$$\begin{aligned} k + A - B + C &= 6 + \frac{1}{2} + 16 + \frac{81}{2} \\ &= 6 + 16 + 41 \\ &= 63 \end{aligned}$$

20. Let's consider,

$$x = \operatorname{cosec} 48^\circ + \operatorname{cosec} 96^\circ + \operatorname{cosec} 192^\circ + \operatorname{cosec} 384^\circ$$

So,

$$x = \operatorname{cosec}(90^\circ - 42^\circ) + \operatorname{cosec} 96^\circ + \operatorname{cosec}(270^\circ - 78^\circ) + \operatorname{cosec}\left(\begin{matrix} 360^\circ \\ +24^\circ \end{matrix}\right)$$

$$= \sec 42^\circ + \operatorname{cosec} 96^\circ - (\sec 78^\circ) + \operatorname{cosec} 24^\circ$$

$$= \frac{1}{\cos 42^\circ} + \frac{1}{\cos 78^\circ} + \frac{1}{\sin 96^\circ} + \frac{1}{\sin 24^\circ}$$

$$= \frac{(\cos 78^\circ - \cos 42^\circ)}{\cos 42^\circ \times \cos 78^\circ} + \frac{(\sin 24^\circ + \sin 96^\circ)}{\sin 96^\circ \times \sin 24^\circ}$$

Further simplify the above,

$$\begin{aligned}
 x &= \frac{-2 \sin 60^\circ \times \sin 18^\circ}{\cos^2 60^\circ - \sin^2 18^\circ} + \frac{2 \sin 60^\circ \times \cos 36^\circ}{\sin^2 60^\circ - \sin^2 36^\circ} \\
 &= \frac{(-2) \left(\frac{\sqrt{3}}{2} \right) \left(\frac{\sqrt{5}-1}{4} \right)}{\left(\frac{1}{2} \right)^2 - \left(\frac{\sqrt{5}-1}{4} \right)^2} + \frac{2 \left(\frac{\sqrt{3}}{2} \right) \left(\frac{\sqrt{5}+1}{4} \right)}{\left(\frac{\sqrt{3}}{2} \right)^2 - \left(\frac{\sqrt{10}-2\sqrt{5}}{4} \right)^2} \\
 &= -2\sqrt{3} + 2\sqrt{3} \\
 &= 0
 \end{aligned}$$

21. Let's consider the expression as,

$$x = \sin^4 \frac{\pi}{8} + \sin^4 \frac{2\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{4\pi}{8} + \sin^4 \frac{5\pi}{8} + \sin^4 \frac{6\pi}{8} + \sin^4 \frac{7\pi}{8}$$

So,

$$\begin{aligned}
x &= \left[\sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{5\pi}{8} + \sin^4 \frac{7\pi}{8} \right] + \left[\sin^4 \frac{2\pi}{8} + \sin^4 \frac{4\pi}{8} \right. \\
&\quad \left. + \sin^4 \frac{6\pi}{8} \right] \\
&= \left[\sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \right. \\
&\quad \left. \cos^4 \left(\frac{\pi}{2} - \frac{5\pi}{8} \right) + \cos^4 \left(\frac{\pi}{2} - \frac{7\pi}{8} \right) \right] + \left[\sin^4 \frac{\pi}{4} + \sin^4 \frac{\pi}{2} \right. \\
&\quad \left. + \sin^4 \left(\frac{\pi}{2} + \frac{\pi}{4} \right) \right] \\
&= \left[\sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} \right] + \left[\sin^4 \frac{\pi}{4} + \sin^4 \frac{\pi}{2} \right. \\
&\quad \left. + \cos^4 \frac{\pi}{4} + \cos^4 \frac{3\pi}{8} \right] \\
&= \left[\left(\sin^4 \frac{\pi}{8} + \cos^4 \frac{\pi}{8} \right) + \left(\sin^4 \frac{3\pi}{8} + \cos^4 \frac{3\pi}{8} \right) \right] + \left[\left(\frac{1}{\sqrt{2}} \right)^4 + 1^4 \right. \\
&\quad \left. + \left(\frac{1}{\sqrt{2}} \right)^4 \right]
\end{aligned}$$

Further simplify the above,

$$\begin{aligned}
x &= \left[\left(1 - 2 \sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8} \right) + \left(1 - 2 \sin^2 \frac{3\pi}{8} \cos^2 \frac{3\pi}{8} \right) \right] + \left[\frac{1}{4} + 1 + \frac{1}{4} \right] \\
&= \left[2 - \frac{1}{2} 4 \sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8} - \frac{1}{2} 4 \sin^2 \frac{3\pi}{8} \cos^2 \frac{3\pi}{8} \right] + \left[\frac{1}{2} + 1 \right] \\
&= 3
\end{aligned}$$

22. Consider the given data,

$$x : y : z = \tan\left(\frac{\pi}{15} + \alpha\right) : \tan\left(\frac{\pi}{15} + \beta\right) : \tan\left(\frac{\pi}{15} + \gamma\right)$$

So,

$$x = k \tan\left(\frac{\pi}{15} + \alpha\right), y = k \tan\left(\frac{\pi}{15} + \beta\right), z = k \tan\left(\frac{\pi}{15} + \gamma\right)$$

Now,

$$\begin{aligned} \frac{z+x}{z-x} \sin^2(\gamma-\alpha) &= \frac{\tan(12^\circ + \gamma) + \tan(12^\circ + \alpha)}{\tan(12^\circ + \gamma) - \tan(12^\circ + \alpha)} \sin^2(\gamma-\alpha) \\ &= \frac{\sin(24^\circ + (\gamma + \alpha))}{\sin(\gamma - \alpha)} \sin^2(\gamma - \alpha) \\ &= [\sin 24^\circ \cos(\gamma - \alpha) + \cos 24^\circ \sin(\gamma + \alpha)] \times \sin 24^\circ \\ &= \sin[\cos(\gamma + \alpha) \sin(\gamma + \alpha)] + \left[\begin{array}{l} \cos 24^\circ \\ + \sin(\gamma + \alpha) \sin \alpha \end{array} \right] \end{aligned}$$

Further simplify the above,

$$\frac{\sin 24^\circ}{2} - (\sin^2 \gamma - \sin^2 \alpha) - \frac{\cos 24^\circ}{2} (\cos^2 \gamma - \cos^2 \alpha)$$

Similarly,

$$\frac{x+y}{x-y} \sin^2(\alpha-\beta) = \frac{\sin 24^\circ}{2} (\sin^2 \alpha - \sin^2 \beta) - \frac{\cos 24^\circ}{2} \begin{pmatrix} \cos^2 \alpha \\ -\cos^2 \beta \end{pmatrix}$$

and,

$$\frac{y+z}{y-z} \sin^2(\beta-\gamma) = \frac{\sin 24^\circ}{2} (\sin^2 \beta - \sin^2 \gamma) - \frac{\cos 24^\circ}{2} \begin{pmatrix} \cos^2 \beta \\ -\cos^2 \gamma \end{pmatrix}$$

From above equations,

$$\frac{z+x}{z-x} \sin^2(\gamma - \alpha) + \frac{x+y}{x-y} \sin^2(\alpha - \beta) + \frac{y+z}{y-z} \sin^2(\beta - \gamma) = 0$$

23. Consider the expression,

$$\sin x \sqrt{4 \cos^2 x} = \frac{2 + x - \{x\}}{1 + x + \{x\}}$$

$$\sin x \sqrt{4 \cos^2 x} = \frac{2 + \{x\}}{1 + \{x\}}$$

$$2 \sin x |\cos x| = \frac{2 + \{x\}}{1 + \{x\}} = y \quad (\text{let})$$

Then,

$$y = \frac{2 + \{x\}}{1 - \{x\}} > 0$$

So,

$$|\cos x| = \cos x, \quad \forall x \in \left[\frac{\pi}{4}, \frac{\pi}{3} \right]$$

Then,

$$2 \sin x \cos x = \frac{2 + \{x\}}{1 - \{x\}}$$

The maximum value of $2 \sin x \cos x = \sin 2x$ is 1 so,

Minimum value of $\frac{2 + \{x\}}{1 - \{x\}}$ (at $\{x = 0\}$) is 2.

24. Consider the expression,

$$f(x) = (\sin^{-1} x)^2 + (\cos^{-1} x)^2$$

Here, $\sin^{-1} x = a$ and $\cos^{-1} x = b$ then,

$$\begin{aligned} f(x) &= a^2 + b^2 \\ &= (a + b)^2 + 2ab \end{aligned}$$

The value of a and b is,

$$\begin{aligned} (\sin^{-1} x + \cos^{-1} x)^2 - 2 \sin^{-1} x \cos^{-1} x &= \frac{\pi^2}{4} - 2 \sin^{-1} x \left(\frac{\pi}{2} - \sin^{-1} x \right) \\ &= \frac{\pi^2}{4} - \pi \sin^{-1} x + 2 (\sin^{-1} x)^2 \end{aligned}$$

For minimum and maximum value,

$$f'(x) = 0$$

$$\pi \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} = 0$$

$$\frac{1}{\sqrt{1-x^2}} [4 \sin^{-1} x - \pi] = 0$$

$$4 \sin^{-1} x - \pi = 0$$

$$x = \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

Thus,

$$f''\left(\frac{1}{\sqrt{2}}\right) = +ve, \text{ So}$$

$$f(x)_{\min} = f\left(\frac{1}{\sqrt{2}}\right)$$

$$\alpha = \frac{\pi^2}{4} - 2\left(\frac{\pi}{4}\right)\left(\frac{\pi}{4}\right)$$

$$\alpha = \frac{\pi^2}{8}$$

$$f(x)_{\max} = f(-1)$$

$$\beta = \frac{\pi^2}{4} - 2\left(-\frac{\pi}{2}\right)(\pi)$$

$$\beta = \frac{5\pi^2}{4}$$

Hence,

$$\begin{aligned} 8(\alpha + \beta) &= 8\left[\frac{\pi^2}{8} + \frac{5\pi^2}{4}\right] \\ &= 8\left[\frac{\pi^2 + 10\pi^2}{8}\right] \\ &= 11\pi^2 \end{aligned}$$

25. For, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

$$\log \sec x = y \text{ (let)}$$

Then,

$$\sec x = e^x$$

$$\cos x = e^y$$

$$\frac{\sinh y}{\cosh y} = \frac{e^y - e^{-y}}{e^y + e^{-y}}$$

$$\frac{\cosh y - \sinh y}{\cosh y + \sinh y} = \frac{e^{-y}}{e^y}$$

$$= \cos^2 x$$

Solve further,

$$\left(\frac{\cosh \frac{y}{2} - \sinh \frac{y}{2}}{\cosh \frac{y}{2} + \sinh \frac{y}{2}} \right)^2 = \cos^2 x$$

$$\frac{\cosh \frac{y}{2} - \sinh \frac{y}{2}}{\cosh \frac{y}{2} + \sinh \frac{y}{2}} = \cos x$$

$$\frac{2 \cosh \frac{y}{2}}{2 \sinh \frac{y}{2}} = \frac{1 + \cos x}{1 - \cos x}$$

$$= \cos^2 \frac{x}{2}$$

Simplify further,

$$\coth \frac{y}{1} = \cot^2 \frac{x}{2}$$

$$= \operatorname{cosec}^2 \frac{x}{2} - 1$$

$$\frac{y}{2} = \coth^{-1} \left(\operatorname{cosec}^2 \frac{x}{2} - 1 \right)$$

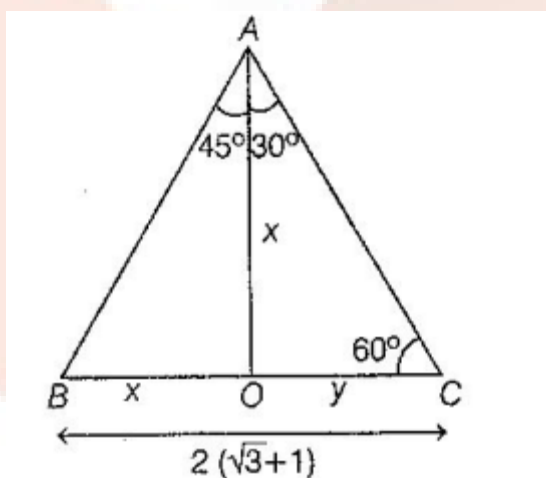
$$y = 2 \coth^{-1} \left(\operatorname{cosec}^2 \frac{x}{2} - 1 \right)$$

26. Consider the given data, In triangle AOC

$$\tan 60^\circ = \frac{x}{y}$$

$$\sqrt{3} = \frac{x}{y}$$

The diagram of triangle is,



From the above figure,

$$x = \sqrt{3}y$$

$$x + y = 2(\sqrt{3} + 1)$$

$$y(\sqrt{3} + 1) = 2(\sqrt{3} + 1)$$

$$y = 2$$

So,

$$x = 2\sqrt{3}$$

Now,

$$\text{area of } \triangle ABC = \text{area of } \triangle AOB + \text{area of } \triangle AOC$$

$$= \frac{1}{2}(x)(x) + \frac{1}{2}(x)(y)$$

$$= \frac{1}{2}x[x + y]$$

$$= \frac{1}{2}(2\sqrt{3})2(\sqrt{3} + 1)$$

$$= 6 + 2\sqrt{3} \text{ sq units}$$

27. It is given that,

$$3a = b + c$$

The perimeter of triangle is given by,

$$s = \frac{a + b + c}{2}$$

So,

$$s = \frac{a + 3a}{2}$$

$$= \frac{4a}{2}$$

$$= 2a$$

Now,

$$\begin{aligned} \cot \frac{B}{2} \cot \frac{C}{2} &= \sqrt{\frac{s(s-b)}{(s-a)(s-c)}} \cdot \sqrt{\frac{s(s-c)}{(s-b)(s-c)}} \\ &= \sqrt{\frac{2a(2a-b)}{(2a-a)(2a-c)}} \cdot \sqrt{\frac{2a(2a-c)}{(2a-b)(2a-c)}} \\ &= \sqrt{\frac{2a(2a-b)}{(2a-a)(2a-c)}} \times \frac{2a(2a-c)}{(2a-b)(2a-c)} \\ &= \sqrt{4} \\ &= 2 \end{aligned}$$

28. Consider the expression,

$$\frac{2r_2r_3}{r_2 - r_1} = r_3 - r_1$$

$$2r_2r_3 = (r_2 - r_1)(r_3 - r_1)$$

$$2 \frac{\Delta}{(s-b)(s-c)} = \left(\frac{\Delta}{(s-b)} - \frac{\Delta}{(s-c)} \right) \left(\frac{\Delta}{(s-c)} - \frac{\Delta}{(s-a)} \right)$$

$$\frac{2\Delta^2}{(s-b)(s-c)} = \Delta^2 \left\{ \frac{s-a-s+b}{(a-b)(s-a)} \right\} \left\{ \frac{s-a-s+c}{(s-c)(s-a)} \right\}$$

Further simplify the above,

$$\frac{2}{(s-b)(s-c)} = \frac{(b-a)}{(s-b)(s-a)} \times \frac{(c-a)}{(s-c)(s-a)}$$

$$= s(s-a)^2 = (b-a)(c-a)$$

$$\frac{2(b+c-a)^2}{4} = (b-a)(c-a)$$

$$b^2 + c^2 + a^2 + 2bc - 2ca - 2ab = 2$$

Then,

$$b^2 + c^2 + a^2 = 2a^2$$

$$a^2 = b^2 + c^2$$

Now,

$$\frac{r_1(r_2 + r_3)}{\sqrt{r_1 r_2 + r_2 r_3 + r_3 r_1}} = \frac{\frac{\Delta}{s-a} \times \Delta \left\{ \frac{1}{s-b} + \frac{1}{s-c} \right\}}{s}$$

$$= \frac{\Delta^2 (2s - b - c)}{s(s-a)(s-b)(s-c)}$$

$$= \frac{\Delta^2 (a + b + c - b - c)}{\Delta^2}$$

$$= a$$

$$= 2R$$

29. From the given vectors,

$$\begin{aligned}\mathbf{PS} &= (4-3)\hat{\mathbf{i}} + (5+2)\hat{\mathbf{j}} + (\lambda+1)\hat{\mathbf{k}} \\ &= \hat{\mathbf{i}} + 7\hat{\mathbf{j}} + (\lambda+1)\hat{\mathbf{k}}\end{aligned}$$

And,

$$\mathbf{PQ} = -\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$$

And,

$$\mathbf{PR} = -4\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$$

Now,

$$\begin{aligned}\mathbf{PQ} \times \mathbf{PR} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -1 & 5 & -3 \\ -4 & 3 & 3 \end{vmatrix} \\ &= \hat{\mathbf{i}}(15+9) - \hat{\mathbf{j}}(-3-12) + \hat{\mathbf{k}}(-3+20) \\ &= 24\hat{\mathbf{i}} + 15\hat{\mathbf{j}} + 17\hat{\mathbf{k}}\end{aligned}$$

Since P, Q, R, S are coplanar so,

$$\begin{aligned}\mathbf{PS} \cdot (\mathbf{PQ} \times \mathbf{PR}) &= 0 \\ [\hat{\mathbf{i}} + 7\hat{\mathbf{j}} + (\lambda+1)\hat{\mathbf{k}}] \cdot [24\hat{\mathbf{i}} + 15\hat{\mathbf{j}} + 17\hat{\mathbf{k}}] &= 0 \\ 24 + 105 + 17(\lambda+1) &= 0 \\ 129 + 17\lambda + 17 &= 0 \\ \lambda &= -\frac{146}{17}\end{aligned}$$

29. Consider the given vectors,

$$\mathbf{OA} = 2\hat{i} + 2\hat{j} + \hat{k}$$

$$\mathbf{OB} = 2\hat{i} + 4\hat{j} + 4\hat{k}$$

So,

$$\begin{aligned} |\mathbf{OA}| &= \sqrt{4 + 4 + 1} \\ &= 3 \\ &= m \quad (\text{let}) \end{aligned}$$

And,

$$\begin{aligned} |\mathbf{OB}| &= \sqrt{4 + 16 + 16} \\ &= 6 \\ &= n \quad (\text{let}) \end{aligned}$$

The angle bisector of $\angle BOA$ intersect the side AB at point P in the ratio $3 : 6 = 1 : 2$.

So,

$$\begin{aligned} \mathbf{OP} &= \frac{2(\mathbf{OA}) + 1(\mathbf{OB})}{3} \\ &= 2\hat{i} + \frac{8}{3}\hat{j} + 2\hat{k} \end{aligned}$$

The magnitude of $\mathbf{OP}$ is,

$$|\mathbf{OP}| = \sqrt{2^2 + \left(\frac{8}{3}\right)^2 + 2^2}$$

$$k = \sqrt{\frac{136}{9}}$$

$$9k^2 = 136$$

30. Consider the expression,

$$\mathbf{a} + x\mathbf{b} + y\mathbf{c} = 0$$

If $\mathbf{a} + \mathbf{b} + \mathbf{c} = 0$ then,

$$\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} \neq 0$$

If $\mathbf{a} + x\mathbf{b} + y\mathbf{c} = 0$

Then,

$$(\mathbf{a} \times \mathbf{b}) = xy(\mathbf{b} \times \mathbf{c}) = y(\mathbf{c} \times \mathbf{a}) = \mathbf{p}$$

So,

$$\mathbf{a} \times \mathbf{b} = \frac{\mathbf{p}}{x}, \mathbf{b} \times \mathbf{c} = \frac{\mathbf{p}}{xy}, \mathbf{c} \times \mathbf{a} = \frac{\mathbf{p}}{y}$$

Now,

$$\begin{aligned}
 \mathbf{a} \times \mathbf{b} - \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} &= \frac{\mathbf{p}}{x} + \frac{\mathbf{p}}{xy} + \frac{\mathbf{p}}{y} \\
 &= \mathbf{p} \left(\frac{x + y + 1}{xy} \right) \\
 &= \left(\frac{x + y + 1}{xy} \right) \times xy \times (\mathbf{b} \times \mathbf{c}) \\
 &= (x + y + 1)(\mathbf{b} \times \mathbf{c})
 \end{aligned}$$

Compare above with $6(\mathbf{b} \times \mathbf{c})$ then,

$$x + y + 1 = 6$$

$$x + y = 5$$

$$x + y - 5 = 0$$

32. From the given data,

$$\begin{aligned}
 \mathbf{AB} \times \mathbf{C} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 3 - \alpha & 0 & 2 - 2\alpha \\ 4 & -1 & 3 \end{vmatrix} \\
 &= \hat{\mathbf{i}}(2 - 2\alpha) - \hat{\mathbf{j}}(9 - 3\alpha - 8 + 8\alpha) + \hat{\mathbf{k}}(-3 + \alpha) \\
 &= (2 - 2\alpha)\hat{\mathbf{i}} + (5\alpha - 1)\hat{\mathbf{j}} + (\alpha - 3)\hat{\mathbf{k}}
 \end{aligned}$$

Compare above with $6\hat{\mathbf{i}} + 9\hat{\mathbf{j}} - 5\hat{\mathbf{k}}$ then,

$$2 - 2\alpha = 6, 5\alpha - 1 = 9 \text{ and } \alpha - 3 = -5$$

$$2\alpha = -4$$

$$\alpha = -2$$

Thus,

$$\begin{aligned}\alpha^2 + \alpha + 5 &= (-2)^2 - 2 + 5 \\ &= 7\end{aligned}$$

33. The shortest distance is given by,

$$\frac{|(\mathbf{a} - \mathbf{c}) \cdot (\mathbf{b} \times \mathbf{d})|}{|\mathbf{b} \times \mathbf{d}|}$$

Now,

$$\begin{aligned}\mathbf{b} \times \mathbf{d} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & -2 & 2 \\ 3 & -2 & -2 \end{vmatrix} \\ &= \hat{\mathbf{i}}(4 + 4) - \hat{\mathbf{j}}(-2 - 6) + \hat{\mathbf{k}}(-2 + 6) \\ &= 8\hat{\mathbf{i}} + 8\hat{\mathbf{j}} + 4\hat{\mathbf{k}}\end{aligned}$$

Then shortest distance is,

$$\begin{aligned}\left| \frac{(10\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}) \cdot (8\hat{\mathbf{i}} + 8\hat{\mathbf{j}} + 4\hat{\mathbf{k}})}{\sqrt{64 + 64 + 16}} \right| &= \left| \frac{80 + 16 + 12}{\sqrt{144}} \right| \\ &= \frac{108}{12} \\ &= 9\end{aligned}$$

34. Consider the expression,

$$\mathbf{r} \cdot \mathbf{a} = 0$$

And,

$$\mathbf{r} \times \mathbf{b} = \mathbf{c} \times \mathbf{b}$$

So,

$$\mathbf{a} \times (\mathbf{r} \times \mathbf{b}) = \mathbf{a} \times (\mathbf{c} \times \mathbf{b})$$

$$\mathbf{r}(\mathbf{a} \cdot \mathbf{b}) - \mathbf{b}(\mathbf{a} \cdot \mathbf{r}) = \mathbf{c}(\mathbf{a} \cdot \mathbf{b}) - \mathbf{b}(\mathbf{a} \cdot \mathbf{c})$$

$$\mathbf{r}(\mathbf{a} \cdot \mathbf{b}) = \mathbf{c}(\mathbf{a} \cdot \mathbf{b}) - \mathbf{b}(\mathbf{a} \cdot \mathbf{c})$$

$$\mathbf{r} = \mathbf{c} - \mathbf{b} \left(\frac{\mathbf{c} \cdot \mathbf{a}}{\mathbf{a} \cdot \mathbf{b}} \right)$$

35. It is given that,

$$\sum x^2 = 2830 \text{ and } \sum x = 170$$

The increase in $\sum x = 10$ and increase in,

$$\begin{aligned} \sum x^2 &= (30)^2 - (20)^2 \\ &= 900 - 400 \\ &= 500 \end{aligned}$$

So,

$$\begin{aligned} \sum x &= 170 + 10 \\ &= 180 \end{aligned}$$

And,

$$\begin{aligned}\sum x^2 &= 2830 + 500 \\ &= 3330\end{aligned}$$

So, the variance is calculated by,

$$\begin{aligned}\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2 &= \frac{3330}{15} - \left(\frac{180}{15}\right)^2 \\ &= 222 - 144 \\ &= 78\end{aligned}$$

36. Consider the given data,

Class Interval	f_i	x_i	$f_i x_i$	$ x_i - 35 $	$f_i x_i - 35 $
0-10	4	5	20	30	120
10-20	6	15	90	20	120
20-30	16	25	400	10	160
30-40	28	35	980	0	0
40-50	16	45	720	10	160
50-60	6	55	330	20	120
60-70	4	65	260	30	120
Total	80		2800		800

The mean is

$$\begin{aligned}\bar{X} &= \frac{\sum f_i x_i}{\sum f_i} \\ &= \frac{2800}{80} \\ &= 35\end{aligned}$$

The mean deviation is calculated as,

$$\begin{aligned}M.D &= \frac{1}{N} \sum_{i=1}^n f_i |x_i - \bar{x}| \\ &= \frac{1}{80} \sum_{i=1}^n f_i |x_i - 35| \\ &= \frac{1}{80} (800) \\ &= 10\end{aligned}$$

37. The is given that A and B each select one number at random from the distinct number $1, 2, 3, \dots, n$ then the probability that the number selected by A is less than selected by B,

$$\begin{aligned}\frac{{}^n C_2}{n \times n} &= \frac{n(n-1)}{2(n \times n)} \\ &= \frac{1009}{2019}\end{aligned}$$

Thus,

$$\frac{n-1}{2n} = \frac{1009}{2019}$$

$$n = 2019$$

Now number of ways selecting a number from 1, 2, 3, ..., 2019 by B is the number immediately next to the number selected by A is 2018. So the required probability is,

$$\frac{2018}{2019 \times 2019} = \frac{2018}{(2019)^2}$$

38. Let the probability of drawing ball from A, B and C be $\frac{1}{3}$.

The required probability is calculated as,

$$\begin{aligned} \frac{1}{3} \times \frac{3}{5} \times \frac{1}{3} \times \frac{2}{6} + \frac{1}{3} \times \frac{2}{5} &= \frac{3}{15} + \frac{2}{18} + \frac{2}{15} \\ &= \frac{1}{5} + \frac{1}{9} + \frac{2}{15} \\ &= \frac{20}{45} \\ &= \frac{4}{9} \end{aligned}$$

39. The probability of green ball is,

$$\begin{aligned}
 P(G) &= \frac{1}{2} \times \frac{2}{6} + \frac{1}{3} \times \frac{3}{6} + \frac{1}{6} \times \frac{1}{6} \\
 &= \frac{1}{6} + \frac{1}{6} + \frac{1}{36} \\
 &= \frac{6+6+1}{36} \\
 &= \frac{13}{36}
 \end{aligned}$$

The probability that green ball drawn from bag C is,

$$\begin{aligned}
 P\left(\frac{C}{G}\right) &= \frac{P\left(\frac{G}{C}\right) \times P(C)}{P(G)} \\
 &= \frac{\frac{1}{6} \times \frac{1}{6}}{\frac{13}{36}} \\
 &= \frac{1}{13}
 \end{aligned}$$

40. Consider given data,

$$P(X = 0) = 3C^3$$

$$P(X = 2) = 5C - 10C^2$$

$$P(X = 4) = 4C - 1$$

Since, $\sum P(X) = 1$ so,

$$3C^3 + (5C - 10C^2) + (4C - 1) = 1$$

$$3C^3 - 10C^2 + 9C - 2 = 0$$

$$(C - 1)(3C^2 - 7C + 2) = 0$$

$$(C - 1)(3C - 1)(C - 2) = 0$$

From above,

$$C = 1, \frac{1}{3}, 2$$

So, $C = \frac{1}{3}$ then,

X	0	2	4
$P(X)$	$\frac{1}{9}$	$\frac{5}{9}$	$\frac{1}{3}$

So, variance is,

$$\begin{aligned}
 \sum X_P^2 - \left(\sum X_P\right)^2 &= \left(0^2 \times \frac{1}{9} + 4 \times \frac{5}{9} + 16 \times \frac{1}{3}\right) - \left(\frac{10}{9} + \frac{4}{3}\right)^2 \\
 &= \left(\frac{20}{9} + \frac{16}{3}\right) - \left(\frac{66}{27}\right)^2 \\
 &= \frac{60 + 144}{27} - \frac{484}{81} \\
 &= \frac{128}{81}
 \end{aligned}$$

41. The required probability is given by,

$P = \text{No white ball} + \text{one white ball} + \text{two white ball}$

$$= \left(\frac{2}{3}\right)^5 + {}^5C_1 \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right)^1 + {}^5C_2 \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)^2$$

$$= \left(\frac{1}{3}\right)^5 [2^5 + 5 \cdot 2^4 + 10 \cdot 2^3]$$

$$= \left(\frac{1}{3}\right)^5 [32 + 80 + 80]$$

Further simplify the above,

$$P = \frac{192}{3^5}$$

$$= \frac{64 \times 3}{3^5}$$

$$= \frac{64}{3^4}$$

$$= \left(\frac{8}{9}\right)^2$$

42. The equations of straight line are,

$$x \sin \theta + (1 - \cos \theta) y = a \sin \theta \quad \text{(I)}$$

And,

$$x \sin \theta - (1 + \cos \theta) y = -a \sin \theta \quad \text{(II)}$$

Subtract equation (II) from (I),

$$(1 - \cos \theta)y + (1 + \cos \theta) = 2a \sin \theta$$

$$y[1 - \cos \theta + 1 + \cos \theta] = 2a \sin \theta$$

$$y = a \sin \theta$$

From equation (I),

$$x \sin \theta + (1 - \cos \theta)a \sin \theta = a \sin \theta$$

$$\sin \theta [x + (1 - \cos \theta)a] = a \sin \theta$$

$$x + a - a \cos \theta = a$$

$$x - a \cos \theta = 0$$

$$x = a \cos \theta$$

Now,

$$x^2 + y^2 = (a \cos \theta)^2 + (a \sin \theta)^2$$

$$x^2 + y^2 = a^2 \cos^2 \theta + a^2 \sin^2 \theta$$

$$x^2 + y^2 = a^2 (\cos^2 \theta + \sin^2 \theta)$$

$$x^2 + y^2 = a^2$$

Thus, the above equation represents a circle.

43. The equation of a line,

$$L: \frac{x}{a} + \frac{y}{b} = 1$$

Rotate the axis with respect to the angle α then,

$$x = x \cos \alpha - y \sin \alpha$$

$$y = x \sin \alpha + y \cos \alpha$$

Then,

$$\frac{x \cos \alpha - y \sin \alpha}{a} + \frac{x \sin \alpha + y \cos \alpha}{b} = 1$$

$$x \left(\frac{\cos \alpha}{a} + \frac{\sin \alpha}{b} \right) + y \left(\frac{\cos \alpha}{b} - \frac{\sin \alpha}{a} \right) = 1$$

Rotate axis, $L' : \frac{x}{p} + \frac{y}{q} = 1$

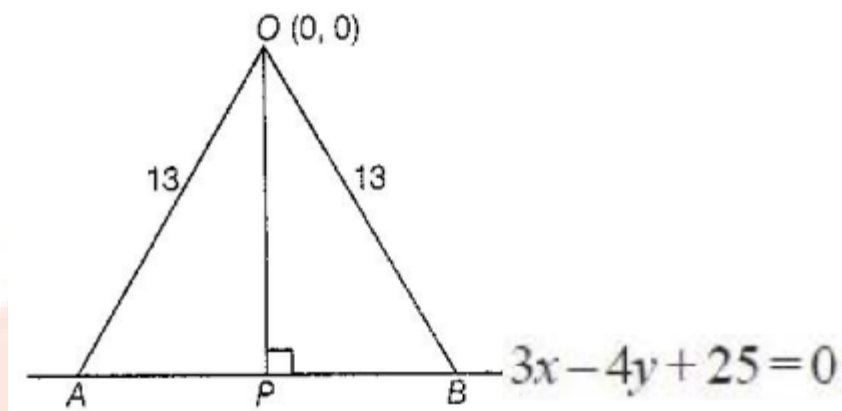
So,

$$\frac{\cos \alpha}{a} + \frac{\sin \alpha}{b} = \frac{1}{p} \text{ and } \frac{\cos \alpha}{a} - \frac{\sin \alpha}{b} = \frac{1}{q}$$

Now,

$$\begin{aligned} \frac{1}{p^2} + \frac{1}{q^2} &= \left(\frac{\cos \alpha}{a} + \frac{\sin \alpha}{b} \right)^2 + \left(\frac{\cos \alpha}{b} - \frac{\sin \alpha}{a} \right)^2 \\ &= \left[\frac{\cos^2 \alpha}{a^2} + \frac{\sin^2 \alpha}{b^2} + \frac{2 \sin \alpha \cos \alpha}{ab} + \frac{\cos^2 \alpha}{b^2} + \frac{\sin^2 \alpha}{a^2} - \frac{2 \sin \alpha \cos \alpha}{ab} \right] \\ &= \frac{\cos^2 \alpha + \sin^2 \alpha}{a^2} + \frac{\cos^2 \alpha + \sin^2 \alpha}{b^2} \\ &= \frac{1}{p^2} + \frac{1}{q^2} \\ &= \frac{1}{a^2} + \frac{1}{b^2} \end{aligned}$$

44. Consider the diagram,



The required distance is given by,

$$\begin{aligned} OP &= \left| \frac{0 + 0 + 25}{\sqrt{3^2 + 4^2}} \right| \\ &= \left| \frac{25}{5} \right| \\ &= 5 \end{aligned}$$

By Pythagoras theorem,

$$AP = PB = 12$$

So,

$$\begin{aligned} \text{Area of } \triangle OAB &= \frac{1}{2} \times 24 \times 5 \\ &= 12 \times 5 \\ &= 60 \end{aligned}$$

45. Consider the line equation,

$$L = 3x - 4y - 8$$

So,

$$\begin{aligned}L_{1,1} &= 3 - 4 - 8 \\ &= -9 < 0\end{aligned}$$

Now,

$$L_{(\alpha, \beta)} > 0$$

Then,

$$\begin{aligned}3x - 4y - 8 &> 0 \\ 3x - 4(-3x) - 8 &> 0 \\ 15x - 8 &> 0 \\ x &> \frac{8}{15}\end{aligned}$$

Thus,

$$\alpha > \frac{8}{15}$$

So,

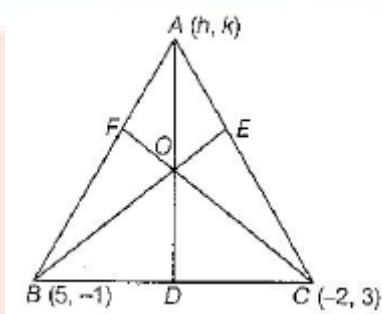
$$\begin{aligned}3x - 4y - 8 &> 0 \\ 5y + 8 &< 0 & [\because 3x = -y] \\ y &< -\frac{8}{5}\end{aligned}$$

Thus,

$$\beta < -\frac{8}{5}$$

46. Consider the condition,
Slope of $BC \times$ slope of $AD = -1$

Then the diagram is,



Now,

$$\frac{3+1}{-2-5} \times \frac{k}{h} = -1$$

$$\frac{k}{h} \times \frac{4}{-7} = -1$$

$$\frac{4k}{-7h} = -1$$

$$4k - 7h = 0 \quad (I)$$

And, Slope of $AB \times$ slope of $CF = -1$ so,

$$M_{AB} \times M_{CF} = -1$$

$$M_{AB} \times \frac{-3}{2} = -1$$

$$M_{AB} = \frac{2}{3}$$

So, the equation of AB is,

$$y + 1 = \frac{2}{3}(x - 5)$$

$$3y + 3 = 2x - 10$$

$$2x - 3y = 13$$

Here $A(h, k)$ lies on AB so,

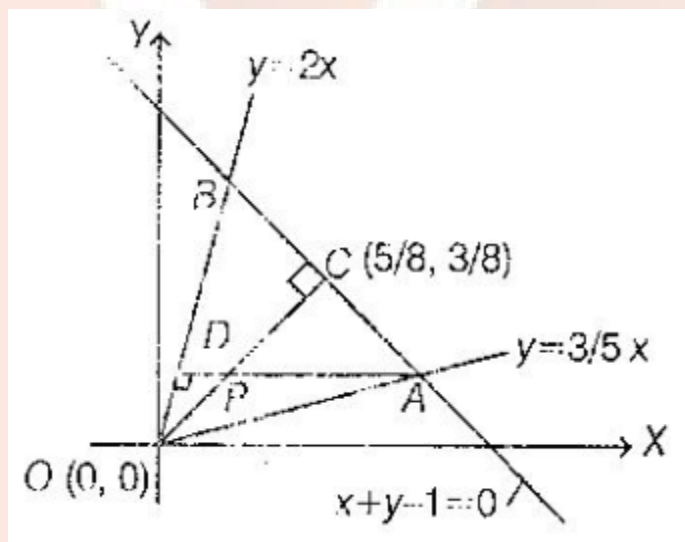
$$2h - 3k = 13 \quad \text{(II)}$$

From equation (I) and (II),

$$h = -4, k = -7$$

Hence third vertex of triangle is $(-4, -7)$.

47. Consider the diagram,



Then,

$$6x^2 - 10xy - 3xy + 5y^2 = 0$$

$$2x(3x - 5y) - y(3x - 5y) = 0$$

$$(2x - y)(3x - 5y) = 0$$

Let the orthocenter be (h, k) so,

Slope of $OP \times$ slope of $AB = -1$

$$\frac{k}{h} \times -1 = -1$$

$$h = k \quad (I)$$

Now,

Slope of $OB \times$ slope of $AD = -1$

$$2 \times \left(\frac{\frac{3}{8} - k}{\frac{5}{8} - h} \right) = -1$$

$$\frac{3}{4} - 2h = h - \frac{5}{8}$$

$$h = \frac{11}{24}$$

Then,

$$k = \frac{11}{24}$$

$$OP = \sqrt{h^2 + k^2}$$

$$\text{Now, } = \sqrt{\left(\frac{11}{24}\right)^2 + \left(\frac{11}{24}\right)^2}$$

$$= \frac{11}{24} \sqrt{2}$$

48. Let,

$$L = y = mx$$

$$L_1 = y = kx$$

$$L_2 = y = -\frac{1}{k}x$$

Now,

$$(y - mx)(y - kx) = 0$$

$$y^2 - ykx - mxy + mkx^2 = 0$$

$$mkx^2 - (k + m)xy + y^2 = 0$$

It is given that,

$$2x^2 + axy + 3y^2 = 0$$

or

$$\frac{2}{3}x^2 + \frac{a}{3}xy + y^2 = 0$$

By comparison,

$$mk = \frac{2}{3}, -(k + m) = \frac{a}{3}$$

Now,

$$(y - mx)\left(t + \frac{x}{k}\right) = 0$$

$$y^2 + \frac{xy}{k} - mxy - \frac{mx^2}{k} = 0$$

$$-\frac{m}{k}x^2 + \left(\frac{1}{k} - m\right)xy + y^2 = 0$$

Thus,

$$2x^2 + xy - 3y^2 = 0$$

or

$$-\frac{2}{3}x^2 - \frac{b}{3}xy + y^2 = 0$$

By the comparison,

$$\frac{-m}{k} = \frac{-2}{3}, \quad -\frac{b}{3} = \frac{1}{k} - m$$

$$m = \frac{2k}{3}, \quad -\frac{b}{3} = \frac{1 - km}{k}$$

Solve equation (I) and (II),

$$m = \frac{2}{3}, \quad k = 1 \text{ and } a = -5, \quad b = -1$$

Then,

$$\begin{aligned} a^2 + b^2 &= 25 + 1 \\ &= 26 \end{aligned}$$

49. The given equation of circle is,

$$S = x^2 + y^2 - 2x - 4y + 3 = 0$$

And,

$$\begin{aligned} p &= (-1)^2 + (1)^2 - 2(-1) - 4(1) + 3 \\ &= 3 \end{aligned}$$

And,

$$\begin{aligned} t &= \sqrt{p} \\ &= \sqrt{3} \end{aligned}$$

Now the circle with centre (p, t^2) is $(3, 3)$.

$$(x - 3)^2 + (y - 3)^2 = r^2$$

The circle passes through origin so,

$$\begin{aligned} (0 - 3)^2 + (0 - 3)^2 &= r^2 \\ r^2 &= 9 + 9 \\ &= 18 \end{aligned}$$

The circle S' will be,

$$(x - 3)^2 + (y - 3)^2 = 18$$

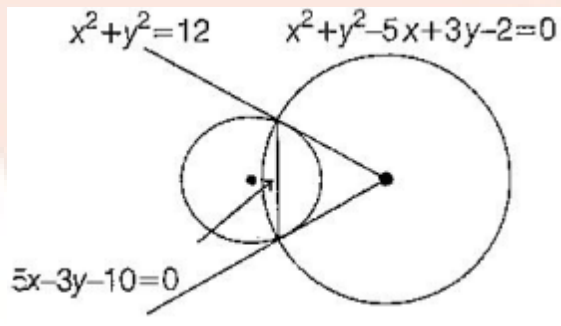
Now, the point $(2, 3)$ w.r.t circle $(x - 3)^2 + (y - 3)^2 = 18$ is,

$$\begin{aligned} (2 - 3)^2 + (3 - 3)^2 - 18 &= 1 - 18 \\ &= -17 < 0 \end{aligned}$$

So, the point $(2, 3)$ lies inside the circle $S' = 0$.

50. Let (h, k) be the intersection point of tangent. Then the chord of the tangent is the common chord of the circles $x^2 + y^2 = 12$ and $x^2 - y^2 - 5x + 3y - 2 = 0$

Consider the diagram,



The equation of the common chord is,

$$5x - 3y - 10 = 0 \quad \text{(I)}$$

The equation of common chord of contact is,

$$hx + ky - 12 = 0 \quad \text{(II)}$$

The equation (I) and (II) represents the same line so,

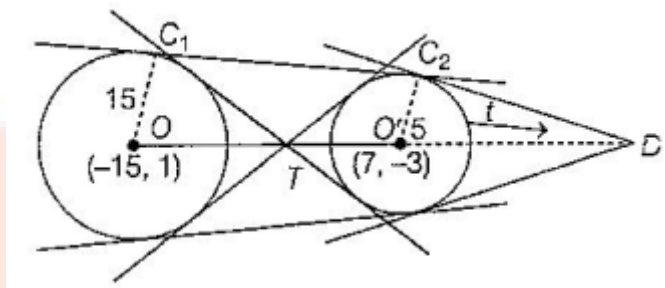
$$\begin{aligned} \frac{h}{5} &= \frac{k}{-3} \\ &= \frac{-12}{-10} \end{aligned}$$

This gives,

$$h = 6, k = \frac{-18}{5}$$

Hence the required point is $\left(6, -\frac{18}{5}\right)$

51. Consider the diagram of the given circles and lines as shown below.



Given circles,

$$C_1 : x^2 + y^2 + 30x - 2y + 1 = 0$$

$$\Rightarrow \text{centre}(O) = (-15, 1)$$

$$\Rightarrow \text{radius}(r_1) = \sqrt{225 + 1 - 1} = 15$$

And,

$$C_2 : x^2 + y^2 - 14x + 6y + 33 = 0$$

$$\Rightarrow \text{centre}(O') = (7, -3)$$

$$\Rightarrow \text{radius}(r_2) = \sqrt{49 + 9 - 33} = 5$$

Since point T divides OO' in the ratio of 15:5 i.e., 3:1 internally.

The co-ordinates of point T is,

$$T = \left(\frac{21 - 15}{4}, \frac{-9 + 1}{4} \right)$$

$$T = \left(\frac{3}{2}, -2 \right)$$

Since point D divides OO' in the ratio of 15:5 i.e., 3:1 externally.

The co-ordinates of point D is,

$$D = \left(\frac{21+15}{4}, \frac{-9-1}{4} \right)$$

$$D = (18, -5)$$

The centre of circle with TD as diameter is,

$$C = \left(\frac{18+3/2}{2}, \frac{-2-5}{2} \right)$$

$$C = \left(\frac{39}{4}, \frac{-7}{2} \right)$$

52. The centers of two circles are $C_1(-\lambda, 0)$ and $C_2(0, -2)$ and their radii are $\sqrt{\lambda^2 - 2}$ and $\sqrt{2}$. So the circles touches each other if,

$$\sqrt{\lambda^2 + 4} = \sqrt{\lambda^2 - 2} + \sqrt{2}$$

$$\lambda^2 + 4 = \lambda^2 - 2 + 2 + 2\sqrt{2}\sqrt{\lambda^2 - 2}$$

$$\sqrt{2} = \sqrt{\lambda^2 - 2}$$

$$\lambda = \pm 2$$

53. The equation of common chord of given circles is,

$$2x + 1 = 0$$

The equation of circle passing through the intersection of given circles is,

$$\begin{aligned}(x^2 + y^2 + 2x + 3y + 1) + \lambda(x^2 + y^2 + 4x + 3y + 2) &= 0 \\ x^2(1 + \lambda) + y^2(1 + \lambda) + (1 + 2\lambda)2x + 3y(1 + \lambda) + 1 + 2\lambda &= 0 \\ x^2 + y^2 + \left(\frac{1 + 2\lambda}{1 + \lambda}\right)2x + 3y + \frac{1 + 2\lambda}{\lambda + 1} &= 0 \dots (I)\end{aligned}$$

Since $2x + 1 = 0$ is diameter of circle so,

$$\begin{aligned}-2\left(-\frac{2\lambda + 1}{1 + \lambda}\right) + 1 &= 0 \\ -4\lambda - 2 + \lambda + 1 &= 0 \\ -3\lambda - 1 &= 0 \\ \lambda &= -\frac{1}{3}\end{aligned}$$

The from equation (I),

$$\begin{aligned}x^2 + y^2 + \left(\frac{1 - \frac{2}{3}}{1 - \frac{1}{3}}\right)2x + 3y + \frac{1 - \frac{2}{3}}{-\frac{1}{3} + 1} &= 0 \\ x^2 + y^2 + \left(\frac{\frac{1}{3}}{\frac{2}{3}}\right)2x + 3y + \frac{\frac{1}{3}}{\frac{2}{3}} &= 0 \\ x^2 + y^2 + x + 3y + \frac{1}{2} &= 0 \\ 2x^2 + 2y^2 + 2x + 6y + 1 &= 0\end{aligned}$$

54. Since, semi latus rectum is the harmonic mean of the segments.

So,

$$\begin{aligned}\text{length of semi latus rectum} &= \frac{2 \times 5 \times 3}{5 + 3} \\ &= \frac{30}{8} \\ &= \frac{15}{4}\end{aligned}$$

So,

$$\begin{aligned}\text{length of latus rectum} &= 2 \times \frac{15}{4} \\ &= \frac{15}{2}\end{aligned}$$

55. Consider the equation of tangent to parabola $y^2 = 4x$,

$$y = mx + \frac{1}{m}$$

The equation passes through $(1, 4)$ so,

$$4 = m + \frac{1}{m}$$

$$m^2 = 4m + 1$$

$$m = \frac{4 \pm \sqrt{12}}{2}$$
$$= 2 \pm \sqrt{3}$$

So, the angle between the slopes is,

$$\tan \theta = \frac{|m_1 - m_2|}{m_1 m_2}$$
$$= \frac{2\sqrt{3}}{1+1}$$
$$= \sqrt{3}$$
$$= \tan \frac{\pi}{3}$$

Thus,

$$\theta = \frac{\pi}{3}$$

56. The tangent to parabola, $y^2 = 4x$ at $(t^2, 2t)$ is,

$$y \cdot 2t = 2(x + t^2)$$

$$yt = x + t^2$$

$$y = \frac{x}{t} + t \quad (I)$$

Normal to the ellipse,

$$4x^2 + 5y^2 = 20$$

or

$$\frac{x^2}{5} + \frac{y^2}{4} = 1 \text{ at } (\sqrt{5} \cos \theta, 2 \sin \theta)$$

So, the slope of normal is,

$$\frac{5y}{4x} = \frac{\sqrt{5}}{2} \tan \theta$$

So, the equation of the line is,

$$y - 2 \sin \theta = \frac{\sqrt{5}}{2} \tan \theta (x - \sqrt{5} \cos \theta)$$
$$y = \frac{\sqrt{5}}{2} \tan \theta \times x - \frac{\sin \theta}{2} \quad \text{(II)}$$

Compare (I) and (II),

$$t = -\frac{\sin \theta}{2}$$

$$\sin \theta = -2t \quad \text{(III)}$$

And,

$$\frac{1}{t} = \frac{\sqrt{5}}{2} \tan \theta$$

$$\sin \theta = \frac{2}{\sqrt{4 + 5t^2}} \quad \text{(IV)}$$

From equation (III) and (IV),

$$2t = \frac{2}{\sqrt{4+5t^2}}$$

$$t^2(4+5t^2) = 1$$

$$5t^4 + 4t^4 = 1$$

57. Consider the equation of ellipse,

$$4x^2 + 9y^2 - 24x + 36y = 0$$

Then,

$$4(x^2 - 6x + 9) + 9(y^2 + 4y + 5) - 36 - 36 = 0$$

$$4(x-3)^2 + 9(y+2)^2 = 72$$

$$\frac{(x-3)^2}{18} + \frac{(y+2)^2}{8} = 1$$

Here required locus is director circle so,

$$(x-3)^2 + (y+2)^2 = 26$$

$$x^2 - 6x + 9 + y^2 + 4 + 4y = 26$$

$$x^2 + y^2 - 6x + 4y = 13$$

$$x^2 + y^2 - 6x + 4y - 13 = 0$$

58. The equation of the chord with midpoint (h, k) to $x^2 + y^2 = 16$

is $T = S_1$

Then,

$$hx + ky - 16 = h^2 + k^2 - 16$$

$$y = -\frac{h}{k}x + \left(\frac{h^2 + k^2}{k}\right)$$

So, the equation of hyperbola is,

$$9x^2 - 16y^2 = 144$$

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

The condition of tangency of line $y = mx + c$ is $a^2m^2 - b^2 = c^2$

So,

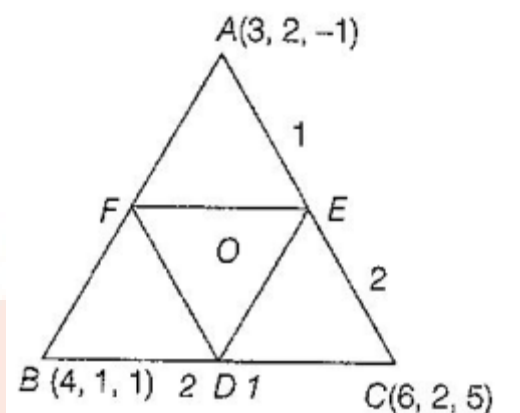
$$16\left(-\frac{h}{k}\right)^2 - 9 = \left(\frac{h^2 + k^2}{k}\right)^2$$

$$16h^2 - 9k^2 = (h^2 + k^2)^2$$

Replace h, k by x, y so,

$$16x^2 - 9y^2 = (x^2 + y^2)^2$$

59. Consider the diagram,



The co-ordinates of point D are,

$$D = \left(\frac{2 \times 6 + 4 \times 1}{3}, \frac{2 \times 2 + 1 \times 1}{3}, \frac{2 \times 5 + 1 \times 1}{3} \right)$$

$$= \left(\frac{16}{3}, \frac{5}{3}, \frac{11}{3} \right)$$

Similarly,

$$E = \left(\frac{12}{3}, \frac{6}{3}, \frac{3}{3} \right)$$

$$F = \left(\frac{11}{3}, \frac{4}{3}, \frac{1}{3} \right)$$

If the centroid of triangle DEF is $O(x, y, z)$, then

$$x = \frac{\frac{16}{3} + \frac{12}{3} + \frac{11}{3}}{3} = \frac{13}{3}$$

$$y = \frac{\frac{12}{3} + \frac{6}{3} + \frac{4}{3}}{3} = \frac{5}{3}$$

$$z = \frac{\frac{11}{3} + 1 + \frac{1}{3}}{3} = \frac{5}{3}$$

So, the coordinates are $\left(\frac{13}{3}, \frac{5}{3}, \frac{5}{3}\right)$.

60. From the given information,

$$\mathbf{OA} = \hat{\mathbf{i}} + 8\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$$

$$\mathbf{OB} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

So,

$$\hat{\mathbf{n}} = \frac{\mathbf{OA} \times \mathbf{OB}}{|\mathbf{OA} \times \mathbf{OB}|}$$

Here,

$$\begin{aligned} \mathbf{OA} \times \mathbf{OB} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 8 & 4 \\ 2 & -3 & 1 \end{vmatrix} \\ &= \hat{\mathbf{i}}(8 + 12) - \hat{\mathbf{j}}(1 - 8) + \hat{\mathbf{k}}(-3 - 16) \\ &= 20\hat{\mathbf{i}} + 7\hat{\mathbf{j}} - 19\hat{\mathbf{k}} \end{aligned}$$

The magnitude of $\mathbf{OA} \times \mathbf{OB}$ is,

$$\begin{aligned} |\mathbf{OA} \times \mathbf{OB}| &= \sqrt{20^2 + 7^2 + (-19)^2} \\ &= \sqrt{20^2 + 7^2 + (-19)^2} \\ &= \sqrt{810} \\ &= 9\sqrt{10} \end{aligned}$$

The unit directional vector is,

$$\hat{\mathbf{n}} = \frac{20\hat{\mathbf{i}} + 7\hat{\mathbf{j}} - 19\hat{\mathbf{k}}}{9\sqrt{10}}$$

So, the direction cosines are,

$$\frac{20}{9\sqrt{10}}, \frac{7}{9\sqrt{10}}, \frac{-19}{9\sqrt{10}} \text{ or } \frac{2\sqrt{10}}{9}, \frac{7\sqrt{10}}{90}, \frac{-19\sqrt{10}}{90}$$

61. The given line is,

$$\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4} = \lambda$$

$$\text{So, } (x, y, z) = (2\lambda + 1, 3\lambda - 1, 4\lambda + 1)$$

The above point is present on the given line so,

$$\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$$

Substitute values,

$$\frac{2\lambda + 1 - 3}{1} = \frac{3\lambda - 1 - k}{2} = \frac{4\lambda + 1}{1}$$

$$\frac{2\lambda - 2}{1} = \frac{4\lambda + 1}{1}$$

$$2\lambda - 2 = 4\lambda + 1$$

$$\lambda = -\frac{3}{2}$$

Now,

$$\frac{3\lambda - 1 - k}{2} = \frac{2\lambda - 2}{1}$$

$$\frac{3\left(-\frac{3}{2}\right) - (-1 - k)}{2} = \frac{2\left(-\frac{3}{2}\right) - 2}{1}$$

$$\frac{-\frac{9}{2} - 1 - k}{2} = \frac{-3 - 2}{1}$$

$$k = \frac{9}{2}$$

62. Let's consider the given expression as,

$$P = \lim_{x \rightarrow \infty} \frac{x^2 (\tan 2x - 2 \tan x)^2}{(1 - \cos 2x)^4}$$

Simplify the above expression,

$$\begin{aligned}
 P &= \lim_{x \rightarrow \infty} \frac{x^2 \left[\left(2x + \frac{(2x)^3}{3} + \frac{2}{15}(2x)^5 + \dots \right) - 2 \left(x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots \right) \right]^2}{\left[1 - \left(1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} + \dots \right) \right]^4} \\
 &= \lim_{x \rightarrow \infty} \frac{4x^8 \left[\left(\frac{4}{3} - \frac{1}{3} \right) + \frac{2}{15}(16x^2 - x^2) + \dots \right]^2}{16x^8 \left[1 + \frac{x^2}{3} + \dots \right]^4} \\
 &= \frac{4}{16} \\
 &= \frac{1}{4}
 \end{aligned}$$

63. The given expression is solved as,

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \left(\frac{6x^2 - \cos 3x}{x^2 + 5} - \frac{5x^3 + 3}{\sqrt{x^6 + 2}} \right) &= \lim_{x \rightarrow \infty} \left(\frac{6 - \frac{\cos 3x}{x^2}}{1 + \frac{5}{x^2}} - \frac{5 + \frac{3}{x^2}}{\left(\frac{|x|^3}{x^3} \right) \sqrt{1 + \frac{2}{x^6}}} \right) \\
 &= 6 + 5 \\
 &= 11
 \end{aligned}$$

64. Consider the function,

$$f(x) = \frac{x-1}{x^3 + 6x^2 + 11x + 6}$$

Simplify the above expression,

$$f(x) = \frac{x-1}{(x+1)(x^2 + 5x + 6)}$$

$$f(x) = \frac{x-1}{(x+1)(x+2)(x+3)}$$

For function to be discontinuous denominator must be 0 so,

$$(x+1)(x+2)(x+3) = 0$$

$$x = -1, -2, -3$$

So, number of discontinuous is R is 3.

65. The given expression is simplified as below,

$$\begin{aligned} \frac{d}{dx} \left(\log \left(\sqrt{x + \sqrt{x^2 + a^2}} \right) \right) &= \left(\frac{1}{\sqrt{x + \sqrt{x^2 + a^2}}} \times \frac{1}{2\sqrt{x + \sqrt{x^2 + a^2}}} \right) \\ &\quad \times \left(1 + \frac{1}{2} \frac{2x}{\sqrt{x^2 + a^2}} \right) \\ &= \frac{(\sqrt{x^2 + a^2} + x)}{2\sqrt{x^2 + a^2} (x + \sqrt{x^2 + a^2})} \\ &= \frac{1}{2\sqrt{x^2 + a^2}} \end{aligned}$$

66. Consider the function,

$$f(x) = \cot^{-1} \left(\frac{x^x - x^{-x}}{2} \right)$$

Let,

$$y = \cot^{-1} \left(\frac{x^{2x} - 1}{2x^x} \right)$$

Put $x^x = \tan \theta$ then,

$$\begin{aligned} y &= \cot^{-1} \left(\frac{\tan^2 \theta - 1}{2 \tan \theta} \right) \\ &= \cot^{-1} (-\cot 2\theta) \\ &= \pi - \cot^{-1} (\cot 2\theta) \\ &= \pi - \theta \end{aligned}$$

This implies,

$$y = \pi - 2 \tan^{-1} (x^x)$$

Differentiate above,

$$\frac{dy}{dx} = \frac{-2}{1 + x^{2x}} \cdot x^x (1 + \log x)$$

So,

$$\begin{aligned}\left.\frac{dy}{dx}\right|_{x=1} &= \frac{-2}{1+(1)^2} \cdot 1^1 (1 + \log 1) \\ &= -\frac{2}{2}(1) \\ &= -1\end{aligned}$$

67. Consider the given information,

$$x = a(t + \sin t) \quad \text{(I)}$$

And,

$$y = a(1 - \cos t) \quad \text{(II)}$$

Differentiate equation (I) w.r.t t then,

$$\frac{dx}{dt} = a(1 + \cos t)$$

Differentiate equation (II) w.r.t t then,

$$\frac{dy}{dx} = a \sin t$$

Then,

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\ &= \frac{a \sin t}{a(1 + \cos t)} \\ &= \frac{\sin t}{(1 + \cos t)}\end{aligned}$$

Now,

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{(1 + \cos t)\cos t - \sin t(-\sin t)}{(1 + \cos t)^2} \times \frac{1}{a(1 + \cos t)} \\ &= \frac{(\cos t + 1)}{a(1 + \cos t)^3} \\ &= \frac{1}{a(1 + \cos t)^2}\end{aligned}$$

$$\text{At } t = \frac{2\pi}{3},$$

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{1}{a\left(\frac{1}{2}\right)^2} \\ &= \frac{4}{a}\end{aligned}$$

68. The given curve is,

$$y^2(x-a) = x^2(x+a)$$

Apply the log on both sides then,

$$\log(y^2(x-a)) = \log(x^2(x+a))$$

$$2\log y + \log(x-a) = 2\log x + \log(x+a)$$

Differentiate both sides w.r.t x ,

$$\frac{2}{y} \frac{dy}{dx} + \frac{1}{x-a} = \frac{2}{x} + \frac{1}{x+a}$$

$$\frac{2}{y} \frac{dy}{dx} = \frac{2}{x} + \frac{1}{x+a} - \frac{1}{x-a}$$

$$= \frac{2(x+a)(x-a) + x(x-a) - x(x+a)}{x(x+a)(x-a)}$$

$$= \frac{2x^2 - 2a^2 - 2ax}{x(x-a)(x+a)}$$

Further simplify the above,

$$\frac{dy}{dx} = \frac{x^2 - a^2 - ax}{x(x-a)(x+a)} x \sqrt{\frac{x+a}{x-a}}$$

$$\text{At } \frac{dy}{dx} = 0,$$

Thus,

$$x^2 - a^2 - ax = 0$$

$$\text{Here, } p = a^2 + 4a^2 > 0$$

So, it has two real roots.

69. Consider the function,

$$f(x) = (2k + 1)x - 3 - ke^{-x} + 2e^x$$

Since $f(x)$ is monotonically increasing for all $x \in R$ so,

$$f'(x) \geq 0$$

$$(2k + 1) + ke^{-x} + 2e^x \geq 0$$

$$e^{-x}(2k + 1)e^x + k + 2e^{2x} \geq 0$$

$$2e^{2x}(e^x + k) + 1(e^x + k) \geq 0$$

Further simplify the above,

$$(2e^x + 1)(e^x + k) \geq 0$$

$$e^x + k \geq 0$$

$$k \geq 0$$

So, the least value of k is 0.

70. Consider the function,

$$f(x) = ax^3 + bx^2 + 11x - 6$$

Above satisfies the Rolle's theorem in $[1, 3]$ so,

$$f(1) = 0 \text{ and } f(3) = 0$$

This implies,

$$a + b + 5 = 0$$

(I)

And,

$$27a + 9b + 27 = 0$$

$$9a + 3b + 9 = 0$$

$$3a + b + 3 = 0 \quad (II)$$

From equation (I) and (II),

$$a + b = -5$$

71. Consider the function,

$f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$ has maximum value at p and minimum value at q so,

$$f'(p) = 0 \text{ and } f'(q) = 0$$

Now,

$$f'(x) = 6x^2 - 18ax + 12a^2$$

Has roots p and q so,

$$\begin{aligned} p + q &= \frac{18a}{6} \\ &= 3a \end{aligned}$$

And,

$$\begin{aligned} pq &= \frac{12a^2}{6} \\ &= 2a^2 \end{aligned}$$

Solving above,

$$p = a \text{ and } q = 2a$$

$$\text{Since } p^2 = q$$

Given that, $p^2 = q$, thus

$$a^2 = 2a$$

$$a^2 - 2a = 0$$

$$a(a - 2) = 0$$

$$a = 2$$

72. Consider the integral,

$$\begin{aligned}\int \cos x \cos 2x \cos 5x dx &= \frac{1}{2} \int 2 \cos x \cos 2x \cos 5x dx \\&= \frac{1}{2} \int \{ \cos(5x + x) + \cos(5x - x) \} \cos 2x dx \\&= \frac{1}{2} \int (\cos 6x + \cos 4x) \cos 2x dx \\&= \frac{1}{4} \int (2 \cos 6x \cos 2x + 2 \cos 4x \cos 2x) dx\end{aligned}$$

Further simplify the above expression,

$$\begin{aligned}\int \cos x \cos 2x \cos 5x dx &= \frac{1}{4} \int (\cos 8x + \cos 4x + \cos 6x + \cos 2x) dx \\&= \frac{1}{4} \left[\frac{\sin 8x}{8} + \frac{\sin 4x}{4} + \frac{\sin 6x}{6} + \frac{\sin 2x}{2} \right] + k \\&= \frac{\sin 2x}{8} + \frac{\sin 4x}{16} + \frac{\sin 6x}{24} + \frac{\sin 8x}{32} + k\end{aligned}$$

On comparison both sides,

$$A = \frac{1}{8}, B = \frac{1}{16}, C = \frac{1}{24} \text{ and } D = \frac{1}{32}$$

Then,

$$\begin{aligned} \frac{1}{B} + \frac{1}{C} &= 16 + 24 \\ &= 40 \end{aligned}$$

And,

$$\begin{aligned} \frac{1}{A} + \frac{1}{D} &= 8 + 32 \\ &= 40 \end{aligned}$$

This implies,

$$\frac{1}{B} + \frac{1}{C} = \frac{1}{A} + \frac{1}{D}$$

73. Consider the given expression,

$$\int e^x \left(\frac{x+2}{x+4} \right)^2 dx = f(x) + c$$

Now,

$$\begin{aligned}\int e^x \left(\frac{x+2}{x+4} \right)^2 dx &= \int e^x \left(\frac{x^2 + 4 + 2x^2}{(x+4)^2} \right) dx \\ &= \int e^x \left(\frac{x}{(x+4)} + \frac{4}{(x+4)^2} \right) dx\end{aligned}$$

Let, $g(x) = \frac{x}{x+4}$ then,

$$g'(x) = \frac{4}{(x+4)^2}$$

So,

$$\begin{aligned}\int e^x \{g(x) + g'(x)\} dx &= e^x g(x) + c \\ &= e^x \left(\frac{x}{x+4} \right) + c\end{aligned}$$

Thus,

$$f(x) = \frac{xe^x}{x+4}$$

74. Consider the given expression,

$$\begin{aligned}\int \frac{dx}{\sin x + \sin 2x} &= \int \frac{dx}{\sin x + \sin 2x} \\&= \int \frac{dx}{\sin x(1 + 2\cos x)} \\&= \int \frac{\sin x dx}{\sin^2 x(1 + 2\cos x)} \\&= \int \frac{-\sin x dx}{(\cos^2 x - 1)(1 + 2\cos x)}\end{aligned}$$

Let $\cos x = t$ then,

$$-\sin x dx = dt$$

So,

$$\int \frac{dt}{(t^2 - 1)(1 + 2t)}$$

By partial fraction,

$$A = \frac{1}{6}, B = \frac{1}{2} \text{ and } C = -\frac{4}{3}$$

Now,

$$\begin{aligned}
\int \frac{dt}{(t^2-1)(1+2t)} &= \int \frac{\frac{1}{6}}{t-1} dt + \frac{1}{2} \int \frac{dt}{t+1} - \frac{4}{6} \int \frac{dt}{\left(t+\frac{1}{2}\right)} \\
&= \frac{\ln(t-1)}{6} + \frac{1}{2} \ln(t+1) - \frac{4}{6} \ln\left(t+\frac{1}{2}\right) + C \\
&= \left[\frac{\ln(\cos x - 1)}{6} + \frac{1}{2} \ln(\cos x + 1) \right] \\
&\quad \left[-\frac{2}{3} \ln\left(\cos x + \frac{1}{2}\right) + C \right] \\
&= \left[\frac{1}{2} \ln|(1 + \cos x)| + \left| \frac{1}{6} \ln(1 - \cos x) \right| \right] \\
&\quad \left[-\frac{2}{3} \ln\left|\cos x + \frac{1}{2}\right| + C \right]
\end{aligned}$$

Further simplify the above expression,

$$\begin{aligned}
&\int \frac{dt}{(t^2-1)(1+2t)} \\
&= \frac{1}{2} \ln(1 + \cos x) + \frac{1}{6} \ln|(1 - \cos x)| - \frac{2}{3} \ln|1 + 2 \cos x| + C
\end{aligned}$$

75. Consider the given expression,

$$I_n = \int \frac{\sin nx}{\sin x} dx \quad (\text{I})$$

And,

$$I_{n-2} = \int \frac{\sin(n-2)x}{\sin x} dx \quad (\text{II})$$

Subtract equation (I) and (II),

$$\begin{aligned} I_n - I_{n-2} &= \int \frac{(\sin nx - \sin(n-2)x)}{\sin x} dx \\ &= \int \frac{2 \sin(n-1)x}{\sin x} dx \\ &= 2 \cos(n-1)x dx \\ &= \frac{2 \sin(n-1)x}{n-1} \end{aligned}$$

This implies,

$$I_6 - I_4 = \frac{2 \sin 5x}{5} \text{ and } I_4 - I_2 = \frac{2 \sin 3x}{3}$$

Now,

$$\begin{aligned} I_2 &= \int \frac{\sin 2x}{\sin x} dx \\ &= \int \frac{2 \sin x \cos x}{\sin x} dx \\ &= 2 \int \cos x dx \\ &= 2 \sin x + c \end{aligned}$$

And,

$$\begin{aligned}
I_6 &= I_4 + 2 \frac{\sin 5x}{5} \\
&= I_2 + 2 \frac{\sin 3x}{3} + 2 \frac{\sin 5x}{5} \\
&= 2 \frac{\sin 5x}{5} + 2 \frac{\sin 3x}{3} + 2 \sin x + c \\
&= \frac{2 \sin 5x}{5} + \frac{2}{3} (3 \sin x - 4 \sin^3 x) + 2 \sin x + c
\end{aligned}$$

So,

$$I_6 = \frac{2}{5} \sin 5x - \frac{8}{3} \sin^3 x + 4 \sin x + c$$

76. Consider the expression,

$$\begin{aligned}
k &= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{1}{n^2} \right) \left(1 + \frac{2^2}{n^2} \right) \dots \left(1 + \frac{n^2}{n^2} \right) \right]^{\frac{1}{n}} \\
\log k &= \lim_{x \rightarrow \infty} \log \left[\left(1 + \frac{1}{n^2} \right) \left(1 + \frac{2^2}{n^2} \right) \dots \left(1 + \frac{n^2}{n^2} \right) \right]^{\frac{1}{n}} \\
&= \lim_{x \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \log \left(1 + \frac{r^2}{n^2} \right) \\
&= \int_0^1 \log(1 + x^2) dx
\end{aligned}$$

Solve further,

$$\begin{aligned}\log k &= \left[\log(1+x^2)x \right]_0^1 - \int_0^1 \log(1+x^2) dx \\&= \log 2 - 2 \int_0^1 \frac{x^2}{1+x^2} dx \\&= \log 2 - 2 \int_0^1 \left(1 - \frac{1}{1+x^2} \right) dx \\&= \log 2 - 2 \left[x - \tan^{-1} x \right]_0^1\end{aligned}$$

Simplify further,

$$\begin{aligned}\log k &= \log 2 - 2 \left[1 - \frac{\pi}{4} \right] \\&= \log 2 - 2 + \frac{\pi}{2} \\&= \log 2 + \frac{\pi}{2} - 2\end{aligned}$$

77. Let's consider the given expression as,

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin^3 x \cos x dx}{\sin^4 x + \cos^4 x} \quad (\text{I})$$

Then,

$$\begin{aligned}
 I &= \int_0^{\frac{\pi}{2}} \frac{\sin^3\left(\frac{\pi}{2} - x\right) \cos\left(\frac{\pi}{2} - x\right)}{\sin^4\left(\frac{\pi}{2} - x\right) + \cos^4\left(\frac{\pi}{2} - x\right)} dx \\
 &= \int_0^{\frac{\pi}{2}} \frac{\cos^3 x \sin x}{\cos^4 x + \sin^4 x} dx \quad \text{(II)}
 \end{aligned}$$

Add equation (I) and (II),

$$\begin{aligned}
 2I &= \int_0^{\frac{\pi}{2}} \frac{\sin^3 x \cos x + \cos^3 x \sin x}{\cos^4 x + \sin^4 x} dx \\
 &= \int_0^{\frac{\pi}{2}} \frac{\sin x \cos x (\sin^2 x + \cos^2 x)}{\cos^4 x + \sin^4 x} dx \\
 &= \int_0^{\frac{\pi}{2}} \frac{\sin x \cos x}{\cos^4 x + \sin^4 x} dx \\
 &= \int_0^{\frac{\pi}{2}} \frac{\sin x \cos x}{\cos^4 x (\tan^4 x + 1)} dx
 \end{aligned}$$

Further simplify the above expression,

$$2I = \int_0^{\frac{\pi}{2}} \frac{\tan x \sec^2 x}{(\tan^4 x + 1)} dx$$

Let $\tan^2 x = t$ then,

$$2 \tan x \sec^2 x dx = dt$$

Now,

$$\begin{aligned} 2I &= \frac{1}{2} \int_0^{\infty} \frac{dt}{t^2 + 1} \\ &= \frac{1}{2} \left[\tan^{-1} t \right]_0^{\infty} \\ &= \frac{1}{2} \left[\tan^{-1}(\infty) - \tan^{-1}(0) \right] \\ &= \frac{1}{2} \times \frac{\pi}{2} \end{aligned}$$

Thus,

$$\begin{aligned} 2I &= \frac{\pi}{4} \\ I &= \frac{\pi}{8} \end{aligned}$$

78. The curve $y = ax^2 + bx$ passes through $(1, 2)$ so,

$$\begin{aligned} 2 &= a(1)^2 + b(1) \\ a + b &= 2 \end{aligned} \tag{I}$$

Now the area under the curve is,

$$\begin{aligned} \int_0^6 (ax^2 + bx) dx &= 108 \\ \left[\frac{ax^3}{3} + \frac{bx^2}{2} \right]_0^6 &= 108 \\ 72a + 18b &= 108 \\ 4a + b &= 6 \end{aligned} \tag{II}$$

From equation (I) and (II),

$$a = \frac{4}{3} \text{ and } b = \frac{2}{3}$$

Then,

$$\begin{aligned} 2b - a &= 2 \times \frac{2}{3} - \frac{4}{3} \\ &= \frac{4}{3} - \frac{4}{3} \\ &= 0 \end{aligned}$$

79. Let,

$$y = Ax^2 + Bx + C$$

Differentiate above w.r.t x then,

$$\frac{dy}{dx} = 2Ax + B$$

Again differentiate,

$$\frac{d^2y}{dx^2} = 2A$$

Again differentiate,

$$\frac{d^3y}{dx^3} = 0$$

This is the required equation.

80. Consider the differential equation,

$$\frac{dy}{dx} + 2y \tan x = \sin x$$

Here, $P = 2 \tan x$ and $Q = \sin x$

So,

$$\begin{aligned}\text{IF} &= e^{\int 2 \tan x dx} \\ &= e^{2 \ln |\sec x|} \\ &= e^{\ln \sec^2 x} \\ &= \sec^2 x\end{aligned}$$

Now the required solution is,

$$y \times (\text{IF}) = \int Q(\text{IF}) dx + c$$

Substitute values.

$$y \times \sec^2 x = \int \sin x \sec^2 x dx + c$$

$$\begin{aligned}y \sec^2 x &= \int \sec x \tan x dx + c \\ &= \sec x + c\end{aligned}$$

At $y = 0$, $x = \frac{\pi}{3}$ so,

$$0 = 2 + c$$

$$c = -2$$

Thus,

$$y \sec^2 x = \sec x - 2$$

$$\begin{aligned} y &= \frac{\sec x}{\sec^2 x} - \frac{2}{\sec^2 x} \\ &= \cos x - 2 \cos^2 x \\ &= \cos x - 2(1 - \sin^2 x) \end{aligned}$$

Further simplify the above,

$$y = \cos x - 2 + 2 \sin^2 x$$

$$y = 2 \sin^2 x + \cos x - 2$$

AP EAMCET-2019-Engineering-21 Apr-Shift-1
Solutions – Physics

81. The value of T is calculated as,

$$\begin{aligned} T &= \sqrt{(\Delta t_1)(\Delta t_2)} \\ &= \sqrt{(2.00)(4.00)} \\ &= 2.828427 \text{ s} \end{aligned}$$

The round off value of T , so $T = 2.83 \text{ s}$.

The expression for the relative error of the product is,

$$\pm \frac{\Delta T}{T} = \pm \left(\frac{1}{2} \frac{\Delta t_1}{t_1} + \frac{1}{2} \frac{\Delta t_2}{t_2} \right)$$

Substitute the given values in the above equation.

$$\begin{aligned} \Delta T &= \pm \left(\frac{1}{2} \frac{\Delta t_1}{t_1} + \frac{1}{2} \frac{\Delta t_2}{t_2} \right) T \\ &= \pm \left(\frac{1}{2} \frac{0.02}{2.00} + \frac{1}{2} \frac{0.02}{4.00} \right) \times 2.8284 \\ &= 0.02121 \text{ s} \end{aligned}$$

The round off value of ΔT , so $\Delta T = 0.02 \text{ s}$

Thus, $T = (2.83 + 0.02) \text{ s}$

82. The expression of the angle between the initial and the final velocities of the particle is,

$$\cos \theta = \frac{\Delta v^2 - v_i^2 - v_f^2}{2v_i v_f}$$

Substitute the given values in the above equation.

$$\cos \theta = \frac{(5)^2 - (\sqrt{5})^2 - (2\sqrt{5})^2}{2(\sqrt{5})(\sqrt{5})}$$

$$\cos \theta = 0$$

$$\theta = 90^\circ$$

83. The expression for the maximum height of the projectile is given by,

$$H = \frac{u^2 \sin^2 \theta}{2g} \dots\dots\dots(1)$$

The expression for the maximum range of the projectile is given by,

$$R = \frac{u^2 \sin 2\theta}{g}$$

The ratio of the maximum height to the range of the projectile is calculated as,

$$\frac{H}{R} = \frac{\frac{u^2 \sin^2 \theta}{2g}}{\frac{u^2 \sin 2\theta}{g}}$$

$$= \frac{\sin^3 \theta}{2 \sin 2\theta}$$

Substitute the given values in the above equation.

$$\frac{H}{R} = \frac{\sin^3 \theta}{2 \sin 2\theta}$$

$$\frac{3}{4} = \frac{\sin^3 \theta}{2 \sin 2\theta}$$

$$3 \sin 2\theta = 2 \sin^3 \theta$$

$$\tan \theta = 3$$

From the geometry of the triangle,

$$\sin \theta = \frac{3}{\sqrt{10}} \dots\dots\dots(2)$$

From the equation (1),

$$\frac{u^2 \sin^2 \theta}{2g} = 3$$

$$u^2 \sin^2 \theta = 3 \times 2g$$

Substitute the values in the above equation.

$$u^2 \left(\frac{3}{\sqrt{10}} \right)^2 = 3 \times 2(10)$$

$$u^2 = \frac{100 \times 2}{3}$$

$$u = 10\sqrt{\frac{2}{3}} \text{ ms}^{-1}$$

84. The expression of the displacement from the equation of the motion is,

$$s = ut + \frac{1}{2}gt^2$$

Substitute the values assuming complete flight.

$$0 = u(2) - \frac{1}{2}(10)(2)^2$$

$$u = 10 \text{ ms}^{-1}$$

The expression of the maximum height is,

$$h = ut - \frac{1}{2}gt^2$$

Substitute the values in the above equation.

$$\begin{aligned} h &= (10)(1) - \frac{1}{2}(10)(1)^2 \\ &= 5 \text{ m} \end{aligned}$$

85. From the graph, the final velocity is calculated as,

$$\begin{aligned} v_f &= \frac{3-3}{8-4} \\ &= 0 \end{aligned}$$

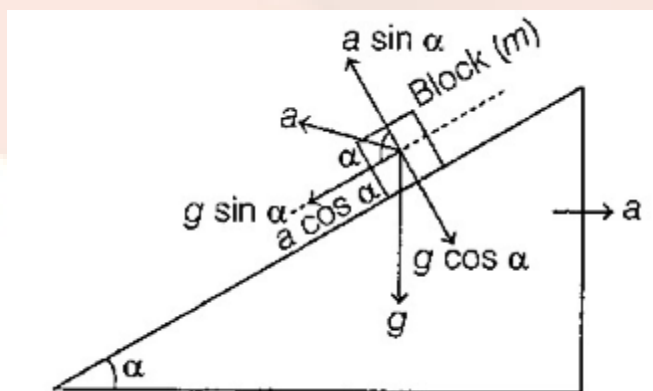
From the graph, the initial velocity is calculated as,

$$\begin{aligned} v_i &= \frac{3-0}{4-0} \\ &= \frac{3}{4} \end{aligned}$$

The impulse is calculated as,

$$\begin{aligned} \text{Impulse} &= m(v_f - v_i) \\ &= 2 \times \left(0 - \frac{3}{4} \right) \\ &= -1.5 \text{ kg} \cdot \text{ms}^{-1} \end{aligned}$$

86. Consider the free body diagram of the block plane system as shown below.



The expression of the downward acceleration is,

$$a_{\text{down}} = g \sin \alpha + a \cos \alpha$$

The expression of the upward acceleration is,

$$a_{\text{up}} = \mu (g \cos \alpha - a \sin \alpha)$$

The condition for the block to remain stationary is,

$$a_{\text{down}} = a_{\text{up}}$$

$$\mu = \frac{g \sin \alpha + a \cos \alpha}{g \cos \alpha - a \sin \alpha}$$

Substitute the given values in above expression,

$$\mu = \frac{10 + 2 \frac{1}{\tan \alpha}}{10 \frac{1}{\tan \alpha} - 2}$$

$$\mu = \frac{10 + 2 \times 5}{10 \times 5 - 2}$$

$$\mu = \frac{20}{48}$$

$$\mu = \frac{5}{12}$$

87. Condition for the minimum potential energy is,

$$U_{\min} = \frac{dU(x)}{dx}$$

$$0 = \frac{d}{dx} \left[\frac{x^2}{2} - x \right]$$

$$0 = x - 1$$

$$x = 1$$

The minimum potential energy is calculated as,

$$\begin{aligned} U_{\min} &= \frac{(1)^2}{2} - 1 \\ &= -\frac{1}{2} \end{aligned}$$

The expression of the kinetic energy is,

$$\begin{aligned} \text{KE} &= \frac{1}{2}mv^2 \\ &= \frac{1}{2}(1)v^2 \\ &= \frac{v^2}{2} \end{aligned}$$

The total mechanical energy of the system is given by,

$$E = U + KE$$

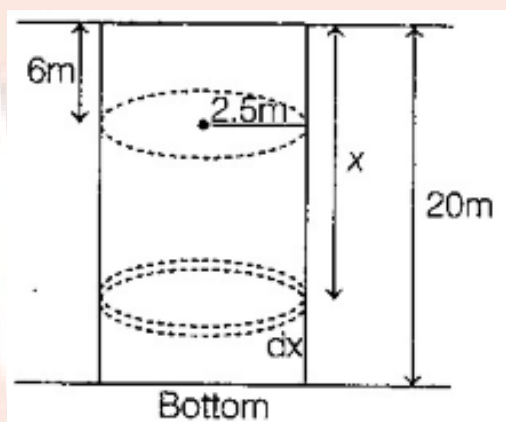
Substitute the given values in the above equation.

$$E = U + KE$$

$$2 = -\frac{1}{2} + \frac{v^2}{2}$$

$$v = \sqrt{5} \text{ m/s}$$

88. Consider the small portion of element dx as shown below.



The mass of the element is,

$$\begin{aligned} dm &= \rho dV \\ &= \rho \pi r^2 dx \end{aligned}$$

The potential energy of the mass dm is calculated as,

$$\begin{aligned} dU &= (dm) gx \\ dU &= g \rho \pi r^2 x dx \end{aligned}$$

Integrate the above equation on both sides.

$$U = g \rho \pi r^2 \int_6^{20} x dx$$

Substitute the given values in the above equation.

$$\begin{aligned}U &= 10 \times 10^3 \times 3.14 \times (2.5)^2 \left[\frac{20^2}{2} - \frac{6^2}{2} \right] \\&= 10^4 \times 3.14 \times (2.5)^2 \times 182 \\&= 35.71 \times 10^6 \text{ J}\end{aligned}$$

The time taken by the pump is calculated as,

$$\begin{aligned}\text{time} &= \frac{\text{work done}}{\text{power}} \\&= \frac{35.71 \times 10^6}{10 \times 746 \times 60} \\&= 80 \text{ min}\end{aligned}$$

89. The magnitude of the radius is calculated as,

$$\begin{aligned}R &= \sqrt{(2)^2 + (1)^2 + (2)^2} \\&= \sqrt{9} \\&= 3 \text{ m}\end{aligned}$$

The magnitude of the force is calculated as,

$$\begin{aligned}F &= \sqrt{(3)^2 + (2)^2 + (-4)^2} \\&= \sqrt{29} \text{ N}\end{aligned}$$

The torque on the flywheel is calculated as,

$$\tau = I\alpha$$

$$FR = \left(\frac{MR^2}{2} \right) \alpha$$

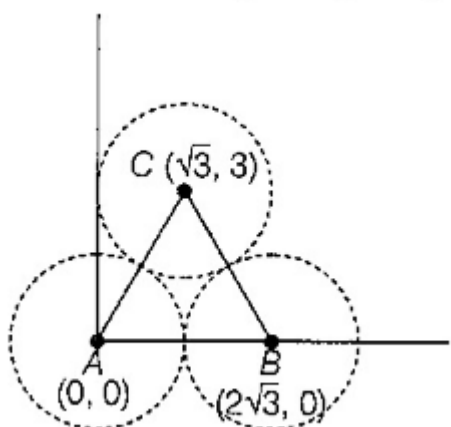
The angular acceleration is,

$$\begin{aligned}\alpha &= \frac{2F}{MR} \\ &= \frac{2\sqrt{29}}{1 \times 3} \\ &= \frac{2}{3}\sqrt{29}\end{aligned}$$

The angular velocity is calculated as,

$$\begin{aligned}\omega &= \omega_0 + \alpha t \\ &= 0 + \frac{2}{3}\sqrt{29} \times (4.5) \\ &= \sqrt{261} \text{ rad s}^{-1}\end{aligned}$$

90. Consider the diagram as shown below,



The expression for the centre of the mass of the spheres is,

$$x_{CM} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$$

The centre of the mass of the spheres along the x axis is calculated as,

$$\begin{aligned}x_{CM} &= \frac{m}{3m}(0 + 2\sqrt{3} + \sqrt{3}) \\&= \frac{3\sqrt{3}}{3} \\&= \sqrt{3}\end{aligned}$$

The centre of the mass of the spheres along the y axis is calculated as,

$$\begin{aligned}y_{CM} &= \frac{m}{3m}(0 + 0 + 3) \\&= 1\end{aligned}$$

The coordinates of the centre of the mass of the sphere is

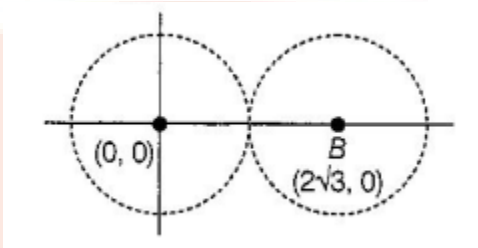
$$C_{CM} = (\sqrt{3}, 1).$$

According to the given condition, the centre of sphere when one sphere is removed is,

$$\begin{aligned}x'_{CM} &= \frac{0 + 2\sqrt{3}}{2} \\&= \sqrt{3}\end{aligned}$$

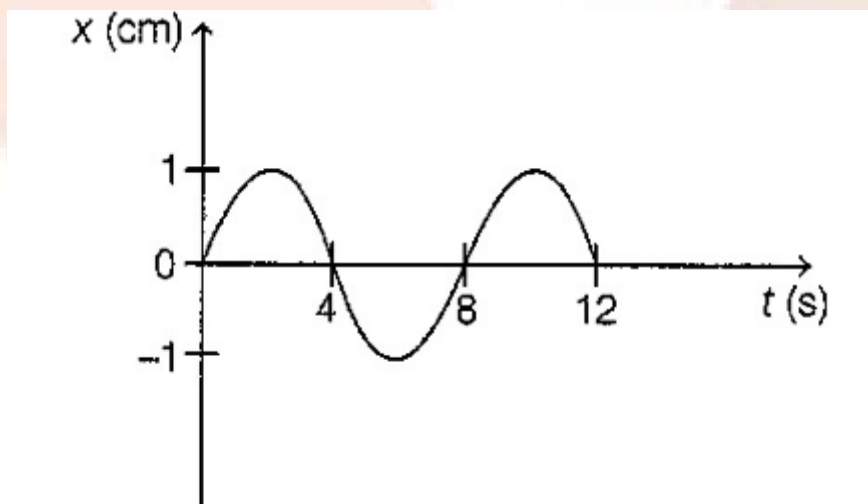
$$y'_{CM} = 0$$

The below diagram is a representation of spheres after the exclusion of one sphere.



The new coordinates of the centre of mass of the spheres is $C'_{CM} = (\sqrt{3}, 0)$, which implies that the centre of the mass is shifted by 1 m in y direction.

91. The diagram of the sine wave function is shown below.



The expression for the angular velocity is given by,

$$\omega = \frac{2\pi}{T}$$

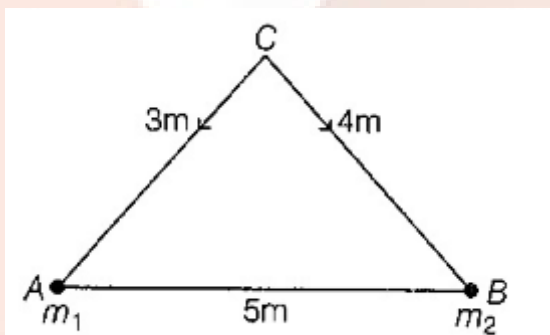
Substitute the given value in the above equation.

$$\begin{aligned}\omega &= \frac{2\pi}{8} \quad (\because T = 8 \text{ s for complete cycle}) \\ &= \frac{\pi}{4} \text{ rad/s}\end{aligned}$$

The acceleration in the simple harmonic motion is calculated as,

$$\begin{aligned}a &= -\omega^2 a \sin \omega t \\ &= -\left(\frac{\pi}{4}\right)^2 \times 1 \times \sin\left(\frac{\pi}{4} \times \frac{4}{3}\right) \\ &= -\frac{\pi^2}{16} \times \frac{\sqrt{3}}{2} \\ &= -\frac{\pi^2 \sqrt{3}}{32} \text{ cm s}^{-2}\end{aligned}$$

92. The below diagram shows the system of the two masses.



The gravitational field intensity due to the mass A is calculated as,

$$\begin{aligned}E_A &= \frac{GM_A}{r_{CA}^2} \\&= G \frac{90}{(3^2)} \\&= 10G \hat{r}_{CA}\end{aligned}$$

The gravitational field intensity due to the mass B is calculated as,

$$\begin{aligned}E_B &= \frac{GM_B}{r_{CB}^2} \\&= G \frac{160}{(4^2)} \\&= 10G \hat{r}_{CB}\end{aligned}$$

The resultant of E_A and E_B is,

$$\begin{aligned}E &= \sqrt{E_A^2 + E_B^2} \\&= \sqrt{(10G)^2 + (10G)^2} \\&= 10\sqrt{2}G\end{aligned}$$

Substitute the value of the G .

$$\begin{aligned}E &= 10\sqrt{2}G \\&= 10\sqrt{2} \times 6.67 \times 10^{-11} \\&= 9.43 \times 10^{-10} \text{ Nkg}^{-1}\end{aligned}$$

93. The expression for the extension of the wire is given by,

$$\Delta L = \frac{FL}{YA}$$

$$\Delta L = \frac{4F}{Y\pi} \times \frac{L}{D^2}$$

$$\Delta L \propto \frac{L}{D^2}$$

Substitute the values given in the option one by one.

For option (1), the extension is calculated as,

$$\begin{aligned}\Delta L &= \frac{4F}{Y\pi} \times \frac{0.5}{(0.5 \times 10^{-3})^2} \\ &= \frac{4F}{Y\pi} \times 2 \times 10^6\end{aligned}$$

Similarly, for option (2), the extension is,

$$\Delta L = \frac{4F}{Y\pi} \times 1.0 \times 10^6$$

For option (3), the extension is,

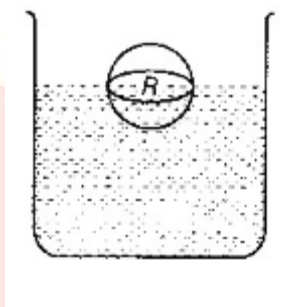
$$\Delta L = \frac{4F}{Y\pi} \times 0.5 \times 10^6$$

For option (4), the extension is,

$$\Delta L = \frac{4F}{Y\pi} \times 0.33 \times 10^6$$

The highest value of extension is associated with the option (1).

94. The liquid drop half immersed in the liquid is shown below.



The force acting on the drop due to surface tension is calculated as,

$$\begin{aligned} F &= Tl \\ &= 10g(\pi D) \end{aligned}$$

The observed weight of the drop is calculated as,

$$\begin{aligned} w &= w_d - w_L \\ &= \rho Vg - \frac{\rho V}{2} g \\ &= \frac{3}{4} \rho Vg \end{aligned}$$

Condition for the equilibrium of ball is,

$$F = w$$

Substitute values,

$$10g\pi D = \frac{3}{4}\rho g \frac{\pi D^3}{6}$$

$$D = \sqrt{\frac{80}{\rho}} \text{ m}$$

95. The expression of the Newton's law of cooling is given by,

$$-\frac{dT}{dt} = k'(T - T_0)$$

Integrate the equation on both sides.

$$\int \frac{dT}{T - T_0} = k' \int dt$$

$$\log_e (T - T_0) = -k't$$

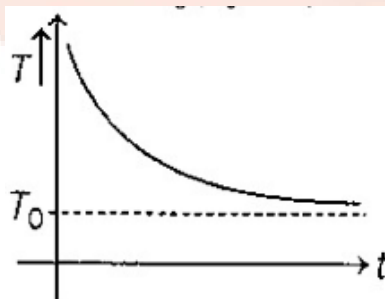
$$\ln(T - T_0) = -k't$$

$$\text{So, } T = e^{-k't} + T_0 \text{ for } t \rightarrow \infty$$

$$\text{And, } T = T_0 \text{ for } T \rightarrow \infty.$$

Hence, the temperature of the body varies exponentially with the time from T to T_0 .

Thus the graph of T verses t is shown below.



96. The expression for the rate of fall of temperature is given by,

$$\frac{\Delta T}{\Delta t} = \frac{\sigma A e (T^4 - T_0^4)}{ms}$$

$$\Delta t = \frac{ms\Delta T}{\sigma A e (T^4 - T_0^4)}$$

Substitute the given values in the given equation.

$$\begin{aligned}\Delta t &= \frac{ms\Delta T}{\sigma A e (T^4 - T_0^4)} \\ &= \frac{(\rho V)C(200\text{K} - 100\text{K})}{\sigma(A)(200^4 - 0^4)} \\ &= \frac{\rho \frac{4}{3} \pi r^3 C \times 100}{\sigma 4\pi r^2 \times (200)^4} \\ &= \frac{1}{48} \frac{\rho r C}{\sigma} \mu\text{s}\end{aligned}$$

97. The expression of the efficiency of the Carnot engine is,

$$\eta = 1 - \frac{T_2}{T_1}$$

For (A), the efficiency is,

$$\eta = 1 - \frac{300}{500}$$
$$= 0.4$$

For (B), the efficiency is,

$$\eta = 1 - \frac{350}{500}$$
$$= 0.3$$

For (C), the efficiency is,

$$\eta = 1 - \frac{400}{800}$$
$$= 0.5$$

For (D), the efficiency is,

$$\eta = 1 - \frac{360}{450}$$
$$= 0.8$$

Thus option (4) is correct.

98. The expression for the kinetic energy is given by,

$$\text{KE} = \frac{1}{2}mv^2$$

Substitute the given values in the above equation.

$$\text{KE} = \frac{1}{2} \times 200 \times 8^2$$
$$= 6400 \text{ J}$$

According to the statement of the question,

$$H = \frac{6400 \times 23}{100}$$
$$= 1472 \text{ J}$$

The rise in temperature of the steel is calculated as,

$$\Delta T = \frac{H}{ms}$$
$$= \frac{1472}{460 \times 0.2}$$
$$= 16 \text{ K}$$

99. The expression for the mean free path is given by,

$$\bar{\lambda} = \frac{kT}{\sqrt{2}\pi d^2 p} \quad \dots\dots (I)$$

The expression for the frequency is given by,

$$\nu = \frac{v}{\lambda} \quad \dots\dots (II)$$

From equation (I) and (II),

$$\nu = \frac{\sqrt{2}\pi d^2 p \times v}{kT}$$

Substitute the given values in the above equation.

$$\begin{aligned}\nu &= \frac{\sqrt{2}(3.14)(10^{-20})(10^5) \times 1380}{(1.38 \times 10^{-23}) \times 314} \\ &= \sqrt{2} \times 10^9 \text{ Hz} \\ &= 1000\sqrt{2} \text{ MHz}\end{aligned}$$

100. The variation of the speed of sound with the temperature is given by,

$$v_t = v_0 \sqrt{\frac{T}{T_0}}$$

Substitute the given values in the above equation.

$$\begin{aligned}v_t &= v_0 \sqrt{\frac{330.75}{300}} \\ &= 1.05v_0\end{aligned}$$

The number of the beats is calculated as,

$$\begin{aligned}n &= n_t - n_0 \\ &= 140(1.05) - 140 \\ &= 7\end{aligned}$$

101. The expression for the apparent frequency for the moving source is,

$$n = \frac{n_0 v}{v - v_s}$$

The condition for $n_{\max}$, $v_s = v_{s\max}$ and for $n_{\min}$, $v_s = -v_{s\max}$.

The expression for $n_{\max}$ is given by,

$$n_{\max} = \frac{n_0 v}{v - v_{s\max}} \dots (1)$$

The expression for $n_{\min}$ is given by,

$$n_{\min} = \frac{n_0 v}{v + v_{s\max}} \dots (2)$$

Divide equation (1) by (2).

$$\frac{1.125 n_{\min}}{n_{\min}} = \frac{v + v_{s\max}}{v - v_{s\max}}$$

$$1.125(v - v_{s\max}) = v + v_{s\max}$$

$$v(0.125) = 2.125 v_{s\max}$$

$$v_{s\max} = \frac{340 \times 0.125}{2.125}$$

$$= 20 \text{ ms}^{-1}$$

The expression for the $v_{s\max}$ is given by,

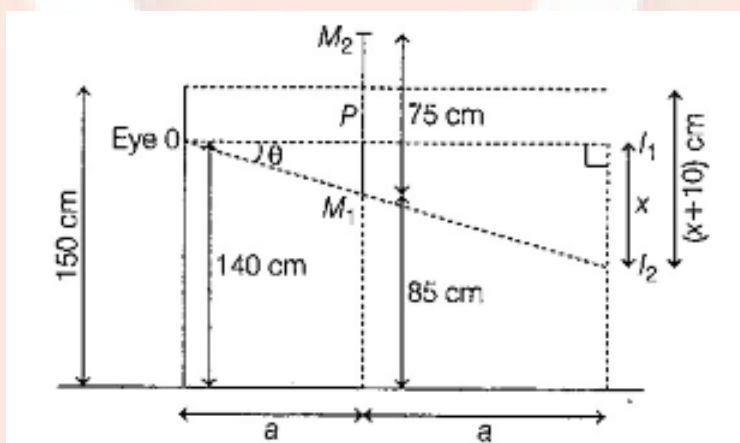
$$v_{s\max} = A \sqrt{\frac{k}{m}}$$

$$k = \frac{v_{s\max}^2 m}{A^2}$$

Substitute the given values in the above equation.

$$\begin{aligned}
 k &= \frac{v_{s_{\max}}^2 m}{A^2} \\
 &= \frac{20 \times 20 \times 0.10}{0.50 \times 0.50} \\
 &= 160 \text{ N m}^{-1}
 \end{aligned}$$

102. The position of a girl in front of a plane mirror is shown below.



From the property of the triangle,

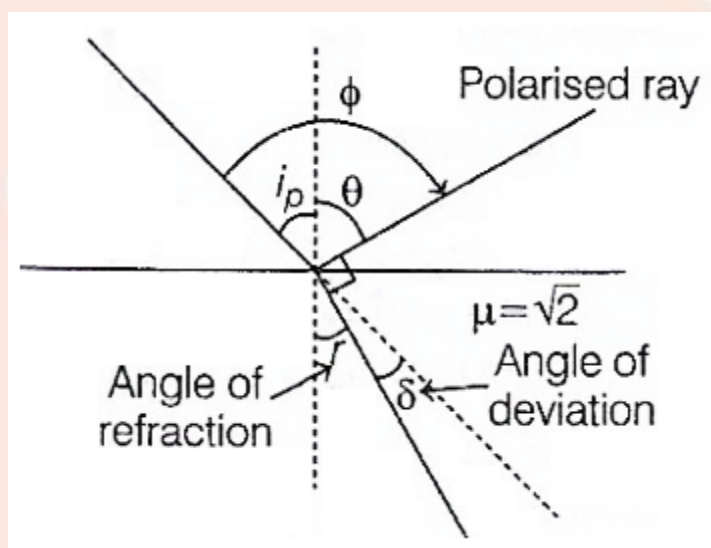
$$\begin{aligned}
 \tan \theta &= \frac{PM_1}{OP} \\
 &= \frac{I_1 I_2}{OP_1} \\
 \frac{x}{2a} &= \frac{140 - 85}{a} \\
 x &= 110 \text{ cm}
 \end{aligned}$$

The maximum height seen by the girl is calculated as,

$$H = 110 + 10$$

$$= 120 \text{ cm}$$

103. The ray diagram of refraction at a plane surface is shown below.



The value of i_p is calculated as,

$$i_p = \tan^{-1} \mu$$

$$= \tan^{-1} \sqrt{2}$$

$$= \sin^{-1} \sqrt{\frac{2}{3}}$$

From Snell's law,

$$\sin i_p = \mu \sin r$$

$$\sin r = \frac{\sin i_p}{\mu}$$

Substitute the given values in the above equation.

$$\sin r = \frac{\sin i_p}{\mu}$$

$$r = \frac{1}{\sqrt{2}} \sqrt{\frac{2}{3}}$$

$$r = \sin^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

The angle of the deviation is calculated as,

$$\begin{aligned} \delta &= i_p - r \\ &= \sin^{-1} \sqrt{\frac{2}{3}} \sin \end{aligned}$$

The angle of reflection is,

$$\theta = 90^\circ - r$$

$$r = 90^\circ - \theta$$

$$\sin^{-1} \frac{1}{\sqrt{3}} = 90^\circ - \theta$$

$$\frac{1}{\sqrt{3}} = \sin(90^\circ - \theta)$$

Solve further,

$$\cos \theta = \frac{1}{\sqrt{3}}$$

$$\theta = \cos^{-1} \frac{1}{\sqrt{3}}$$

The angle between incident and completely polarised light is calculated as,

$$\begin{aligned}\phi &= i_p + \theta \\ &= \sin^{-1} \sqrt{\frac{2}{3}} + \cos^{-1} \left(\frac{1}{\sqrt{3}} \right) \\ &= 2 \sin^{-1} \sqrt{\frac{2}{3}}\end{aligned}$$

104. The expression for the electric field intensity on the axis is given by,

$$\begin{aligned}E_{\text{axis}} &= \frac{2kP}{r^3} \\ 4 &= \frac{2kP}{r^3} \\ \frac{kP}{r^3} &= 2\end{aligned}$$

The expression for the electric field intensity on the equatorial axis is given by,

$$E_{eq} = \frac{k'P}{r_1^3}$$

Here, $r_1 = 2r$ and $k' = \frac{k}{4}$

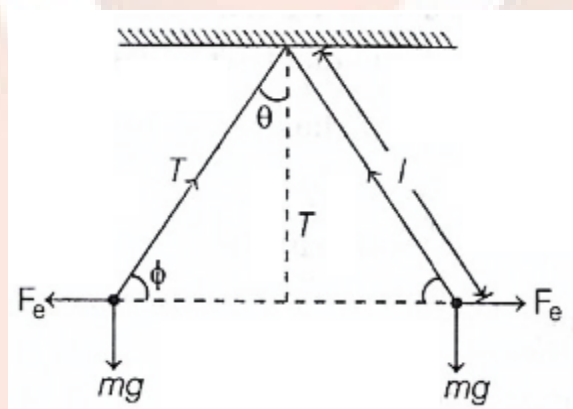
So,

$$E_{eq} = \frac{kP}{4 \times 8r^3}$$

Substitute 2 for $\frac{kP}{r^3}$ in the above equation.

$$\begin{aligned} E_{eq} &= \frac{2}{4 \times 8} \\ &= \frac{1}{16} \text{ NC}^{-1} \end{aligned}$$

105. Consider the below diagram.



$$F_e = \frac{kq^2}{r^2} \dots (1)$$

$$r = 2l \cos \theta$$

$$\tan \theta = \frac{F_e}{mg}$$

For the value of $\theta = 45^\circ$ and $\phi = 45^\circ$.

$$r = \frac{2l}{v^2}$$

$$= \sqrt{2}l$$

Substitute the value of r in equation (1).

$$F_e = \frac{kq^2}{2l}$$

From the geometry of the triangle,

$$\tan 45^\circ = \frac{kq^2}{2lmg}$$

$$mg = \frac{kq^2}{2l} \dots (2)$$

For the value of $\theta = 30^\circ$ and $\phi = 60^\circ$.

$$r = 2l \times \frac{1}{2}$$

$$= l$$

Similarly,

$$F'_e = \frac{kq^2}{l^2}$$

So,

$$\tan 30^\circ = \frac{k'q^2}{l^2 - m'g}$$

Assuming that the sphere is suspended in a liquid of density $\frac{2}{3}d$.

The observed weight of the body is calculated as,

$$m' = V \left(d - \frac{2d}{3} \right)$$

$$= \frac{m}{3}$$

Thus,

$$\tan 30^\circ = \frac{3k'q^2}{l^2 mg}$$

$$mg = \frac{3\sqrt{3}k'q^2}{l^2} \dots (3)$$

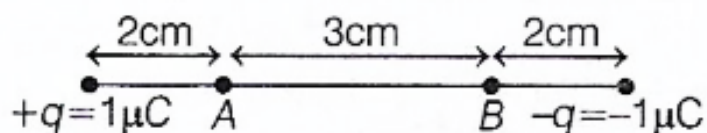
From equation (2) and (3),

$$3\sqrt{3}k' = \frac{k}{2}$$

$$k' = \frac{k}{\epsilon_r}$$

$$\epsilon_r = 6\sqrt{3}$$

106. Consider the below diagram.



The potential at the point A is calculated as,

$$\begin{aligned}
 V_A &= V_{A^{+q}} + V_{A^{-q}} \\
 &= \frac{9 \times 10^9 \times 10^{-6}}{2 \times 10^{-2}} - \frac{9 \times 10^9 \times 10^{-6}}{5 \times 10^{-2}} \\
 &= 9 \times 10^5 \left[\frac{1}{2} - \frac{1}{5} \right]
 \end{aligned}$$

The potential at B is calculated as,

$$\begin{aligned}
 V_B &= V_{B^{+q}} + V_{B^{-q}} \\
 &= 9 \times 10^5 \left[\frac{1}{5} - \frac{1}{2} \right]
 \end{aligned}$$

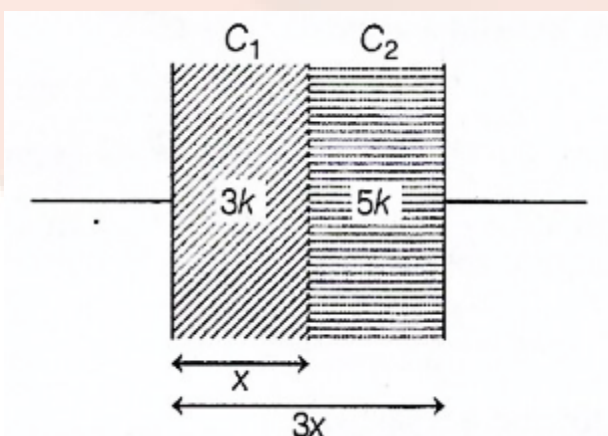
The potential difference is calculated as,

$$V_A - V_B = 5.4 \times 10^5 \text{ V}$$

107. The expression for the capacitance of a parallel plate is,

$$C = \epsilon_r \cdot \frac{\epsilon_r A}{d}$$

The capacitor is shown below.



Here, $d_0 = 3x$, $d_1 = x$, $\varepsilon_1 = 3k$, $d_2 = 2x$ and $\varepsilon_2 = 5k$

The capacitance across the plates is calculated as,

$$C_1 = 3k \frac{\varepsilon_0 A}{x} \text{ and } C_2 = 5k \frac{\varepsilon_0 A}{2x}$$

For the combination of the capacitor in the series.

$$V_1 = \frac{Q}{C_1} \text{ and } V_2 = \frac{Q}{C_2}$$

Substitute the values in the above equation.

$$V_1 = \frac{\theta x}{3k\varepsilon_0 A} \text{ and } V_2 = \frac{\theta 2x}{5k\varepsilon_0 A}$$

The ratio of the potential difference is calculated as,

$$\begin{aligned} \frac{V_1}{V_2} &= \frac{\frac{\theta x}{3k\varepsilon_0 A}}{\frac{\theta 2x}{5k\varepsilon_0 A}} \\ &= \frac{5}{3 \times 2} \\ &= \frac{5}{6} \end{aligned}$$

108. The conductivity of the metal decreases with the increase in the temperature whereas the conductivity of semiconductor increases with the increase in the temperature.

So, the conductivity of aluminium decreases with the increase in the temperature whereas the conductivity of the silicon increases with the increase in the temperature. Hence, assertion (A) is incorrect but the reason (R) is correct.

109. The expression for the heat transfer through the conduction wall is given by,

$$\frac{dQ}{dt} = kA \frac{(T_1 - T_0)}{x}$$

Substitute the given values in the above equation.

$$\begin{aligned} P &= \frac{kA \times 4.184 \times (T_1 - T_0)}{x} \\ &= \frac{k6a^2 \times 4.184 \times 10^3}{x} \\ &= \frac{4 \times 10^{-4} \times 6 \times (60^2) \times 10^3 \times 4.184}{0.1} \\ &= 361.49 \text{ W} \end{aligned}$$

The resistance is calculated as,

$$\begin{aligned}P &= \frac{V^2}{R} \\R &= \frac{V^2}{P} \\&= \frac{400 \times 400}{361.49 \times 10^3} \\&= 0.4426 \Omega\end{aligned}$$

110. The expression of the net resistance of circuit is,

$$R_0 = R + \frac{G \times S}{G + R} \dots (1)$$

According to the given condition,

$$R_0 = G \dots (2)$$

From the equation (1) and (2),

$$G = R + \frac{G \times S}{G + S}$$

$$R = G - \frac{GS}{G + S}$$

$$R = \frac{G^2}{G + S}$$

111. The expression for the time period of revolution of proton in the magnetic field is given by,

$$T_p = \frac{2\pi m_p}{eB} \dots (1)$$

The expression for the time period of revolution of α particle in the magnetic field is given by,

$$T_\alpha = \frac{2\pi m_p}{qB} \dots (2)$$

The ratio of the time period of the proton and α particle is calculated as,

$$\frac{T_p}{T_\alpha} = \frac{1}{2}$$

$$T_\alpha = 2T_p$$

112. The expression for the torques in the magnetic field is given by,

$$\tau = MB \sin \theta$$

For $\theta = \theta_1$.

$$\tau_1 = MB \sin \theta_1 \dots (1)$$

Similarly,

$$\begin{aligned} \tau_2 &= MB \sin \theta_2 \\ &= MB \sin 2\theta_1 \end{aligned}$$

According to the given condition,

$$\begin{aligned}
 \tau_2 &= \tau_1 + \tau_1 \times \frac{41.4}{100} \\
 &= 1.414\tau_1 \\
 &= \sqrt{2}\tau_1
 \end{aligned}$$

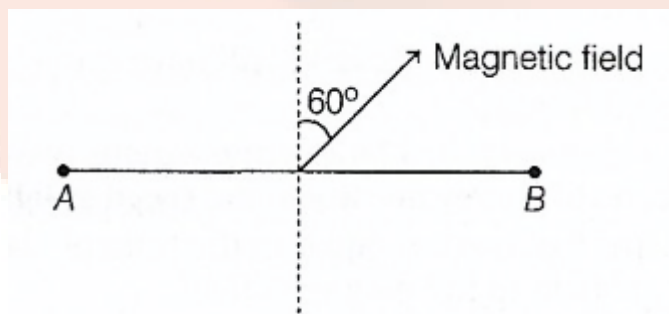
So,

$$\sqrt{2}\tau_1 = MB \sin 2\theta_1 \dots (2)$$

From equation (1) and (2),

$$\begin{aligned}
 \frac{1}{\sqrt{2}} &= \frac{\sin \theta_1}{\sin 2\theta_1} \\
 \sin 2\theta_1 &= \sqrt{2} \sin \theta_1 \\
 2 \sin \theta_1 \cos \theta_1 &= \sqrt{2} \sin \theta_1 \\
 \cos \theta_1 &= \frac{\sqrt{2}}{2} \\
 \theta_1 &= 45^\circ
 \end{aligned}$$

113. Consider the below diagram.



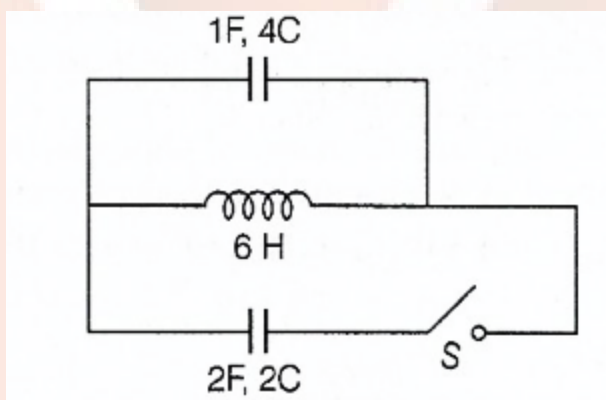
The expression for the emf induced in the metal rod is given by,

$$\begin{aligned}e_{AB} &= V_A - V_B \\&= l \cdot (\mathbf{v} \times \mathbf{B}) \\&= lvB \sin(90^\circ - 60^\circ)\end{aligned}$$

Substitute the given values in the above equation.

$$\begin{aligned}V_A - V_B &= 50 \times 10^{-2} \times 8 \times 2 \times \sin 30^\circ \\&= 8 \times \frac{1}{2} \\&= 4 \text{ V}\end{aligned}$$

114. Consider the below arrangement of the electrical circuit.



The energy in 1F capacitor is calculated as,

$$\begin{aligned}E_1 &= \frac{1}{2}CV^2 \\&= \frac{1}{2}C\left(\frac{Q}{C}\right)^2 \\&= \frac{1}{2}\frac{Q^2}{C} \\&= \frac{1}{2} \times \frac{4^2}{1} \\&= 8 \text{ J}\end{aligned}$$

Similarly,

$$\begin{aligned}E_2 &= \frac{1}{2}\frac{Q^2}{C} \\&= \frac{2^2}{2 \times 2} \\&= 1 \text{ J}\end{aligned}$$

The total energy is calculated as,

$$\begin{aligned}E_0 &= E_1 + E_2 \\&= 8 + 1 \\&= 9 \text{ J}\end{aligned}$$

The common potential of the capacitors is calculated as,

$$\begin{aligned}V_{common} &= \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} \\&= \frac{1 \times 4 + 2 \times 1}{1 + 2} \\&= \frac{6}{3} \\&= 2 \text{ V}\end{aligned}$$

For the new arrangement of the circuit the energy is calculated as,

$$\begin{aligned}E_1 &= \frac{1}{2} C_1 V_{common}^2 \\&= \frac{1}{2} \times 1 \times 4 \\&= 2 \text{ J}\end{aligned}$$

Similarly,

$$\begin{aligned}E_2 &= \frac{1}{2} C_2 V_{common}^2 \\&= \frac{1}{2} \times 2 \times 4 \\&= 4 \text{ J}\end{aligned}$$

From the conservation of the energy, the energy stored in the inductor is calculated as,

$$\begin{aligned}E_L &= E_0 - (E_1 + E_2) \\&= 9 - (2 + 4) \\&= 3 \text{ J}\end{aligned}$$

115. The energy of the electromagnetic wave is equally distributed between electric and magnetic field.

Thus, option (2) is correct.

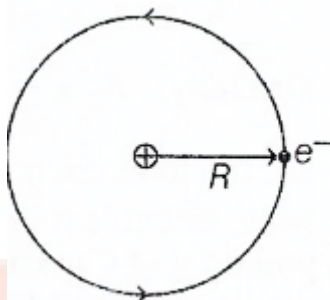
116. The maximum energy of photoelectron is calculated as,

$$\begin{aligned}E_{\max} &= \frac{h\nu_{\max}}{e} \\&= \frac{6.6 \times 10^{-34} \times (3.6 \times 10^{15} + 1.2 \times 10^{15})}{1.6 \times 10^{-19}} \\&= 19.8 \text{ eV}\end{aligned}$$

The maximum kinetic energy is calculated as,

$$\begin{aligned}KE_{\max} &= E_{\max} - W_0 \\&= 19.8 - 2.35 \\&= 17.45 \text{ eV}\end{aligned}$$

117. The magnetic movement of the electron is shown below.



The expression for the magnetic moment is,

$$M = iA$$

$$= \frac{e\omega}{2\pi} \pi R^2$$

Substitute the value of current in the above equation.

$$M = \frac{ev}{2R} R^2$$

$$= \frac{evR}{2}$$

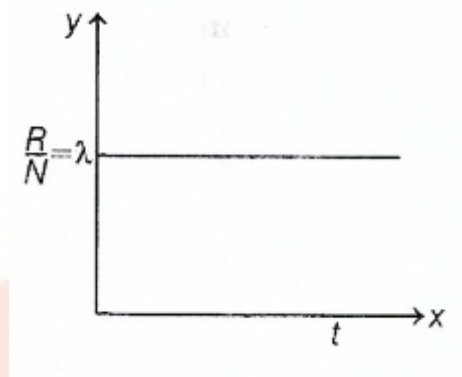
$$= \frac{eL}{2m}$$

Simplify further using the relations $L = mvr = \frac{nh}{2\pi}$.

$$M = \frac{enh}{2\pi m}$$

$$M \propto n$$

118. Consider the below diagram.



The statement of the law of the radioactive decay is,

$$\frac{dN}{dt} = -\lambda N$$

$$R = -\lambda N$$

$$R = -\lambda$$

Thus the correct graph is shown in option (4).

119. The maximum energy of the photon on the visible spectrum is calculated as,

$$\begin{aligned} E_{\max} &= h\nu \\ &= \frac{6.6 \times 10^{-34} \times 7 \times 10^{14}}{1.6 \times 10^{-19}} \\ &= 2.88 \text{ eV} \end{aligned}$$

The minimum energy of the photon on the visible spectrum is calculated as,

$$\begin{aligned}
 E_{\max} &= h\nu \\
 &= \frac{6.6 \times 10^{-34} \times 4 \times 10^{14}}{1.6 \times 10^{-19}} \\
 &= 1.65 \text{ eV}
 \end{aligned}$$

Which implies the correct range of the energy is 1.7 eV to 3 eV .

120. The expression of the maximum distance of line of sight mode is,

$$d_m = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

Substitute the given value in the above equation.

$$40 = \sqrt{2 \times 6.4 \times 10^3 \times 20 \times 10^{-3}} + \sqrt{2 \times 6.4 \times 10^3 \times h_R \times 10^{-3}}$$

$$40 = 16 + \sqrt{2 \times 6.4 \times 10^3 \times 10^{-3} \times h_R}$$

$$2 \times 6.4 \times h_T = 24 \times 24$$

$$h_T = 45 \text{ m}$$

AP EAMCET-2019-Engineering-21 Apr-Shift-1
Solutions – Chemistry

121. The expression of energy is given by,

$$E_n = -2.18 \times 10^{-18} \cdot \frac{Z^2}{n^2}$$

For Helium ion, $Z = 2$, $n = 1$

Therefore,

$$\begin{aligned} E_n &= -2.18 \times 10^{-18} \cdot \frac{(2)^2}{(1)^2} \\ &= -8.72 \times 10^{-18} \text{ J} \end{aligned}$$

For Lithium ion, $Z = 3$, $n = 3$

Therefore,

$$\begin{aligned} E_n &= -2.18 \times 10^{-18} \cdot \frac{(3)^2}{(3)^2} \\ &= -2.18 \times 10^{-18} \text{ J} \end{aligned}$$

122. Symbol l represents value of sub-shell and symbol n represent number of sub-shell and symbol m_l represent the value of orientation of the orbital.

Given that,

$$n = 3, l = 1 \text{ and } m_l = -1$$

It means an orbital is present in p-shell of 3rd shell and have orientation value -1.

Therefore, only one orbital is possible for the given set of values of n , l , and m_l .

123. Z represents number of protons in a nucleus.

Thus, for an isoelectronic species,

Size of anion > atom > cation .

It means the species with more electrons w.r.t. protons in nucleus is of larger size that is, has smaller value of Z .

Therefore, $Z - 2$ has least number of protons, whereas $Z - 1$ has maximum number of protons.

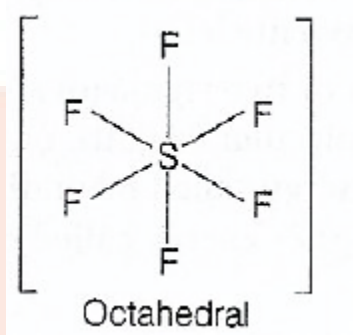
Thus, the order of size for isoelectronic species is

$$Z - 2 > Z > Z + 1$$

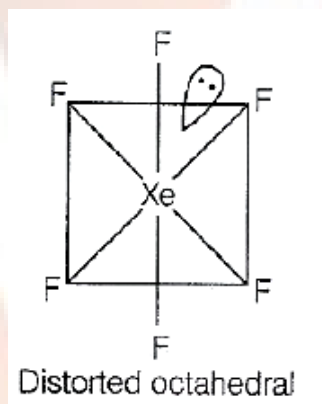
124. The arrangement of electron pairs around the central atom is octahedral and shape is not octahedral and can be identified with the help of presence of lone pair of electrons and bond of electrons associated with the central atom.

The structure of given species are as follows.

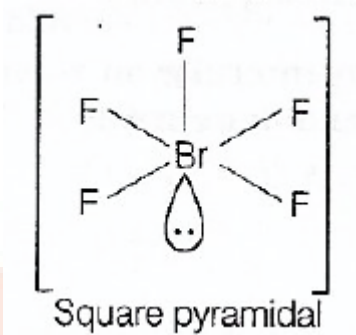
In SF_6 , the central atom sulphur has 6 valence electrons, which are bonded with 6F-atoms, thus have octahedral shape.



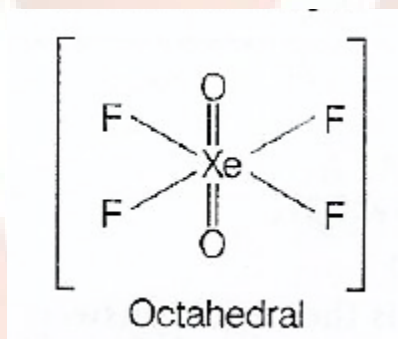
In XeF_6 , central atom, Xe has 8 valence electrons, out of which 6 are bonded with 6 F-atoms and one *lp* of electron is associated with Xe-atom. Thus, has distorted octahedral shape.



In BrF_5 , central atom (Br) has 7 valence electrons, out of which 5 are bonded with 5 F-atoms and one *lp* of electron is associated with Br-atom having square pyramidal shape.



In XeO_2F_4 , central atom, Xe has 8 valence electrons, out of which 4 are bonded through 4-sigma bonds with 4F-atoms and two oxygen atoms each having one sigma bond and one pi bond is associated with Xe-atom, Thus, has octahedral shape.



Thus, option (3) is correct.

125. According to molecular orbital theory:

I. σ^* -orbital are formed by the subtraction of two wave functions say ψ_A and ψ_B .

Therefore, $\sigma^* = \psi_A - \psi_B$.

II. σ^* -orbital are formed when two orbital's are in

same phase and thus, do not have nodal plane in between two nuclei.

III. Combination of two sigma atomic orbital's

given two molecular orbital's, out of which one is of lower energy called a bonding orbital and other is of higher energy called c? anti-bonding orbital.

Hence, (I) and (III) are the correct statements and option (4) is the correct answer.

126.. The boiling point of a liquid is that temperature at which its vapor pressure becomes equal to the atmospheric pressure. Weaker is the attractive force between the molecules of given species, more is its volatile nature and smaller is its boiling point.

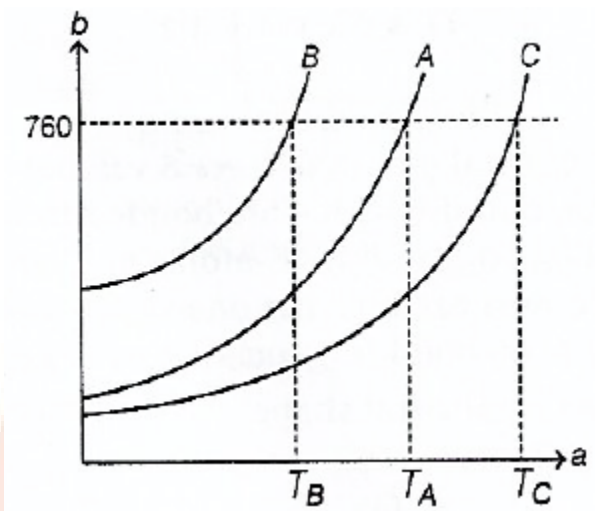
$$\text{Attractive force} \propto \frac{1}{\text{Volatile nature}},$$

$$\text{Volatile nature} \propto \text{Vapour pressure}$$

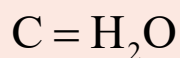
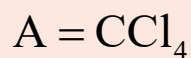
And,

$$\text{Temperature for boiling point} \propto \frac{1}{\text{Volatile nature}}$$

Now draw perpendicular on x-axis from point B, A, and C and we have $T_C > T_A > T_B$.



Therefore,



127. Use the following expression to calculate the concentration of HCl (M_1).

$$M_1 V_1 = M_2 V_2$$

Substitute the values.

$$\begin{aligned}M_1 &= \frac{M_2 V_2}{V_1} \\&= \frac{0.1 \times 20}{30} \\&= 0.066 \text{ M}\end{aligned}$$

Calculate the final volume of HCl as follows,

$$\begin{aligned}M_1 V_f &= M_3 V_3 \\V_f &= \frac{M_3 V_3}{M_1} \\&= \frac{0.2 \times 30}{0.066} \\&= 90 \text{ mL}\end{aligned}$$

128. Calculate the temperature difference as follows,

$$\begin{aligned}\Delta T &= 60^\circ\text{C} - 40^\circ\text{C} \\&= 20^\circ\text{C}\end{aligned}$$

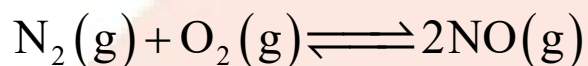
Calculate the number of moles of Al as follows,

$$\begin{aligned}n &= \frac{w}{M} \\&= \frac{54}{27} \\&= 2 \text{ mol}\end{aligned}$$

Calculate the heat required as follows,

$$\begin{aligned}Q &= nC_s \Delta T \\&= 2 \times 24 \times 20 \\&= 960 \text{ J}\end{aligned}$$

129. Given,



And,

$$[\text{N}_2] = [\text{O}_2]$$

The expression of equilibrium constant is given by,

$$\begin{aligned}K_c &= \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]} \\&= \frac{[\text{NO}]^2}{[\text{N}_2]^2}\end{aligned}$$

Substitute the values.

$$\begin{aligned}0.5625 &= \frac{[3 \times 10^{-3}]^2}{[\text{N}_2]^2} \\[\text{N}_2]^2 &= 16 \times 10^{-6} \\&= 4 \times 10^{-3}\end{aligned}$$

130. Use the following expression to calculate the solubility.

$$K_{sp} = x^x \cdot y^y \cdot S^{x+y}$$

Substitute the values.

$$3.2 \times 10^{-11} = 2^2 \cdot 1^1 \cdot S^{2+1}$$

$$S^3 = \frac{3.2}{4} \times 10^{-11}$$

$$= 8 \times 10^{-12}$$

$$S = 2 \times 10^{-4}$$

131. Calculate the Normality of H_2O_2 .

$$\begin{aligned} \text{Normality of } H_2O_2 &= \frac{\text{Volume strength}}{\text{Equivalent weight}} \\ &= \frac{20}{5.6} \\ &= 3.57 \text{ N} \end{aligned}$$

The reaction is in acidic medium for $KMnO_4$ so, valence factor will be 5.

The normality of $KMnO_4$ is,

$$N_{KMnO_4} = \text{Molarity} \times 5$$

Use the following expression.

$$N_1 V_1 (H_2O_2) = N_2 V_2 (KMnO_4)$$

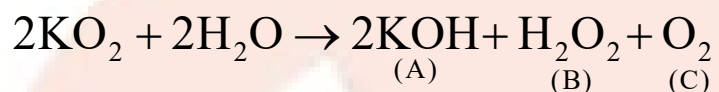
Substitute the values.

$$3.57 \times V_1 = 0.02 \times 5 \times 500$$

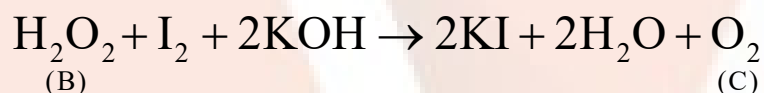
$$V_1 = \frac{50}{3.57}$$

$$= 14.0 \text{ mL}$$

132. When KO_2 reacts with water it forms A, B, and C as follows,



When B reacts with iodine in basic medium as follows,

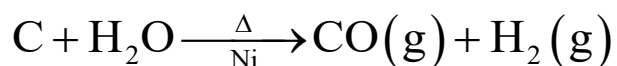


133. The Ga_2O_3 can reacts with both acid and base so, it is amphoteric oxide.

The dimer of aluminium chloride has three $\text{Al}-\text{Cl}$ bridge bonds.

The melting point of Boron is 2349 K so, it is very hard refractory solid of high melting point.

134. When steam (H_2O) is passed over coke (C), it gives CO and H_2 as follows,

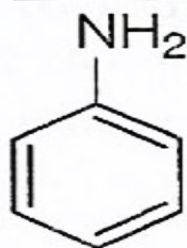


135. The correct match is as follows,

A) Insecticide	Na_3AsO_3
B) $\text{K}_2\text{Cr}_2\text{O}_7$ / 50% H_2SO_4	COD
C) Bleaching of cloths and paper	H_2O_2
D) Eye Irritant	PAN

136. The group in which nitrogen contains double bond and present in hetrocyclic ring cannot be estimated by Kjeldahl's method.

The structure of Aniline is as follows,

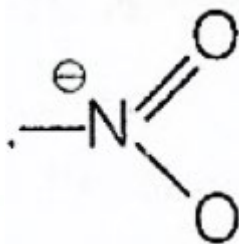


The nitrogen present in this structure does not contain double bond so it can be estimated by Kjeldahl's method.

The structure of Azobenzene possesses, $-\text{N}=\text{N}-$ group.

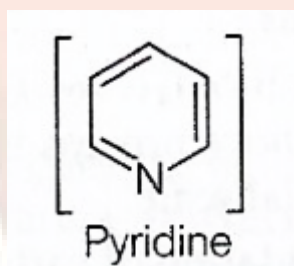
The nitrogen present in this structure contains double bond so it cannot be estimated by Kjeldahl's method.

The structure of Nitrobenzene is as follows,



The nitrogen present in this structure contains double bond so it cannot be estimated by Kjeldahl's method.

The structure of Pyridine is as follows,

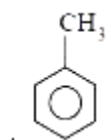


In this structure nitrogen is present in hetrocyclic ring so it cannot be estimated by Kjeldahl's method.

Thus, the correct option is (1).

137. The structure will have maximum number of no bond resonance structures if more H-atoms bonded with C-atoms which are bonded with the benzene ring.

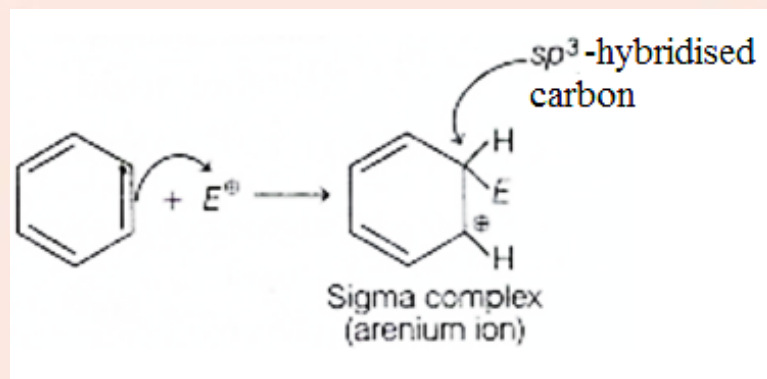
In structure,



The C-atom bonded with the ring has 3 H-atoms.

Thus, the correct option is (4).

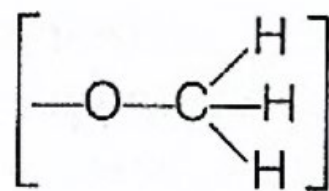
138. (II) σ complex is the intermediate substance in electrophilic substitution of benzene is correct statement it forms arenium ion as intermediate when electrophile attack on benzene ring.



Benzene on Friedel-Crafts alkylation with n-propyl chloride gives isopropyl benzene is correct statement.

139. The functional groups which contains double bond in its structure and C-atom, which is bonded with the ring has more electronegative oxygen atom bonded with it is meta directing product.

In structure,



There is no double bond and O-atom, which bonded with the ring has less electronegative C-atom bonded with it. Therefore, it does not give meta directing product.

140. Calculate the edge length (a).

$$\begin{aligned}a &= \frac{4}{\sqrt{3}} r \\&= \frac{4}{\sqrt{3}} (173.2 \times 10^{-10} \text{ cm}) \\&= 400 \times 10^{-10} \text{ cm}\end{aligned}$$

Calculate the volume as follows.

$$\begin{aligned}V &= a^3 \\&= (400 \times 10^{-10} \text{ cm})^3 \\&= 6.4 \times 10^{-23} \text{ cm}^3\end{aligned}$$

141. The expression of relative lower in vapour pressure of solution is given by,

$$\frac{\Delta p}{p^0} = \chi_B = \frac{n_B}{n_A + n_B}$$

Calculate the number of moles of solvent A and solute B.

$$n_A = \frac{360}{18}$$

$$= 20$$

$$n_B = \frac{w_B}{M_B}$$

$$= \frac{10}{M_B}$$

Now,

$$5 \times 10^{-3} = \frac{\frac{10}{M_B}}{20 + \frac{10}{M_B}}$$

$$5 \times 10^{-3} = \frac{10}{20M_B + 10}$$

$$20M_B + 10 = 2000$$

$$M_B = 99.5 \text{ g mol}^{-1}$$

142. Use the following expression to calculate the value of x.

$$p = iCRT = i \times \frac{w}{M} \times \frac{1}{V} \times RT$$

$$2.463 = \frac{1.8 \times x \times 0.082 \times 300}{120 \times 2.5}$$

$$x = \frac{2.463 \times 120 \times 2.5}{1.8 \times 0.082 \times 300}$$

$$\approx 16.6 \text{ g}$$

143. The expression of free energy is given by,

$$[\Delta G^0] = -nFE^0$$

Thus,

For $n = 1$,

$$\begin{aligned} -\Delta G_1^0 &= nE_1^0 \\ &= E_1^0 \end{aligned}$$

$$\begin{aligned} -\Delta G_2^0 &= nE_2^0 \\ &= E_2^0 \end{aligned}$$

For $n = 2$,

$$\begin{aligned} -\Delta G_3^0 &= nE_3^0 \\ &= 2E_3^0 \end{aligned}$$

For reaction $M^{2+}(aq) + 2e^- \rightarrow M(s)$

Therefore,

$$-\Delta G_3^0 = (-)\Delta G_1^0 + (-)\Delta G_2^0$$

$$\begin{aligned} 2E_3^0 &= (E_1^0 + E_2^0) \\ &= 0.15 + 0.5 \end{aligned}$$

$$E_3^0 = 0.325 \text{ V}$$

144. Calculate the values of K_1 and K_2 .

$$K_1 (\text{at } 300 \text{ K}) = \frac{0.693}{50}$$

$$= 0.014 \text{ s}^{-1}$$

$$K_1 (\text{at } 400 \text{ K}) = \frac{0.693}{10}$$

$$= 0.07 \text{ s}^{-1}$$

Calculate the activation energy of the reaction as follows,

$$\log \frac{K_2}{K_1} = \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_1 \cdot T_2} \right]$$

Substitute the values.

$$\log \left(\frac{0.07}{0.014} \right) = \frac{E_a}{2.303 \times 8.314} \left[\frac{400 - 300}{400 \times 300} \right]$$

$$\log(5) = \frac{E_a}{2.303 \times 8.314} \left[\frac{100}{400 \times 300} \right]$$

$$E_a = 0.7 \times 19.15 \times 1200$$

$$= 16.08 \text{ kJ mol}^{-1}$$

145. The extent of adsorption decreases with an increased in temperature.

$$\text{Adsorption} \propto \frac{1}{\text{temperature}}$$

146. The composition of German silver is,

Zinc (Zn) = about 35%

Copper (Cu) = about 50%

Nickel (Ni) = about 15%

The composition of Brass is,

Zinc (Zn) = about 30%

Copper (Cu) = about 70%

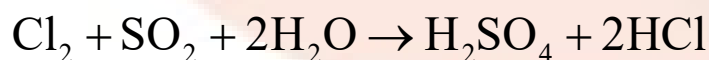
Iron, Fe, is not present in both of them.

147. $\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$ is correct reaction and its occur in two steps.

$2\text{NaNO}_2 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{NO}_2 + \text{H}_2\text{O}$ is also the correct reaction because HCl can decompose salts like NaNO_2 to NaCl, NO_2 and H_2O .

148. For first case,

The reaction is as follows,



Calculate the oxidation number of sulphur in H_2SO_4 .

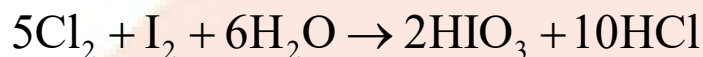
$$2 + x + 4(-2) = 0$$

$$x + 2 - 8 = 0$$

$$x = +6$$

For second case,

The reaction is as follows,



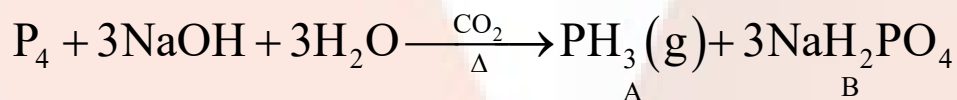
Calculate the oxidation number of I in HIO_3 .

$$1 + x + 3(-2) = 0$$

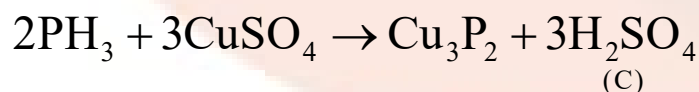
$$x + 1 - 6 = 0$$

$$x = +5$$

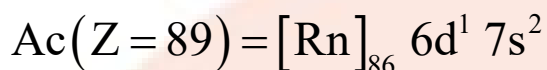
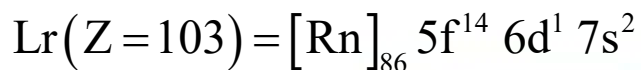
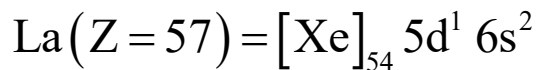
149. When white phosphorous is heated with concentrated NaOH in inert atmosphere of CO_2 , the products obtained are as follows,



When (A) is bubbled into aqueous CuSO_4 solution, products obtained are as follows,



150. The electronic configuration of each element is as follows,



Therefore, La, Ac and Th do not have f-electrons.

151. The expression of energy difference is given by,

$$\Delta_0 = E_{\text{Higher}} - E_{\text{Lower}}$$

$$= E_{e_g} - E(t_{2g})$$

Substitute the values.

$$1000 = E_{e_g} - (-400)$$

$$1000 = E_{e_g} + 400$$

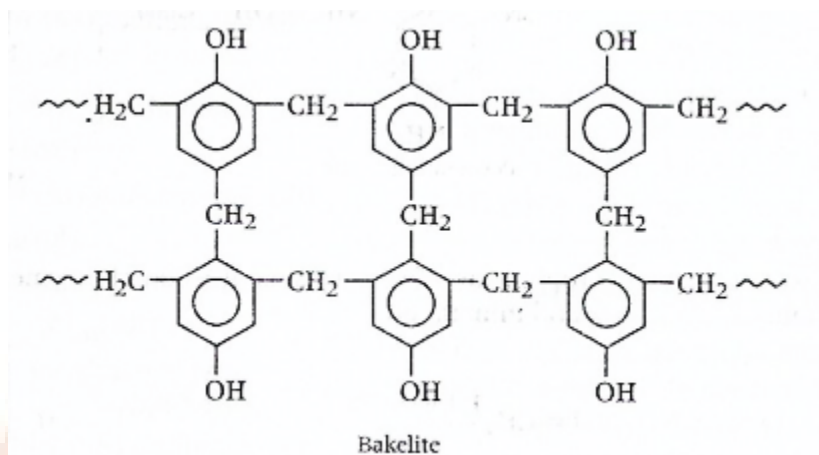
$$E_{e_g} = 1000 - 400$$

$$= 600 \text{ kJ/mol}$$

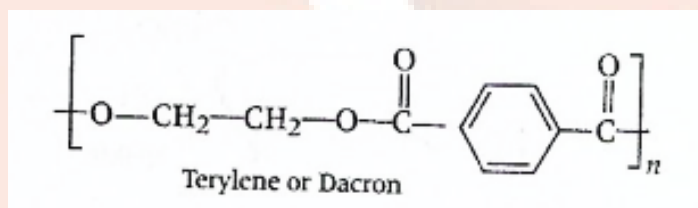
152. The condensation polymers are Bakelite, Melamine, and Dacron.

The structure of these polymers is as follows,

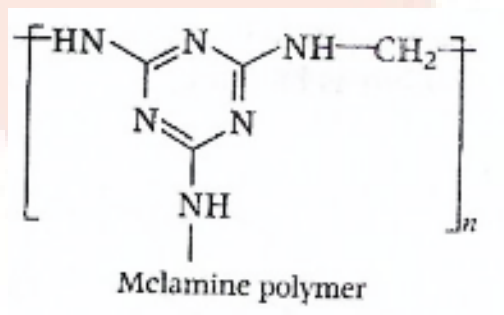
Bakelite,



Dacron,



Melamine,

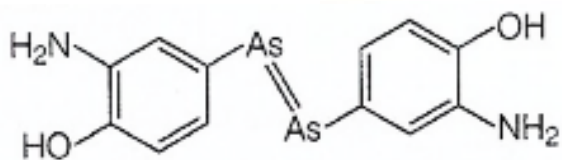


153. The source of vitamin B₆ is milk and it causes deficiency disease known as convulsions.

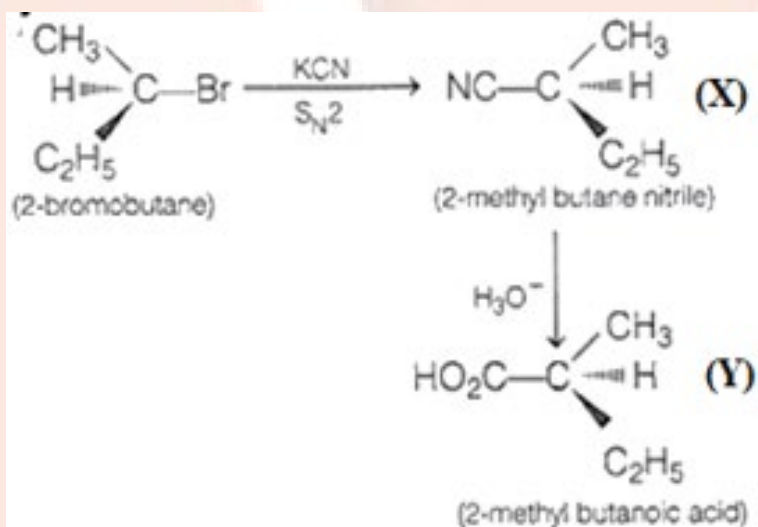
Therefore, the correct option is 1.

154. The $-\text{As} = \text{As} -$ bond is present in Salvarsan.

The structure of Salvarsan is as follows:



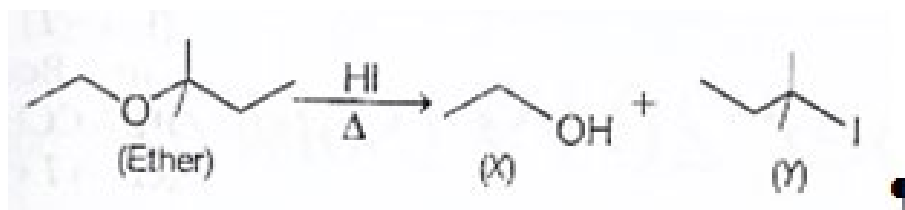
155. The product (X) and (Y) formed are as follows:



156. The correct pair is as follows,

Sets	Property
(I) $\text{NCCH}_2\text{COOH} > \text{FCH}_2\text{COOH} > \text{H}_3\text{CCH}_2\text{COOH}$	acidity
(II) $\text{H}_3\text{COCH}_2\text{CH}_3 < \text{H}_3\text{CCH}_2\text{CHO} < \text{H}_3\text{CCH}_2\text{CH}_2\text{OH}$	boiling points

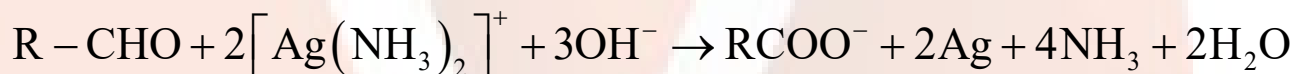
157. The products (X) and (Y) formed are as follows,



In the presence of 3° carbon atom, the group $-I$ combines via S_N1 mechanism (Z).

158. When an aldehyde ($RCHO$) reacts with Tollens reagent

$[Ag(NH_3)_2]^+$, the product formed is as follows,



159. The correct sequence of formation of the given species is

4, 1, 2, 3.

160. The reagents (X), (Y) and (Z) are shown below.

